

GUYANA | EXPLORATION TO PRODUCTION

THE GUYANA SEQUENCE

Petroleum Governance, Environmental Stewardship, Natural Resource
Fund Design and Lessons for Emerging Producers

Independent research by Ragunauth Ramsaroop

Working research edition | 27 September 2026

Expanded research compendium with ten research parts, technical workbooks and a documented source register.
The final 520-560 substantive-page flagship and external expert review remain pending.

Source evidence cut-off: 27 September 2026. Observed 2026 financial data are through June unless indicated.

Contents

Executive research brief	23
Audit status	23
Purpose	23
Verified sequence	23
September 2026 evidence snapshot	24
The distinction most useful to a new producer	24
Institutional subjects for detailed scrutiny	24
Comparator lenses	25
Implementation workbook for a newly discovering jurisdiction	25
Publication acceptance conditions	25
Substantive manuscript	28
Abstract	28
Method and definitions	28
1. Before discovery: institutions, geology and the timing problem	29
2. The discovery-to-first-oil window, 2015-2019	30
What early fund establishment demonstrates, and what it does not	31
3. From first oil to large-scale production, 2019-September 2026	31
4. The Natural Resource Fund as a fiscal institution	32
Cash management is not public investment evaluation	33
5. Fiscal contract oversight and the cost bank	34
6. Environmental law, liability and offshore operations	35
7. Forest conservation, carbon finance and the fossil-fuel coexistence problem	36
8. Local value, social outcomes and accountability	37
9. Five international comparator designs and limits	38
10. Transferable administrative sequence and adverse cases	39
Adverse case A: a long low-price interval	40
Adverse case B: newly approved development replenishes the cost bank	40
Adverse case C: a major offshore spill	40

Adverse case D: overheating without fiscal insolvency	40
Adverse case E: public confidence breaks despite disclosure	40
11. Initial findings and publication boundaries	41
Technical appendix A. Analytical definitions and worked examples	41
Technical appendix B. The newcomer decision record	42
Technical appendix C. Questions for external reviewers	42
References	43
Transferable country handbook	44
THE NEW-PRODUCER HANDBOOK	44
A staged institutional workbook for an emerging oil-producing country or island state	44
1. Start with a resource and institutional map rather than a model law	44
2. Exploration stage: six documentary systems	45
3. After discovery: the contract and approval decision file	46
4. The first-oil readiness rehearsal	47
5. Designing a fund: alternatives rather than a copied withdrawal percentage	48
6. The petroleum revenue transparency ledger	49
7. Local participation and the value-added question	50
8. Indigenous peoples, coastal livelihoods and public participation	51
9. Environmental assurance across the full project lifecycle	52
10. Public investment and durable service outcomes	53
11. Five-year observation schedule	53
12. Ten cross-agency exercises for the implementation team	54
13. What Guyana's case does not establish	55
14. Compact reusable templates	56
New producer implementation blueprint	58
New Producer Implementation Blueprint	58
A configurable exploration-to-production governance system derived from the Guyana case ⁵⁸	
How to use this blueprint	58
Phase 0: Before a commercial discovery	58

A. Legal and institutional map	58
B. Geological information governance	58
C. Environmental baseline	59
D. Fiscal readiness	59
Phase 1: Commercial discovery	60
Gate 1: Independent discovery review	60
Gate 2: Contract and fiscal review	60
Gate 3: Sovereign revenue architecture	60
Phase 2: Development approval	61
Technical approval packet	61
Environmental decision packet	61
Public revenue packet	61
Phase 3: Twelve months before first oil	62
Revenue rehearsal	62
Emergency rehearsal	62
Public investment readiness	63
Phase 4: First oil through rapid scale-up	63
Quarterly production-to-cash statement	63
Cost-audit programme	63
Local-value ledger	64
Environmental performance ledger	64
Phase 5: Production maturity	64
Fiscal transition	64
Economic transition	65
Resource depletion plan	65
A special module for small islands	65
Marine concentration risk	65
Land and labour limits	65
Fiscal scale	66

Natural-capital differences	66
Disaster overlap	66
Twenty-four-month government action calendar after a commercial discovery	66
Five-year scorecard after first oil	67
Red flags	67
Final adaptation rule	68
Comparative cases and transfer limits	69
Comparative Cases Matrix	69
How Guyana should be compared without creating a political ranking	69
Comparative research protocol	69
Ghana	69
Norway	69
Timor-Leste	70
Suriname	70
Small-island adaptation	70
Expanded flagship research dossiers	71
PART I RESEARCH DOSSIER	72
Before the Discovery: Guyana's Geological, Institutional, Environmental and Fiscal Foundations	72
Purpose of Part I	72
Chapter 1 research dossier	72
Guyana before offshore petroleum	72
1.1 Economic starting point	72
1.2 Commodity and extractive experience	73
1.3 Geography and infrastructure constraints	74
Chapter 2 research dossier	74
A century of petroleum exploration	74
2.1 Exploration did not begin in 2015	74
2.2 Public investment in geological knowledge	75

2.3 The 1986 statutory foundation	75
2.4 Maritime boundaries as a petroleum-governance issue	75
2.5 The Liza-1 discovery record	76
2.6 Exploration evidence gaps	76
Chapter 3 research dossier	77
Environmental and natural-capital foundations before oil	77
3.1 Environmental law predates first oil	77
3.2 The 2009 Low Carbon Development Strategy	77
3.3 Why natural capital belongs in a petroleum case study	78
3.4 Coastal exposure	78
3.5 Environmental baseline for offshore petroleum	79
Chapter 4 research dossier	80
The state-capacity baseline before the petroleum windfall	80
4.1 Capacity is an institutional system	80
4.2 Legal and fiscal capabilities before commercial production	80
4.3 The discovery-to-first-oil capacity race	80
4.4 The Natural Resource Fund as a pre-first-oil milestone	81
4.5 Measuring institutional adaptation	81
Part I exhibit programme	82
Exhibit I-1. Petroleum exploration chronology	82
Exhibit I-2. Pre-oil institutional map	82
Exhibit I-3. Environmental and natural-capital baseline	83
Exhibit I-4. Before/after institutional capability matrix	83
Exhibit I-5. What a newcomer should do before discovery	84
Part I source package currently available	84
Part I next-source requirements	84
Part I research boundary	85
PART II RESEARCH DOSSIER	86
Discovery, Contractual Architecture, Cost Recovery and Revenue Assurance	86

Research objective	86
Chapter 5 research dossier	86
Liza 2015 and the information shock	86
Chapter 6 research dossier	87
The 2016 Stabroek petroleum agreement	87
6.1 Contract crosswalk required	87
6.2 Headline fiscal mechanics	89
6.3 Effective date and recoverability	89
Chapter 7 research dossier	89
Cost recovery, accounting and the cost bank	89
7.1 Proposed audit taxonomy	90
Allocation across blocks	90
Timing	90
Duplicate or erroneous charges	90
General and administrative costs	90
Emergency and environmental costs	90
Valuation	90
Affiliate transactions	90
Decommissioning	90
7.2 The cost bank	90
7.3 Audit status categories	91
Chapter 8 research dossier	91
Later model terms and changing information	91
Chapter 9 research dossier	92
Contract governance for a newcomer	92
Before negotiation	92
During negotiation	92
Before signature	93
After signature	93

Initial documentary findings from government-published audit material	94
Contract chapter exhibits	94
Exhibit II-1. Agreement chronology	94
Exhibit II-2. Fiscal waterfall	94
Exhibit II-3. Cost-bank bridge	95
Exhibit II-4. Audit exception register	95
Exhibit II-5. Agreement-generation crosswalk	95
Required quantitative tests	95
Source package currently available	96
Remaining Part II evidence gates	96
Publication boundary	96
PART III RESEARCH DOSSIER	97
The Pre-First-Oil Window, First Revenue and Production Scale-Up	97
Research objective	97
Chapter 10 research dossier	97
Discovery to development decision	97
Development-decision evidence	98
Chapter 11 research dossier	98
The pre-first-oil governance window	98
Petroleum administration	99
Fiscal	99
Sovereign revenue	99
Environmental	99
Transparency	99
Chapter 12 research dossier	100
First oil and the first revenue cycle	100
Chapter 13 research dossier	100
FPSO-by-FPSO production expansion	100
Liza Destiny	101

Liza Unity	101
Prosperity	101
ONE GUYANA	101
Project table to build	101
Chapter 14 research dossier	101
Production data quality	101
Five measures that must not be confused	102
Nameplate/design capacity	102
Daily production	102
Monthly daily average	102
Annual average	102
Cargo lift	102
Production and revenue bridge	103
Layer A. Physical production	103
Layer B. Commercial lifting	103
Layer C. Public cash	103
Production-scale regulatory pressure	103
New-producer blueprint extracted from Part III	104
Required Part III exhibits	104
Remaining Part III evidence gates	105
Publication boundary	105
PART IV RESEARCH DOSSIER	106
The Natural Resource Fund, Fiscal Absorption, Public Investment and National Wealth	106
Research objective	106
Chapter 15 research dossier	106
NRF law through time	106
Legal crosswalk fields	106
Operational agreement	107
Chapter 16 research dossier	107

Reconstructing fund cash flows	107
Annual reconciliation standard	108
Chapter 17 research dossier	108
Saving, spending and intergenerational equity	108
Intergenerational framework	109
International comparison	109
Chapter 18 research dossier	110
The non-oil primary balance and absorption	110
Absorption constraints	110
Chapter 19 research dossier	111
Public investment management	111
Project lifecycle	111
Outcome categories	111
Lifecycle liabilities	112
Chapter 20 research dossier	112
National balance-sheet accounting	112
Assets	112
Financial	112
Produced capital	112
Human capital	113
Natural capital	113
Liabilities	113
Petroleum depletion	113
NRF dashboard design	114
Petroleum receipts	114
Fund performance	114
Legal withdrawal	114
Public expenditure	114
Adverse fiscal scenarios	115

Price shock	115
Production interruption	115
Market loss	115
Construction inflation	115
Natural disaster	116
New-producer blueprint extracted from Part IV	116
Required Part IV exhibits	116
Remaining Part IV evidence gates	117
Publication boundary	117
PART V RESEARCH DOSSIER	118
Environmental Governance, Offshore Risk, Financial Responsibility and Climate Coexistence	
Research objective	118
Chapter 21 research dossier	118
The environmental authorization system	118
Public participation	119
Chapter 22 research dossier	120
Project-by-project environmental evidence	120
Liza Phase 1 as an operating example	121
Uaru as a later permit-generation example	121
Chapter 23 research dossier	121
Oil-spill prevention, preparedness and response	121
Response resources	122
Worst-case discharge modelling	122
National command	123
Chapter 24 research dossier	123
Liability, insurance and financial assurance	123
Court decisions	123
Financial-stress framework	124
Chapter 25 research dossier	124

Coastal, fisheries and biodiversity exposure	124
Fisheries	125
Biodiversity	125
Chapter 26 research dossier	125
LCDS, forest finance and petroleum	125
Petroleum and forest accounting boundaries	126
Chapter 27 research dossier	126
Energy transition and emissions accounting	126
Measurement hierarchy	127
Cumulative environmental governance	127
New-producer blueprint extracted from Part V	128
Required Part V exhibits	128
Remaining Part V evidence gates	129
Publication boundary	129
PART VI RESEARCH DOSSIER	130
Local Content, Domestic Value, Workforce, Household Welfare and Indigenous/Hinterland Outcomes	130
Research objective	130
Chapter 28 research dossier	131
Local content law and implementation	131
Reporting evolution	131
Chapter 29 research dossier	132
Domestic value added versus local procurement	132
Case A: imported item, local reseller	132
Case B: imported inputs, local fabrication	132
Case C: local service	132
Case D: foreign-controlled local operation	132
Domestic value-added formula	132
Ownership verification	133

Chapter 30 research dossier	133
Workforce and skills	133
Skills pipeline	134
Training outcomes	134
Chapter 31 research dossier	134
Household welfare and distribution	134
Welfare dashboard	134
Price effects	135
Chapter 32 research dossier	135
Amerindian and hinterland development	135
Consent and participation	136
Chapter 33 research dossier	137
Social licence, grievance and participation	137
Evidence versus sentiment	137
Longitudinal local-content evidence	138
Local-value audit sample	138
Distributional analysis	139
New-producer blueprint extracted from Part VI	139
Required Part VI exhibits	140
Remaining Part VI evidence gates	140
Publication boundary	141
PART VII RESEARCH DOSSIER	142
Economic Transformation, Public Investment, Energy, Diversification and Competitiveness ¹⁴²	
Research objective	142
Chapter 34 research dossier	142
Growth beyond headline GDP	142
GDP decomposition	143
Why oil GDP requires separate treatment	143
Non-oil growth	143

Labour market	144
Chapter 35 research dossier	144
Infrastructure transformation	144
Infrastructure classes	144
Transport	144
Water and drainage	144
Social	145
Productive	145
Project register fields	145
Lifecycle cost	146
Chapter 36 research dossier	146
Energy security and gas-to-energy choices	146
Power-system metrics	146
Gas-to-energy evaluation	147
Demand-side management	147
Chapter 37 research dossier	148
Private-sector development and diversification	148
Diversification metrics	148
Linkages	148
Chapter 38 research dossier	149
Inflation, competitiveness and Dutch-disease channels	149
Indicator set	149
Dutch-disease channels	150
Policy response framework	150
National wealth conversion framework	150
IMF 2026 evidence relevant to Part VII	151
Required Part VII exhibits	151
New-producer blueprint extracted from Part VII	152
Remaining Part VII evidence gates	152

Publication boundary	153
PART VIII RESEARCH DOSSIER	154
Transparency, Cost Audits, Procurement, Parliamentary/Judicial Oversight and Public Data	154
Research objective	154
Chapter 39 research dossier	154
EITI and extractive disclosure	154
Validation timeline to build	154
Disclosure quality	155
Timeliness	156
Chapter 40 research dossier	156
Cost audits and revenue assurance	156
Revenue assurance institutions	156
Audit-cycle metric	157
Chapter 41 research dossier	157
Procurement and public-spending oversight	157
Supreme audit institution	158
Project-audit register	158
Chapter 42 research dossier	159
Parliament, judiciary and institutional checks	159
Parliamentary petroleum-revenue trail	159
Judicial review	159
Chapter 43 research dossier	160
Statistics and public data	160
Data-quality inventory	160
Petroleum	160
Fiscal	161
Economic	161
Social	161
Environmental	161

Reproducibility standard	162
Accountability architecture	162
Layer 1. Rule	162
Layer 2. Measurement	162
Layer 3. Disclosure	162
Layer 4. Independent challenge	162
Layer 5. Correction	162
Layer 6. Outcome	162
Accountability case studies required	163
New-producer blueprint extracted from Part VIII	163
Required Part VIII exhibits	163
Remaining Part VIII evidence gates	164
Publication boundary	164
PART IX RESEARCH DOSSIER	165
Comparative Petroleum Governance and Transferability	165
Research objective	165
Chapter 44 research dossier	165
Norway	165
Transferability limits	166
Chapter 45 research dossier	166
Ghana	166
Mechanism comparison	167
Transferability	167
Chapter 46 research dossier	167
Timor-Leste	167
Core comparison questions	168
Chapter 47 research dossier	168
Suriname	168
Chapter 48 research dossier	169

Additional producer mechanisms	169
Trinidad and Tobago	169
Senegal	169
Namibia	169
Mozambique	169
Botswana	170
Chapter 49 research dossier	170
Island and small-state adaptation	170
Marine concentration	170
Infrastructure concentration	170
Labour scale	170
Fiscal concentration	171
Hurricane and climate exposure	171
Blue economy and natural capital	171
Comparative mechanism matrix	172
Transferability decision tree	172
Resource condition	172
Fiscal condition	172
Institutional condition	172
Social condition	172
Environmental condition	173
Legal condition	173
Comparative evidence standards	173
Required Part IX exhibits	174
New-producer conclusion from comparison	174
Remaining Part IX evidence gates	174
Publication boundary	175
PART X	176
New Producer Operating Manual	176

Stage 0	176
Before a commercial discovery	176
Objective	176
0.1 Legal map	176
0.2 Acreage and geological-data register	177
0.3 Environmental baseline	177
0.4 Fiscal options	178
0.5 Expert capacity	178
0.6 Maritime boundaries	179
Stage 0 gate	179
Stage 1	180
After a commercial discovery	180
Objective	180
1.1 Independent resource review	180
1.2 Negotiation decision file	180
1.3 Fiscal model	181
1.4 Environmental decision file	181
1.5 Decommissioning	182
Stage 1 gate	182
Stage 2	182
First-oil readiness	182
Objective	182
2.1 Meter and production chain	182
2.2 Sales and entitlement chain	182
2.3 Sovereign revenue chain	183
2.4 Emergency drill	183
2.5 Public transparency rehearsal	183
Stage 2 gate	184
Stage 3	184

First year of production	184
Objective	184
Month 1	184
Quarter 1	185
Quarter 2	185
Quarter 3	185
Quarter 4	186
Stage 4	186
Years 2 to 5	186
Objective	186
Annual institutional capacity review	186
Sovereign fund	186
Public investment	187
Local value	187
Social baseline	187
Environmental system	188
Stage 4 gate	188
Crisis playbooks	188
A. Oil price falls sharply	188
B. Production outage	189
C. Major offshore spill	189
D. Cost-audit dispute	190
E. Construction overheating	190
F. Natural disaster plus oil-revenue shock	191
Island-state adaptation module	191
Marine economy	191
Single-point infrastructure	191
Labour	191
Revenue concentration	192

Publication and data blueprint	192
Legal register	192
Operational register	192
Fiscal register	193
Environmental register	193
Social/local register	193
24-month pre-first-oil action calendar	193
Months 24-18	193
Months 18-12	194
Months 12-6	194
Months 6-3	194
Months 3-0	194
Five-year dashboard	195
Final decision principles extracted from the Guyana study	195
Acceptance checklist for a country using this manual	196
NATURAL RESOURCE FUND RECONCILIATION WORKBOOK	197
Why this workbook exists	197
Control equation	197
Verified early-year audit observations	197
2020	197
2021	198
Withdrawal history verified from Bank of Guyana quarterly reporting	199
June 2025 fund snapshot	199
Cash-receipt chronology control	199
2026 H1 government reporting bridge	200
Annual master table	200
Withdrawal-rule legal crosswalk to build	201
Public-investment separation	201
Audit controls	201

Publication rule	202
ENVIRONMENTAL PERMIT AND RISK REGISTER	203
Control rule	203
Project register	203
Liza Phase 1 permit chronology	203
Uaru permit controls	204
Oil Pollution Act 2025 status control	204
Financial assurance register	205
Environmental incident taxonomy	206
Marine ecological evidence	206
Spill response stress tests	207
Scenario A: controlled surface release	207
Scenario B: subsea loss of well control	207
Scenario C: adverse weather plus spill	207
Scenario D: transboundary movement	208
Evidence hierarchy for environmental claims	208
Remaining environmental evidence gates	208
Publication boundary	209
MODEL METHODS AND LIMITATIONS	210
Purpose	210
Contract waterfall	210
Why a percentage of gross sales is not a permanent government take	210
Contract sensitivity dataset	211
Combined production shock dataset	211
Sovereign-fund stress model	211
Why the fund model is separate from the NRF reconciliation	212
Model review gates	212
Reproducibility	213
Interpretation rule	213

Executive research brief

Read alongside MANUSCRIPT_WORKING_DRAFT.md, SOURCE_REGISTER.md, ADVERSARIAL_PUBLICATION_AUDIT.md and CLAIM_SOURCE_MATRIX.md. Edition 1.1 has completed an internal adversarial review cycle. Independent external expert review remains pending.

Audit status

Edition 1.1 is an internally audited publication candidate, not an externally peer-reviewed or independently validated publication. The internal audit identified and corrected publication-status, source-coverage, legal-scope, geopolitical-risk and PDF-design weaknesses. High-risk legal characterization remains deliberately narrow where an authenticated judgment, commencement instrument, insurance instrument or later dispute determination has not been located in the public sources reviewed. Guyana's 2026 EITI Validation remains ongoing at the evidence cut-off. The Bank of Guyana's audited 2025 NRF statements and current 2026 monthly reporting are now part of the evidence register. [S88-S89, S96]

Purpose

Reconstruct the institutional and commercial sequence behind Guyana's transition to a large offshore petroleum producer. Test where its history supplies documentary evidence for another country preparing for first oil, and where forest endowment, geological scale, fiscal terms and local political institutions limit direct transfer. The intended audience is senior officials, new petroleum regulators, central banks, finance ministries, environmental agencies, civil-society researchers and offshore investors.

Verified sequence

Guyana has decades of exploration history. Liza was announced as commercially discovered in May 2015. The Stabroek petroleum agreement was entered in June 2016 and the official block register shows an October 2016 effective date. The first Natural Resource Fund law received assent on 23 January 2019, before oil production commenced in December 2019. The 2021 statute replaced the first framework and 2024 legislation changed withdrawal limits. The fourth operating FPSO, Yellowtail, began production in August 2025. Four FPSOs were operating during the first half of 2026. This chronology separates pre-oil institutional decisions from subsequent modifications. [S01-S05, S11]

September 2026 evidence snapshot

The September 2026 Ministry of Finance mid-year report, describing activity through June, records 162 crude lifts, of which 21 represented government profit oil. During the half-year, government received USD 1,778.6 million in profit-oil proceeds and USD 218.4 million in royalties. The Natural Resource Fund stood at USD 4.2942 billion on June 30 after USD 1.02 billion of first-half withdrawals. The ministry revised its projected full-year 2026 petroleum deposits to USD 6.4976 billion. The latter number is a projection rather than revenue already received. Government estimated overall real GDP growth of 33.3% and non-oil growth of 10.1% for the first half; GDP estimates do not directly show changes in household poverty. [S07, S10]

The distinction most useful to a new producer

The 2016 Stabroek PSA and Guyana's later model PSA do not have the same fiscal terms. For 2016 Stabroek, the published administrative summary describes a 2% royalty, a cost-recovery ceiling of 75% and equal sharing of profit oil. Under the government revenue authority's simplified binding-cap example, 12.5% profit oil plus 2% royalty yields 14.5% of gross revenue. The figure changes as recoverable costs change. The later model for new licensing announced a 10% royalty, 65% maximum cost recovery and 10% corporate tax. The two regimes must be analysed separately, along with the information available at the time each was negotiated. [S02, S14-S15]

Institutional subjects for detailed scrutiny

1. Early-law timing. The NRF predates first oil, but the institution changed over time. Assess legislation, actual reporting, parliamentary authorization, investment stewardship and expense outcomes individually.
2. Fiscal and contract administration. Petroleum volumes, government lifts, sales, cost-bank reductions, royalties, cash deposits and budget transfers are different measures. Cost-oil dispute resolution and audit resources matter alongside contract percentages.
3. Environmental assurance. Project-specific EIAs and permits, monitoring, emergency assets, enforceable insurance, operator liability and cross-border response are separate protections. The 2025 pollution act was enacted, but the 2026 mid-year report described implementation regulations and commencement as ongoing work. The May 2026 Court of Appeal decision on financial guarantees requires analysis of the full judgment and underlying instruments.
4. Economic absorption. A high fiscal balance does not establish that roads, hospitals, schools, housing and flood defenses meet service and maintenance targets. IMF staff in July 2026 called for long-run fiscal guideposts, spending-outcome monitoring, procurement compliance and stronger cost-oil audits.
5. Social and community outcomes. Gross domestic supplier spend is not local value added. Old poverty-survey results cannot demonstrate current welfare. Indigenous village allocations require programme-level and locally grounded verification.
6. Transparency and institutional participation. The EITI Board launched Guyana's delayed second validation in May 2026 and recorded concerns about confidence in the independence of civil-society representative selection. The final paper will capture independent stakeholder responses and the final validation decision if published by the cut-off.

7. Parallel climate systems. Guyana's 2009 LCDS and Norway partnership preceded oil. Carbon-credit contracts, payments and village benefit-sharing require separate accounting from the NRF and domestic petroleum emissions. Forest results cannot be transferred mechanically to reef-based or low-forest island states.

Comparator lenses

Ghana's 2011 revenue law offers distinct stabilization and heritage funds and public publication requirements. Norway's fund documents describe transferring net state petroleum cash into a large investment fund and a long-run real-return-linked fiscal guideline. Timor-Leste's 2005 Petroleum Fund provides another small-state legal design. Suriname's GranMorgu was still under construction in September 2026, with first oil planned for 2028, making the country a relevant pre-production comparison rather than a second operating case. No design is treated as an automatic policy choice for another country. [S24-S27]

Implementation workbook for a newly discovering jurisdiction

Exploration: Record basin geology and uncertainty, licensing authority and contract disclosure, ownership, environmental baseline, marine response capabilities, available technical expertise and prospective contingent liabilities.

Commercial discovery: Build independent resource and development assessment, model fiscal terms on common assumptions, identify costs and the risk allocation actually negotiated, obtain project-specific environmental assessment, map nearby fisheries and communities, establish audit rights and publish authorized agreements subject to legitimate protections.

Before first oil: Test metering and valuation, recoverable-cost records, royalty invoicing, financial custody, parliamentary procedures, periodic public reporting, spill drills, grievance channels and beneficial-ownership records. Determine budget contingencies without booking forecast oil prices as guaranteed receipts.

First two years of production: Publish reconciled lifts, realized prices, royalty and profit-oil cash, fund balances, parliamentary withdrawals, government project execution and all material environmental incidents. Check whether procurement and supplier development result in verified domestic value.

Scaling: Stress-test price changes, new project cost-bank additions, maintenance obligations, delayed infrastructure and major spill financial exposure. Compare national growth with independently sourced real household outcomes.

Publication acceptance conditions

Complete the full multi-chapter manuscript and original, reproducible exhibits. Independently reconcile NRF period accounts, obtain the Court of Appeal judgment and original insurance and permitting instruments, resolve the legal commencement status of the 2025 spill act, incorporate final EITI 2026 status, document Indigenous stakeholder evidence, check page-specific citations and have relevant experts challenge fiscal, legal, environmental and social conclusions. Count only substantive PDF pages, inspect charts and citations, and describe independent external review as pending unless completed.

Primary citations are maintained in SOURCE_REGISTER.md and the high-risk claim crosswalk in CLAIM_SOURCE_MATRIX.md. The 500-plus-page document is a research compendium and publication candidate. External peer review, authenticated resolution of open legal evidence, and documented responses to specialist reviewers remain separate release gates.

2026 observation snapshot

2015	May Liza commercial discovery	[S01]
Jan 2019	NRF statute assented before first oil	[S04]
Dec 2019	Liza Phase 1 first oil	[S01]
Aug 2025	Yellowtail became fourth producing FPSO	[S11]
Jun 2026	End-June NRF cash: USD 4.2942 billion	[S07]
Jun 2026	H1 authorized withdrawals: USD 1.02 billion	[S07]

Dates refer to distinct events. Production capacity is not actual annual output. Fund cash is not contracted or forecast future revenue.

Worked fiscal exhibit: simplified 2016 Stabroek waterfall

Using the Guyana Revenue Authority's illustrative gross-revenue convention, royalty is separate and borne by the contractor group. A 75% cost-recovery cap is a maximum, not a permanent claim on every barrel. Values below are illustrative units out of 100, not observed 2026 receipts. [S02, S15]

Recovery used	State profit oil	State royalty	Combined state units
75% (cap binding)	12.5	2.0	14.5
50% (illustrative)	25.0	2.0	27.0
20% (illustrative)	40.0	2.0	42.0

Substantive manuscript

Abstract

Guyana's offshore petroleum expansion offers an unusually compressed record of resource discovery, contractual negotiation, institutional preparation, sovereign revenue management and rapid commercial production. Liza was announced as a commercial discovery in 2015. The Natural Resource Fund was established by legislation in January 2019 before the first offshore crude was produced in December of the same year. Four floating production, storage and offloading facilities were operating by the first half of 2026. At June 2026 the government reported a USD 4.2942 billion NRF cash balance after USD 1.02 billion of withdrawals in the first half. These observations are meaningful, but on their own they establish neither an optimal petroleum agreement nor durable welfare gains. [S01, S04, S07]

This research reconstructs events in the order they happened. It distinguishes pre-existing foundations, policy instruments established before first oil, later reforms and results still under evaluation. It examines revenue accounting, the 2016 Stabroek production sharing agreement, environmental impact assessments, oil-spill legislation, public financial management, labour and domestic suppliers, forest protection and climate finance. It then maps administrative and technical decision points for another prospective producer. The paper treats a new island state differently from a forested continental state, and a non-commercial discovery differently from an operating offshore field.

The evidence is mixed. Formal fiscal and reporting institutions preceded first oil. Government and operator data document rapid expansion and growing petroleum revenues. IMF staff in July 2026 also called for stronger oil-sector audit capacity, better procurement and public-enterprise oversight, more complete household statistics and a firmer long-run fiscal guide. The EITI Board opened a delayed 2026 validation while stating concern about the independence and stakeholder confidence of the civil-society selection process. The legal framework for pollution preparedness expanded with a 2025 act, but the June 2026 government report described regulations and commencement as work still underway. These are central facts rather than footnotes. [S07, S08, S16, S19]

A transferable case is not a list of admired policies. It is a documented sequence of institutional choices, conditions required for operation, observable results, alternative interpretations and identified points of failure.

Method and definitions

The unit of analysis is Guyana's petroleum and related national governance system from exploratory activity to 27 September 2026. The offshore Stabroek project provides the production case, but the ecological and development analysis covers Guyana as a whole. The temporal cut-off is essential: September 2026 projections about year-end output and revenue are not presented as observations.

Five evidence categories are kept apart. Legal evidence consists of texts enacted, brought into force and amended. Administrative evidence consists of official licence, permit, audit and oversight documents. Financial evidence consists of audited or dated management accounts, with

explicit cash-period boundaries. Operating evidence consists of production and commissioning records. Social and ecological evidence requires outcome data rather than expenditure or project announcements. Statements from government and companies are attributed to their authors. Independent multilateral assessments and civil-society concerns are reported with their own evidentiary limitations.

Throughout, a production capacity figure means engineering capacity, not an actual monthly mean. A crude export receipt is not public revenue. Government profit-oil lifts are not identical to sales received in the same month. A fund deposit is not a parliamentary withdrawal, and the fund's cash balance is not equivalent to the public investment portfolio. Contractual value for future carbon-credit purchases is not current cash. National GDP is not a household consumption survey. These separations prevent the apparent precision of high-level figures from concealing incompatible denominators.

1. Before discovery: institutions, geology and the timing problem

Guyana's petroleum history did not begin in 2015. Onshore work dates to the early twentieth century; offshore exploration began in the middle of that century. Seismic surveys and drilling by successive companies preceded the commercial Liza discovery. ExxonMobil's retrospective 2023 report places offshore exploration attempts in the late 1950s and describes more than forty wells drilled in the basin over decades. This history matters because a country rarely controls the exact timing of an economically viable discovery. Dry wells, capital cycles, technical interpretation and petroleum prices determine much of the interval between exploration and commercial development. [S01]

Guyana also entered the modern oil period with an existing economic and ecological agenda. Its original Low Carbon Development Strategy began in 2009. A parallel Guyana-Norway results-based forest agreement began in the same year. The government subsequently reported USD 227 million earned for forest climate services during 2009-2015. A monitoring, reporting and verification system for the forest sector developed before first oil. These were pre-petroleum policy institutions and a separate financial stream, not products of the petroleum fund. [S22]

This chronology prevents two analytical mistakes. First, attributing every later institution to the petroleum discovery erases existing public administration, regulatory experience, forest monitoring and the long history of commodity production. Second, projecting Guyana's forest-carbon results onto a small island with few forests would confuse the national resource endowment with a universally available funding instrument. The transferrable administrative lesson is the practice of mapping existing technical capacity before a commercial discovery, not the replication of Guyana's exact endowment.

In an exploration-stage country, the relevant decision documents include the geological data policy, licence register, basin environmental baseline, fiscal-risk inventory, ministerial and regulatory division of responsibilities, spill-planning jurisdiction, beneficial-ownership requirements and public disclosure rules. Each is meaningful before a first commercial find. A government may procure legal and reservoir expertise through transparent short-term appointments instead of creating a large permanent petroleum bureaucracy without proven

reserves. Whether to establish a state producer, auction acreage, negotiate directly or reserve blocks requires a jurisdiction-specific legal and fiscal assessment. The case study will therefore map decision rights and procedural safeguards rather than prescribe one commercial model.

Guyana is also climatically distinctive. Much of its population and capital stock sit on a low coastal plain subject to flooding, while large forested and hinterland regions have different infrastructure, service-delivery and land-governance needs. An offshore development changes the fiscal envelope without eliminating these geographic constraints. An independent comparison must measure whether new capital expenditure reduces lifecycle exposure to flooding, power insecurity and maintenance costs. Building a new asset is an input; a working, resilient asset delivering service over time is the intended outcome.

The period before discovery is thus analytically separate from 2015-2019. A useful exploration chronology should identify not only when a well discovered hydrocarbons but also when the state gathered geological information, published licence terms, defined ownership, set environmental baselines and strengthened its ability to challenge an operator's commercial and technical submissions. Without that baseline, governments and external observers struggle to separate inherited institutional capacity from later improvements.

2. The discovery-to-first-oil window, 2015-2019

The Liza discovery announced in May 2015 changed the commercial outlook for the offshore Stabroek block. On June 27, 2016 the government and the consortium entered a petroleum agreement, with a stated October 7, 2016 effective date in the official block register. Different public records refer to signature, agreement and effective dates. Those dates should appear separately in the final chronology. The project reached first oil in December 2019. The roughly four-year interval after discovery is the institutional preparation window studied here. [S01, S02, S03]

The 2016 Stabroek agreement reflects the conditions and bargaining positions of a frontier offshore development at the time it was signed. Any contractual comparison needs the exact agreement text, hydrocarbon uncertainty at signature, oil-price conditions, deepwater capital intensity, exploration obligations, relinquishment provisions and environmental commitments. Retrospective government revenues and later geological success do not automatically establish the value a counterparty placed on acreage before commercial certainty. Conversely, early commercial uncertainty does not remove the need to document fiscal options or expose a government to unreviewed recoverable costs.

The public Stabroek agreement has a 2% royalty, up to 75% monthly cost-recovery ceiling, and an equal division of remaining profit oil. The Guyana Revenue Authority explains that when the ceiling binds, a simplified revenue calculation gives 12.5% of gross sales as government profit oil plus a 2% royalty, or 14.5% under its stated accounting approach. That is a conditional illustration, not an unchanging government entitlement. When permitted recoverable costs use less than the cap, the profit-oil pool increases. Approved future development expenditures may replenish the cost bank. Audit treatment, allowable expenditure and valuation matter as much as headline sharing percentages. [S02, S07, S15]

The historical test of the contract therefore has at least four separate components: returns relative to the information available at signing, compliance with the signed terms, effectiveness of subsequent cost verification, and the state's ability to use later experience in different acreage awards. Public debate frequently merges these into a single judgment. This manuscript keeps them separate and will not declare a country-level winner or a contract-level overall score.

A relevant reform sequence is observable. The government later announced model terms for future offshore bidding with a 10% royalty, a 65% cost-recovery ceiling and a 10% corporate tax. Profit oil remains shared equally after cost recovery under the model. These later terms are distinct from the continuing Stabroek agreement and must never be back-applied when calculating historical or forecast revenues from the producing block. The published model documents and any final signed new licence terms require an agreement-by-agreement crosswalk. [S14]

What early fund establishment demonstrates, and what it does not

The Natural Resource Fund Act received assent on January 23, 2019, almost eleven months before first oil in December 2019. The Ministry of Finance's 2024 NRF annual report confirms the date. This is a material pre-production institutional milestone: the state adopted a statutory vehicle before recurring offshore oil cash arrived. Yet formal creation alone says little about later spending efficiency, distributional outcomes or the adequacy of subsequent withdrawal rules. Those questions require separate assessment. [S04]

The 2019 establishment also did not mean the rules remained unchanged. The 2021 law replaced the 2019 act. The withdrawal provisions changed again on February 6, 2024. A new country examining this case should distinguish statutory creation, operational readiness, independent monitoring, transparent publication, integration with annual budgets and revision procedures. A law on paper is the first link in a longer accountability chain.

3. From first oil to large-scale production, 2019-September 2026

Production commenced at Liza Phase 1 in December 2019. Liza Phase 2 followed in 2022, Payara in November 2023 and Yellowtail on August 8, 2025. Yellowtail's ONE GUYANA FPSO joined the Destiny, Unity and Prosperity vessels and took combined installed capacity past 900,000 barrels per day. ExxonMobil announced a daily production milestone of 900,000 barrels in November 2025. That one-day milestone is distinct from annual average production, and installed capacity alone never establishes actual output. [S01, S11, S12]

The national petroleum production chart extends through July 31, 2026. The Ministry of Finance's 2026 mid-year report says production from the four FPSOs exceeded 900,000 barrels per day during the first half. The same report describes the fifth Uaru project as under construction with start-up targeted for the final quarter of 2026. Future development output should remain in a prospective column until a dated commissioning announcement and verified operating record are obtained. [S07, S13]

Production is not a single variable. An operating dossier needs the count and size of producing vessels, uptime, reservoir performance, flaring events, methane monitoring, water injection,

transport logistics, crude grades, lifting dates, export prices, cost-bank balances and the share delivered to the state. The government recorded 162 crude lifts during January-June 2026, including 21 government profit-oil lifts from the four producing units. Yet receipts in the same period included payments for three lifts occurring in late 2025, while payments for some mid-2026 lifts fall later. An analytical table pairing production dates directly with cash receipts without that timing adjustment would misstate the relationship. [S07]

Guyana reported USD 1,778.6 million in government profit-oil sales receipts and USD 218.4 million in royalties during the first half of 2026. End-June NRF cash stood at USD 4,294.2 million after USD 1,020 million in withdrawals during the period and including reported interest. These figures come from the government's September 2026 publication. Independent reconciliation with Bank of Guyana monthly and quarterly releases and external audit statements is part of the final source gate. [S07]

The speed of commissioning is commercially significant but creates linked challenges. Regulatory personnel need sufficient specialist expertise and access to data as vessel numbers multiply. Shipping and aviation capacity grow along with occupational risk. Contractor spending can expand faster than qualified local supplier capacity. Power demand, roads and port assets face supply-chain constraints. Large monthly tax and petroleum receipts expand budget options while also increasing the consequences of weak procurement or poor project sequencing. None of these effects is captured by a barrels-per-day graph.

The final paper will therefore present a development-by-development table with first-oil dates, sanctioned costs as of the applicable date, FPSO investment-basis capacity, actual quarterly output, permitted emissions and flaring, environmental reporting and the regulator's approval and inspection milestones. An approved project not yet producing will have blank realized-output cells rather than copied operator forecasts.

4. The Natural Resource Fund as a fiscal institution

The conceptual purpose of a natural-resource fund is to separate volatile extractive cash inflows from the immediate authorization of public expenditure. In Guyana's case, the fund was legally established before first oil. The 2021 replacement statute specified a governance structure including a board, revised accountability arrangements and withdrawal rules. The 2024 amendment increased the statutory withdrawal envelope. A claim of good fiscal practice requires analysis of each version and its observed effects rather than treating the act as an immutable institution. [S04, S05]

The fund receives the government's defined petroleum revenue streams and investment income. Withdrawals move money to the consolidated fund under parliamentary budget authorization. The public account must allow a reader to reconstruct each stock-and-flow identity:

Opening fund cash + petroleum receipts + investment income + other recognized movements - authorized transfers = closing fund cash.

Each item needs an observation period, currency and treatment of accruals or late receipts. The initial paper uses the figures disclosed for June 2026 rather than extrapolating a final year-end

outcome. The Ministry of Finance reported USD 4.2942 billion in cash at end-June after USD 1.02 billion in first-half transfers. Its revised 2026 estimate raised full-year petroleum deposits to USD 6.4976 billion, from the much lower January budget assumption. This updated full-year estimate remains a projection, influenced in the ministry's explanation by oil prices and reduced use of the cost-recovery ceiling. [S06, S07]

The revision illustrates two fiscal hazards. First, a large increase in projected receipts can be mistaken for an equally large increase in prudent current-year spending space. The withdrawal law and parliamentary budget framework are separate from the projection. Second, a rapid decline in historical cost recovery could create short-term windfall receipts that do not recur in the same amount after a new field enters development. The 2026 mid-year report expressly notes that costs from additional projects may increase the cost bank again. [S07]

IMF staff's July 2026 statement recognized the accumulation of substantial NRF balances and Guyana's debt position, while proposing consideration of a longer-run operational fiscal guide based on the non-oil primary balance. Staff argued that unexpected gains under sustained high oil prices should be examined against the economy's ability to absorb investment and called for stronger monitoring of public spending outcomes and public-enterprise risks. These are attributed external assessments. They do not prove one single withdrawal formula suitable for all emerging producers. [S08]

Cash management is not public investment evaluation

A fund statement answers where petroleum cash is and how money moves. A public investment system answers which projects are funded, procured, built, independently inspected, maintained and used. The two systems are linked but not interchangeable. A budget line for a hospital is not a verified improvement in service quality. A construction contract is not a functional road that survives flooding. A scholarship allocation is not a measured increase in skills or labour-force productivity.

The transfer framework should therefore include a separate public investment register. Each large project would record a published business case, procurement method, contract variations, delivery date, capital and lifecycle cost, disaster exposure, expected users and independent completion evidence. Where data are absent, the final manuscript will mark an evidence gap instead of inferring success or failure. Even with high revenues, capital-intensive programmes compete for engineers, surveyors, procurement specialists, nurses, teachers, land and imported construction inputs.

For a potential new producer, the administrative sequence to document includes the statutory revenue recipient, published deposit statements, a withdrawal authorization chain, parliamentary scrutiny, an external audit schedule and the project-level outcome ledger. The precise spending-versus-saving rule is a sovereign policy choice shaped by development needs, reserve life, debt, population, climate exposure and macroeconomic conditions. This research describes alternative designs and their trade-offs without assigning a universal answer.

5. Fiscal contract oversight and the cost bank

Production sharing is often explained as a royalty followed by a cost-oil allocation and a split of the remainder. That summary is insufficient for technical oversight. A credible government revenue forecast requires a model of recoverable cost categories, timing and approval of development expenditures, production valuation, commodity prices, crude quality adjustments, operating expenses, currency and transfer-pricing controls, decommissioning treatment and dispute procedures.

Under the 2016 Stabroek framework as summarized by the Guyana Revenue Authority, the simplified case in which the 75% cap is fully used allocates 75 of each 100 normalized revenue units to cost recovery, 12.5 to government's half of remaining profit oil and 12.5 to the operator group. A separate 2-unit royalty produces the familiar 14.5-unit combined government royalty/profit share under the authority's calculation. This is an explanatory model rather than a substitute for detailed contract accounting. In other periods the ceiling need not bind. [S02, S15]

At a hypothetical 50-unit cost share with a 100-unit revenue base, a simple 50/50 remainder gives the state 25 units of profit oil plus 2 units of royalty. At a hypothetical 20-unit share, the same pedagogical formula gives 40 plus 2. Actual payments need the contract's detailed oil-volume and valuation procedures, and this illustration excludes tax treatment, timing, carried costs and cash-versus-kind settlement. No reader should interpret 27% or 42% as an observed 2026 fiscal take or a forecast.

The 2026 mid-year report reported accelerated cost recovery as the historical cost bank was reduced and projected a substantially higher number of government lifts than expected in January. The report warns that new projects enter the cost bank and that clearing historical costs does not create a permanent government claim to half of gross sales. Such cautions belong in the main text because the choice of production denominator changes the interpretation of public revenue. [S07]

Contract administration also depends on verifiable audit capacity. The July 2026 IMF mission noted ongoing efforts to resolve cost-oil audits, including arbitration, and emphasized timely resolution, specialist audit staffing and the quality of government oversight. An administrative case study must distinguish allegations made in disputes, expenditures accepted as recoverable, auditor adjustments proposed, adjustments finally collected and open proceedings. A proposed disallowance is not recovered public revenue. [S08]

The later model fiscal terms illustrate a second issue: the government's bargaining position and acceptable return vary across the discovery cycle. The announced model's 10% royalty and 65% maximum cost recovery, together with equal residual profit oil and a 10% corporate tax, differ from the 2016 block agreement. A comparison should show gross expected government receipts, investor breakeven and downside geological outcomes under common assumptions; it should not infer comparative welfare from two headline royalty rates. [S14]

6. Environmental law, liability and offshore operations

Petroleum regulation should begin with an accurate inventory of harms, affected populations, potential transboundary effects and the physical capacity to respond. For a coastal state the issues include coastal water and fisheries, tourism, biodiversity, emergency maritime aviation, ocean currents, offshore worker safety, greenhouse-gas emissions, shore-based waste and possible liability beyond national waters. Guyana's EPA operates a formal environmental authorization process for projects with environmental impacts. The existence of a permit is a necessary legal record, not proof that its terms have been independently monitored or that every risk is extinguished. [S21]

The sequence should distinguish environmental assessment before project approval, permit conditions during construction, real-time operating monitoring, incident reporting, post-incident enforcement, insurance coverage and financial recourse. It should also distinguish public access to raw monitoring results from operator-authored summaries. A complete research exhibit would list each operating FPSO, its applicable environmental permit and environmental impact assessment, permitted flaring limits, reporting frequencies, available inspections, observed incidents and documented remediation.

Guyana enacted the Oil Pollution Prevention, Preparedness, Response and Responsibility Act in 2025. It addresses oil-spill prevention, planning, national response and financial responsibility. The 2026 mid-year report, covering January-June, says implementing regulations were still being drafted and that commencement was expected by the end of 2026. At the 27 September cut-off this report does not establish that the entire statute was already operational. The final legal chapter must check the Official Gazette and commencement instruments before moving a provision from 'enacted' to 'in force'. [S19, S07]

Financial assurance illustrates why precise legal language matters. In May 2026 the Court of Appeal overturned a 2023 High Court order requiring an unlimited guarantee under an offshore permit. The government's description of the appellate decision says the EPA had lawful discretion over the amount of financial assurance and that the High Court had improperly substituted its own decision for the regulator's judgment. A contemporary independent report also distinguishes an operator's continuing liability from the amount of security furnished in advance. Neither point, alone, resolves the empirical question of whether funds available following a catastrophic release would meet cleanup, compensation and ecological restoration costs. An authenticated judgment and actual policy documents are required for final legal analysis. [S20]

An offshore assurance model has multiple layers. Direct operator resources, insurance, any parent-company guarantee, rapid-access emergency cash, contracted response assets, reciprocal regional assistance and state contingency planning differ in legal trigger, latency, currency, scope and enforceability. They should not be merged into a headline figure without examining policy exclusions, aggregate limits, arbitration, who controls claim settlement and conditions after a multi-jurisdiction spill. For island and continental states alike, cross-border current modelling and neighbours' ability to respond are material planning inputs.

A future final chapter will include a reproducible response drill template: discovery-to-notification time, mobilization time, dispersant approval procedures, mechanical recovery resources,

sensitive-habitat priority maps, fishery-closure authority, independent incident sampling, medical surveillance and claims processing. Without disclosed observed drill performance, the template remains a prospective audit instrument rather than a finding about current operational readiness. Forest conservation and offshore pollution are separate environmental systems and must be evaluated on their own evidence.

7. Forest conservation, carbon finance and the fossil-fuel coexistence problem

Guyana's 2009 LCDS predated oil production by a decade. Its results-based forest partnership with Norway covered 2009-2015 and the 2024 LCDS update reported USD 227 million earned in that period. In 2022 the country introduced an expanded LCDS 2030 and arranged a multiyear sale of ART TREES jurisdictional forest credits to Hess. The published minimum purchase commitment is USD 750 million, with specific prices and volumes by credit vintage. The policy states that 15% of associated receipts are dedicated to eligible Indigenous village-led plans, and the remaining share finances wider LCDS priorities. A contract for future delivery is distinct from credits already issued, transferred, paid for and spent. [S22, S23]

This history permits analysis of two distinct resource-finance channels: extractive revenue deposited in the NRF and ecosystem-service payments managed through LCDS financial structures. A combined national wealth view is useful for climate and budget choices, but it would be inaccurate to describe all carbon receipts as oil revenue or add contracted future carbon sales to cash in the NRF. Oil production is a fossil-fuel supply activity; forest-credit issuance depends on independently specified carbon-accounting baselines, monitoring, additionality and leakage treatment. One does not mechanically cancel the other. Any claim of national carbon neutrality requires economy-wide emissions and accounting safeguards, not a simple subtraction of a contracted offset volume from petroleum exports.

The transferable question is whether a resource-producing state maintains independently verifiable environmental accounting for different assets. In Guyana's case, the forest monitoring and results-payment experience is relevant to land-use data quality and international partnership design. Its applicability to another jurisdiction depends on ecosystem type, forest tenure, historical deforestation, Indigenous rights, corresponding adjustments and carbon buyer eligibility. A reef-based island has different biodiversity and marine accounting issues. Carbon-market prices, certification methods and buyer eligibility are also vulnerable to change.

LCDS policy statements about village allocations, sea defenses, electricity investment and community livelihoods need outcome-level tests. A national transfer proves expenditure authorization, but independent review would examine arrival in designated accounts, locally agreed development plans, procurement documentation, functioning assets and households reached. Consultation must identify which communities participated, which declined and how grievances were addressed. A strategy can carry the language of inclusion while still requiring village-level verification.

IMF staff in July 2026 highlighted Guyana's market-based forest conservation, disaster-resilience investments and energy transition, while also identifying needs for stronger official statistics and procurement. These assessments do not eliminate the analytical tension between increasing fossil

output and the global energy transition. The final analysis will report domestic operational emissions separately from combustion of exported oil, land-use emissions and carbon-credit issuance. A policy commitment and a measured atmospheric outcome belong in different columns. [S08]

8. Local value, social outcomes and accountability

Guyana's Local Content Act was passed in December 2021. The law establishes a formal route for specified Guyanese participation and reporting by petroleum operators and suppliers. The 2026 mid-year report estimates roughly USD 466.2 million in procurement across sectors prioritized for Guyanese participation during the first half of the year, while government work on possible expansion of the law's scheduled categories was still underway. The number measures gross procurement in designated sectors. It does not independently reveal domestic value added, local beneficial ownership, wage quality, profits retained, product standards or the effect on non-petroleum employers. [S05, S07]

The final local-value chapter will disaggregate contractors, service categories and ownership. It will use an explicit distinction between a registered Guyanese supplier and domestic production content. Examples include port services, welding, catering, transport, engineering, aviation, environmental monitoring and professional services. An imported piece of equipment resold by a locally registered firm generates different domestic value from an engineered product manufactured and maintained in Guyana. The public data should be audited against supplier certification and customs records where legally available, with confidentiality and commercial rights preserved.

Rapid resource expansion also changes wages and the availability of teachers, nurses, civil servants and smaller-firm engineers. Government's 2026 mid-year report discusses technical and professional skills shortages and potential execution delays. Fiscal transfers and construction can raise activity while pushing against constraints in local land, machinery and skilled labour. An evidence-led study needs salary indexes, migration measures, training completions and occupational vacancy data by region. National job announcements or scholarships alone do not establish net employment and long-run earnings effects. [S07]

Social outcomes are especially susceptible to mistaken timing. The World Bank's country overview reports an older national poverty estimate based on 2019 data and explicitly notes the absence of recent comparable poverty data in that published summary. It would be misleading to treat that figure as the current 2026 poverty rate. An updated household survey, real regional consumption, nutrition, access to safe water, school completion and hinterland service coverage offer a stronger basis for later conclusions. The final paper will identify missing indicators rather than invent a post-oil poverty trajectory. [S10]

Transparency has several layers: formal contract publication, beneficial ownership, production volumes, government receipts, cost-oil audits, independent regulation, public participation and spending outcomes. Guyana has published several petroleum documents and EITI national reports. Its December 2025 EITI country report covered the 2023 fiscal year. The time lag matters for a rapidly expanding industry. [S18]

The EITI Board's May 2026 decision launched a full validation starting May 15 following a delay and stated concern about a lack of stakeholder confidence in the independence of civil-society representatives' selection process. This is a documented disagreement about the quality of participation, not grounds for this manuscript to pronounce on the legitimacy of a political actor. The final report will present the EITI decision, views of the multi-stakeholder parties and subsequent verified developments without asserting the outcome of an unfinished evaluation. [S16, S17]

More frequent digital publication can improve timeliness but will not replace independent challenge. A public resource ledger becomes useful when readers can reconcile production, sale prices, contract obligations, royalty payments, NRF deposits and authorized budget transfers. That ledger should carry dataset versions, source URLs and correction notices. A complete digital publication should be intelligible to citizens and technically testable by researchers.

9. Five international comparator designs and limits

Ghana's Petroleum Revenue Management Act 2011 established a stabilization fund and a heritage fund, along with publication and parliamentary accountability provisions. This provides a comparator for separating short-run expenditure stabilization from long-run financial savings and for setting statutory reporting frequencies. Comparing statutes alone would not show which country delivered better outcomes: data on implementation, withdrawals, public debt and actual welfare need to be paired with each legal instrument. [S24]

Norway's Government Pension Fund Global receives the state's net petroleum cash flow and sits within a fiscal framework linked to an estimated 3% long-term real return. Its 2025 official white paper discusses a guideline for keeping normal-year withdrawals below that rate and maintaining buffers for adverse years. Norway has a different institutional history, population, capital market environment and mature stock of financial assets. A new producer cannot translate its numerical fiscal guideline without modelling local infrastructure deficits, expenditure absorption, debt and reserve uncertainty. [S25]

Timor-Leste established a Petroleum Fund by law in 2005 and amended that framework in 2011. Its early fund architecture offers a comparison for a small developing state facing large prospective petroleum receipts. The final paper will add verified 2026 data on petroleum-field production, fund withdrawals, budget reliance, and development outcomes before drawing conclusions about the performance of its particular formula. A statutory rule and its fiscal sustainability are separate propositions. [S26]

Suriname offers a regional pre-first-oil comparison. TotalEnergies' September 19, 2026 update reports continuing construction at the GranMorgu offshore development, with first oil planned for 2028. It is therefore a prospective comparison, not another operating case. The administrative question concerns which Guyanese preparations could be examined before Suriname's commissioning and which differ under its national oil company, contract terms, marine environment and institutional structure. [S27]

Small island states must be handled on their own terms. A prospective producer with no tropical forest sector cannot finance development through Guyana's precise jurisdictional forest-credit

structure. A shallow-water or nearshore field presents a different spill-response geography from deepwater Stabroek operations. Tourism and reef exposure, narrow fiscal revenue bases, hurricane risk, desalination, land scarcity and cross-border response agreements change the feasibility of proposed institutions. The resulting handbook will be a configurable set of decision records, not a claim that one nation's contract and fund law suit every geology or society.

10. Transferable administrative sequence and adverse cases

A sequence-based framework begins with the premise that sovereign choices are made under different degrees of uncertainty. During exploration, the uncertainty is commercial: hydrocarbons may never be discovered in economic quantities. After a discovery, the challenge is negotiating rights and environmental protections before irrecoverable infrastructure is commissioned. At first oil, financial accounting and incident reporting need to work reliably. During scaling, the bottleneck often shifts toward public investment capacity, cost-oil review, labour, supplier quality, maintenance and exposure to global prices.

An early-country workbook would document the following actions and evidence without deciding the jurisdiction's political choices: establish a public chronology of licence and environmental authorities; inventory all fiscal obligations and contingent liabilities; map marine and coastal ecosystem baselines; obtain independent reservoir, legal, accounting and spill expertise; publish contract versions and define justified confidentiality exceptions; describe who verifies volume and price; and simulate an offshore emergency with available regional partners. Each item has a named evidence document and an owner, rather than a generic assertion of preparedness.

The second workbook starts when a commercial discovery is established. It records the development proposal's production range, cost envelope, decommissioning plan, environmental alternatives and labour forecast. It creates a source-and-use-of-funds bridge under different oil prices and differentiates operator spending from direct government expenditure. It identifies whether fiscal rules would operate before or after first cash receipts, which body publishes fund statements and which institution investigates alleged audit anomalies. Political decisions about contract models, ownership structures or spending ratios remain open; the workbook presents their quantifiable constraints.

A first-oil readiness audit would test whether field meters, cargo valuation, royalty invoices, cost-bank records, central bank reconciliations, incident-response roles and legislative reporting function under realistic load. A regulator should not assume every new FPSO offers the same risk profile or environmental permit conditions. A simple flowchart linking the operator's field data to the revenue authority, finance ministry, NRF custodian and parliamentary appropriation creates a traceable basis for independent scrutiny.

Adverse case A: a long low-price interval

A sustained fall in realized crude prices compresses gross sales, royalties and profit oil. If economic activity depends on imported equipment, construction can also slow. A test should compare projected authorized spending against revised legally available fund transfers, fiscal buffers, project postponement clauses, debt service and protection of essential operations. The same price shock has different effects when the PSA cost cap binds and after capital costs decline. Never assume a fixed state share of gross revenue across all phases.

Adverse case B: newly approved development replenishes the cost bank

A field reaches high production and government receipts rise as historical development costs decline. The operator then proceeds with further sanctioned projects. New recoverable expenditures affect the profit-oil calculation. A model based on a single high-receipt period overstates durable public cash if it ignores this cycle. Preserve fields, vintages, cost categories and contractual cross-charging constraints separately.

Adverse case C: a major offshore spill

The study assumes a serious release during adverse sea conditions and asks which legal provisions are in force, who orders emergency action, when cash becomes available, which state pays upfront, what coverage exclusions apply, how claims enter foreign jurisdictions and how independent evidence is preserved. The hazard exercise tests institutions, not an asserted probability of catastrophe. The final paper requires independent review of actual emergency plans and contractually enforceable insurance terms.

Adverse case D: overheating without fiscal insolvency

Even with a rising financial fund balance, public construction demand may exceed available contractors, engineers and aggregate supply. Prices and wages rise, project delays accumulate and the non-oil economy loses competitiveness. The July 2026 IMF statement highlights absorption, procurement and monitoring concerns as questions for ongoing policy review. The stress test therefore tracks real input costs and completed units of service, not budgeted nominal spending alone. [S08]

Adverse case E: public confidence breaks despite disclosure

An extractive revenue portal posts thousands of data rows while representatives of affected communities question the independence of stakeholder selection or regulators' responses. Mechanical disclosure does not substitute for meaningful access to information, independent representation and a process for correcting data. The EITI Board's May 2026 statement makes the distinction concrete. The audit would trace how complaints are recorded, decisions documented and adverse findings disclosed. [S16]

11. Initial findings and publication boundaries

Five verifiable facts structure the emerging case. The first commercial discovery preceded first oil by roughly four years. A Natural Resource Fund statute was assented to before first production. Subsequent fund governance and withdrawal rules changed in 2021 and 2024. Four offshore FPSOs were producing by the first half of 2026, with growth in reported profit-oil receipts and substantial cash held in the NRF. Guyana also has a distinct pre-petroleum low-carbon and forest-finance history that continued alongside expanding oil production. [S01, S04, S07, S22]

The next layer concerns institutional performance rather than chronology. IMF staff documents continuing needs in fiscal anchors, public investment monitoring, procurement, petroleum audit capacity, statistics and public-enterprise oversight. EITI 2026 validation is an ongoing process with publicly recorded stakeholder concerns. Environmental policy includes expanded statutory architecture whose commencement and practical implementation require careful verification. These issues are integral to a useful international case study. [S07, S08, S16, S19]

What an emerging producer can extract from the evidence is a disciplined research method: establish the historical sequence, date each legal change, distinguish entitlement from receipt and expenditure, record ecological hazards at the licence level, publish cross-checkable financial reconciliations, separate climate assets from fossil-fuel accounting, and verify public-service outcomes independently of announcements. Those administrative questions apply across different policy choices. Specific contract terms, saving formulas and energy strategies remain jurisdiction-dependent decisions.

This is a first research manuscript, not a finished monograph. Before final publication the author must complete annual fund reconciliations, contractual and judicial text review, independent environmental and social evidence, a dated 2026 EITI outcome check, quantitative scenarios, comparative-country datasets, external technical review and page-specific reference verification. The proposed 140-160 substantive pages are a future acceptance gate rather than a page count claimed here.

Technical appendix A. Analytical definitions and worked examples

A1: Stock-flow reconciliation. Let F_0 be opening fund cash, D be period petroleum deposits, I be recognized cash investment income, O be other permitted cash flows and W be lawful disbursements. Then $F_1 = F_0 + D + I + O - W$. Disclose transaction timing, currency, investment valuation and whether interest has already been included in D . The Ministry's June 2026 fund balance was USD 4.2942 billion, after USD 1.02 billion in first-half withdrawals. An annual account cannot be assembled by combining that figure with full-year projected deposits without dates. [S07]

A2: Production-to-cash bridge. Petroleum units produced in month M differ from saleable units, cargo lifted in month M and receipts recognized by government in month M . A dated lift ledger should identify cargo, lift date, entitlement owner, grade, price determination, settlement currency, invoice date, deposit date and exchange rate. In H1 2026 the government's 21 lifts included late-recognition effects from 2025. [S07]

A3: Simplified production-sharing example. Normalize a period's gross sales revenue to 100. In the binding 75% ceiling case, assign 75 to illustrative cost recovery and split 25 profit units equally. The state receives 12.5 profit units, plus the separately calculated 2% royalty described by the GRA, giving 14.5 combined units. At a 50% illustrative actual cost share, the split of remaining 50 units yields 25 state profit units plus royalty. At 20%, the corresponding illustration yields 40 plus royalty. These simplified examples omit detailed PSA adjustments and are neither observed 2026 figures nor revenue forecasts. [S02, S15]

A4: Illustrative oil-price sensitivity. A purely hypothetical field selling 1,000,000 barrels per day for 365 days produces 365,000,000 annual barrels. A USD 10 per barrel realized price change moves gross oil sales by USD 3.65 billion before operating expenses, royalty, recoverable costs, fiscal terms, discounting or cash timing. Applying that change directly to the NRF would be incorrect. The case-study model must instead use contract-specific entitlement and fund law.

A5: Capacity versus actual production. Nameplate daily capacity multiplied by 365 describes a theoretical annual maximum before downtime and operational constraints. Actual barrels are the integral of daily measured output, adjusted for reporting boundaries. Government monthly data and independently reported operator annual results should be reconciled.

A6: Carbon versus petroleum cash. A forest-credit agreement describes volume, price, vintage, payment conditions, registry issuance and benefit-sharing obligations. Petroleum cash follows royalty and profit-oil provisions. These may be reported on the same national fiscal page only if separately attributed. Contracted Hess purchase value is not the same thing as realized forest-carbon receipts. [S23]

Technical appendix B. The newcomer decision record

For every proposed instrument, the handbook has eight fields: legal authority; public purpose and measurable output; geological or ecological dependency; evidence required before adoption; implementing institution and budget; independent reviewer; disclosure frequency; and evidence that would trigger revision or suspension. Entries will cover the petroleum licence, environment permit, benefit-sharing instrument, revenue fund, procurement contract, local supplier registry, public investment project and transboundary spill protocol.

Technical appendix C. Questions for external reviewers

- Did the paper apply the legal text in force on the relevant date?
- Did the cost-oil model follow the original executed PSA, including valuation and royalty provisions?
- Did any model count contracted credit sales as cash already received?
- Were unpublished operator assumptions mistakenly treated as regulator-confirmed data?
- Were spill-insurance limits, legal liability and access to immediately liquid compensation distinguished?

- Do Indigenous communities' documented positions appear beyond official summaries?
- Did any poverty estimate use survey data older than the observation year claimed?
- Are state receipts, deposits, parliamentary transfers and spending outputs separately traceable?
- Does the case transfer instruments without recognizing island-state ecological and administrative limits?
- Has an independent expert performed the review, or is a review merely proposed?

References

Full URLs, source types, dates, numerical facts and remaining challenges are recorded in SOURCE_REGISTER.md (S01-S27). Citations inside the final PDF must point to exact pages/sections where available. This edition cites the public original sources by source ID and deliberately does not represent external peer review as completed.

Transferable country handbook

THE NEW-PRODUCER HANDBOOK

A staged institutional workbook for an emerging oil-producing country or island state

Independent companion annex to The Guyana Sequence | Ragunauth Ramsaroop | 27 September 2026

This handbook is a transferability instrument, not a claim that Guyana's history is a universal model. It offers defined questions, document templates and alternative institutional structures from published country cases. Sovereign authorities remain responsible for policy, legislation, fiscal choices, treaty obligations and public consultation. Statements of Guyana's actual history refer to source IDs in SOURCE_REGISTER.md.

1. Start with a resource and institutional map rather than a model law

The first analytical task for a prospective petroleum state is to describe the country before petroleum income changes its financial structure. Guyana's 2009 Low Carbon Development Strategy, earlier commodity economy and decades of offshore exploration predated the first major commercial discovery. The country did not begin building every relevant technical or financial capability only after oil arrived. An island without large tropical forests or comparable offshore acreage has a different starting set of options. [S01, S22]

An initial country profile records offshore geology and the uncertainty of commercial reserves, coastal population and infrastructure, current export composition, public debt, annual tax revenue, national disaster losses, fisheries and tourism dependence, existing environmental agencies, maritime boundaries, regulatory staff, procurement integrity processes, public finance reporting and access to international specialist advice. These are separate datasets. A positive resource discovery changes the geological column but does not erase pre-existing environmental or fiscal exposure. The profile should state the latest observation year for every variable. Missing observations are recorded as missing.

The profile also identifies the institutional decision-makers whose legal mandates may overlap. Typical actors include a parliament, a ministry responsible for resources, finance authorities, a revenue agency, an environmental regulator, a national audit institution, a central bank, a coast guard or maritime authority, regional government, Indigenous or customary-rights institutions, professional regulators and an independent judiciary. Actual institutions differ under civil-law, common-law, hybrid and constitutionally decentralized jurisdictions. The workbook records the current legal mandate rather than silently assigning a new one.

A useful opening deliverable is a two-page public evidence map. The first page identifies the legal source, date of enactment and commencement, administering authority and current

implementing regulations. The second page identifies data held by each institution, reporting lag, legal publication restrictions, unresolved boundary disputes and any independent review process. If existing law does not authorize the proposed activity, the gap is visible before budgets or investment agreements assume implementation.

A separate social and ecological baseline sets the denominator against which later claims will be tested. A coastline with dense tourism and reef ecosystems has a different vulnerability profile from Guyana's low coastal plain and broad forested interior. Climate adaptation, coastal fishery economics, drinking-water resilience and household service levels require direct measurement. Petroleum is a potential revenue source, not a substitute for obtaining the baseline.

2. Exploration stage: six documentary systems

The exploration stage has no guaranteed development or receipts. Guyana's experience includes decades of surveys and unsuccessful wells before Liza. A new state considering costly petroleum-specific institutions should record whether they are scaled to known commercial activity or to uncertain potential. In a country with limited human resources, time-bound specialist contracts and regional sharing agreements are among the institutional forms to compare against a permanent bureaucracy. Neither arrangement removes the need for retained institutional memory or independent conflict-of-interest screening. [S01]

The first documentary system is geological. A searchable block map links the licensing unit, coordinates, bathymetry, historical well records, independent reserve interpretation and uncertainty classification. All petroleum-in-place, contingent-resource and commercially recoverable-reserve statements use explicit definitions. Geological prospects are not parliamentary assets until commerciality is established and sales cash materializes.

The second system is fiscal. Each licence award or proposed agreement has a versioned record of fiscal instruments, stabilization clauses, obligations carried by the contractor, state obligations, allowable costs, relinquishment, abandonment and disputed-area exposure. Where parties negotiate confidentially, the public register states the legal basis and duration of nondisclosure and identifies which post-award terms enter the public domain. Tender dates and the grounds for different procurement methods are traceable.

The third system is ecological. Before drilling, the regulator receives baseline observations of water quality, seabed habitats, fisheries seasons, migration patterns, coastal protected areas, maritime traffic and prevailing currents. For transboundary risks, the record identifies neighboring states and the existing international response agreements. Geological survey and drilling permit terms should not be conflated with consent for a commercial producing field.

The fourth system is technical independence. A national authority can appoint external petroleum engineers, transaction counsel, public accountants and environmental scientists, subject to published conflict controls, confidentiality protections and knowledge-transfer requirements. The record lists each adviser, scope, deliverable, price, tenure, supervision and training obligation. Independent review is not established solely by naming an international consultancy; the published decision file needs evidence of the questions advisers examined and competing options considered.

The fifth system is public participation and rights. A dated register describes legally relevant consultations with fisher groups, coastal settlements, Indigenous and customary-rights holders, tourism representatives and other affected communities. Participation evidence records who received notice, technical materials and translation support, the matters raised and the regulator's reasoned response. National constitutional and treaty obligations determine how consultation, consent, compensation and review operate. The register should not expose personal data or confidential cultural sites.

The sixth system is contingency. A prospective exploration state maps the institutional owner of offshore emergency notification, regional deployment, oil-spill equipment, public-health coordination, aviation and maritime rescue, and access to claims assistance. At this stage a tabletop exercise may demonstrate gaps more clearly than a general emergency-policy statement. Its report identifies response time, jurisdiction and unresolved procurement, not a presumed ability to contain an unmodelled release.

These six systems can be prepared without predicting that commercial oil will be discovered. In an unsuccessful exploration outcome, their data and procedures may still inform geological archives, marine science and public procurement. The financial case for building them at a particular scale remains country-specific.

3. After discovery: the contract and approval decision file

A major commercial discovery introduces a different information problem. The state needs independent evidence about the field's technical and financial viability, while the operator controls substantial reservoir and project-cost information. The period between Guyana's 2015 Liza discovery and December 2019 first oil illustrates why agreements, environmental decisions and fiscal arrangements should be dated separately. Guyana's first NRF law was enacted during this pre-production period. [S01, S02, S04]

The development decision file begins with a technical summary listing verified discovered resources, development phases, forecast decline, required subsea facilities, FPSO or platform capacity, export route, power requirements, drilling schedule and decommissioning concept. A high production forecast must remain labelled as an operator case unless independently confirmed. Where the geology is uncertain, the file records alternative development sequences and their effect on public revenue rather than presenting one central production curve as an inevitable trajectory.

The commercial file compares the actual investment risks allocated by each contract form under the assumptions prevailing when signed. Production-sharing agreements, royalties and taxation, conventional licences and concession-tax systems, service contracts and state equity offer different combinations of upfront funding, price exposure, cost audit, abandonment liability and public oversight. The workbook does not presume any one form is appropriate for a given jurisdiction. It requires internally consistent cash flows, investor-return assumptions and explicit responsibility for failure to reach commercial production.

The fiscal model maintains separate input ledgers for price, recoverable costs, operating costs, investment timing, production, depreciation where relevant, state equity, royalties, profit sharing,

corporate taxation, bonuses and decommissioning. The output ledger distinguishes gross petroleum sales, contractor receipts, the state's profit oil, royalty payments, other taxes, NRF deposits, investment income and money later appropriated through the budget. Each ledger must reconcile on a dated basis.

Guyana's signed 2016 Stabroek agreement includes a 2% royalty, a 75% cost-recovery ceiling and equal sharing of profit oil. Its later model announced a 10% royalty, a 65% recovery ceiling and 10% corporate tax for different licensing circumstances. The older contract's cost bank and the later model should occupy separate scenarios in any comparison. A 2026 cash outcome is not enough to reconstruct the entire risk and bargaining position at the date of the first agreement. [S02, S14, S15]

The environmental approval file is linked to but institutionally distinct from the financial model. A field's commercial value does not establish that mitigation, marine response, compensation, monitoring or financial assurance are adequate. The file identifies each applicable environmental impact assessment, alternatives considered, legal right to inspect and halt operations, monitoring data ownership, baseline methods, spill modelling assumptions and required independent studies.

The public record should preserve dissenting technical views. A regulator's reasoned decision has different evidentiary status from an operator's response to comments, a government announcement or a court ruling. Where litigation is pending, a risk register distinguishes the order in force, the relief requested and legal contingencies still unresolved. Guyana's May 2026 appeal concerning environmental financial assurance demonstrates why final manuscripts must read authenticated judgments instead of relying on competing summaries. [S20]

4. The first-oil readiness rehearsal

First oil often arrives before the initial public development model has stabilized. Under a production-sharing arrangement, cost recovery, crude lifting schedules and sales settlement affect the precise timing of government cash. A national oil accounting system must therefore function during commissioning, not several years later when petroleum receipts already dominate fiscal reporting. Guyana's H1 2026 figures demonstrate the importance of this distinction: some payments received in the first half corresponded to cargoes lifted near the end of 2025. [S07]

The rehearsal opens with a fictitious cargo. The operator supplies a metered production record and government entitlement calculation. The designated fiscal authority checks the volume, crude quality adjustment, eligible costs and royalty; the sales entity checks invoicing and settlement. The central bank confirms receipt. The finance ministry reconciles the money to the NRF statement. The parliamentary record then tracks any authorized transfer from the fund to the consolidated budget. Each step carries a date, accountable official, audit trail and route for a discrepancy.

No example needs to use confidential, live commercial numbers. A deliberately fictional first exercise can contain a missing invoice, late shipment, wrong exchange rate, duplicated cost invoice and incorrect posting of investment income as oil sales. The purpose is to test whether the participating institutions identify and correct differences without an opaque manual adjustment. Material issues found in exercises should appear in a remediation log with a revised protocol and retest date.

At the same time, a parallel environmental drill begins with a hypothetical subsea discharge in poor visibility and a second simulated warning from a neighboring maritime authority. The exercise records the emergency command structure, notification triggers, helicopter and vessel mobilization, sampling, shoreline protection priorities, public communication, legal evidence preservation and claims intake. A state with limited equipment may document formal agreements for shared regional response. The record should distinguish an agreement on paper from a tested mobilization time.

A third rehearsal concerns access to information. Citizens, parliamentarians and researchers try to trace a fictional cargo from the project's regulatory record to the public revenue ledger and verify an appropriation against the published budget. If essential data are lawfully confidential, the portal should specify why and what aggregated fields remain public. A portal with attractive graphics but irreconcilable periods or undefined terms is not a substitute for usable records.

The rehearsals create three different types of evidence: account reconciliation, operating-response capability and public accessibility. None is a measure of the ultimate success of an oil development, but each identifies specific institutional functions before first-year petroleum revenue becomes an ordinary budget assumption.

5. Designing a fund: alternatives rather than a copied withdrawal percentage

Guyana established a statutory Natural Resource Fund before first oil, replaced the first law in 2021 and changed withdrawal provisions in 2024. Ghana's 2011 revenue framework has dedicated stabilization and heritage components. Timor-Leste's Petroleum Fund dates to 2005 with a 2011 amendment. Norway has a mature global fund with a long-run real-return-linked fiscal guideline. The relevant analytical insight is the range of statutory and fiscal architectures, not a ranking of countries or a single numerical withdrawal formula. [S04, S24, S25, S26]

A country examining a resource fund first documents the problem the fund addresses. Annual petroleum cash is exposed to price and production changes. Underground reserves cannot be valued as risk-free cash. Fiscal law often needs a legal method for receiving proceeds, reporting investment performance, transferring a defined amount to the budget and exposing disbursements to ordinary procurement and audit rules. Different development needs, debt service, age structure, exchange-rate arrangements and climate liabilities alter the budget and savings trade-offs.

One analytical model separates stabilization assets from a long-term savings portfolio, with a published transfer rule for each. A second holds a single sovereign portfolio and ties normal-year spending to an estimate of sustainable returns. A third adopts a formula linked to a defined share of verified revenue or assets within constitutional budget procedures. A fourth hybrid model retains a legal spending envelope but adds a multi-year expenditure ceiling, emergency exception and a clear independent disclosure process. Each model creates different accounting, forecasting and governance demands.

The workbook requires a country-specific estimate of the usable cash buffer under a joint commodity-price and natural-disaster stress event. It identifies the unit in which spending obligations arise, the currency and liquidity of fund investments, the risk of simultaneous

domestic recession, and a credible re-entry path after an emergency exception. If a rule permits extra withdrawals during a disaster, the document states who declares the exception, the amount and duration, oversight route and subsequent repayment or recalibration mechanism where applicable.

The core fund reconciliation is ordinary accounting. Opening cash, new petroleum deposits, investment return, other permitted adjustments and withdrawals together explain closing cash. Any transfer into the budget must reconcile to both the fund's statement and the government's budget execution statement. If royalties appear in oil-sale proceeds, they should not be added again under a different financial label. Cash received for prior-year liftings needs a visible timing bridge. One annual press release is not enough to replace account-level controls.

External audits establish whether statements are fairly prepared under an identified accounting standard. They do not establish whether the government selected or delivered the most valuable public projects. That second question requires public investment appraisal, procurement oversight, engineering completion evidence and independent service-delivery evaluation. Guyana's large fund cash balance in June 2026 and the IMF mission's July comments about investment monitoring illustrate the need to keep the two questions separate. [S07, S08]

A useful funding-options dossier therefore contains side-by-side legislation, liquidity simulations, administrative costs, governance responsibilities, spending plans, unknowns and public consultation records. The sovereign authority chooses the design under its constitutional and fiscal system. Publication of alternative assumptions allows later researchers to reconstruct the decision.

6. The petroleum revenue transparency ledger

An extractive revenue website serves different audiences. Parliament needs statutory compliance and account reconciliations. A central bank needs cash management and portfolio controls. Tax authorities need expenditure verification and settlement records. Local communities need understandable information about revenues, distribution programmes and environmental incidents. Researchers need data definitions and historical versions. A public ledger therefore needs more than a current total.

The minimal petroleum portion lists the project and contract identifier, fiscal instrument, observation period, quantity and unit, price basis, lift date, government entitlement, invoice period, payment date, receiving account and audit status. Project-specific records with legally protected commercial information may be aggregated, but the scope and legal basis of aggregation should be documented. An entity-level register records beneficial-ownership disclosures and changes subject to applicable privacy and security law.

The fund portion lists opening assets, new deposits by category, investment interest or income, valuation adjustment, lawful budget transfers and ending cash. Each published table includes its accounting basis and revision history. The budget portion separately identifies appropriations, procurement contract values, actual spending, verified physical completion, maintenance allocations and beneficiary results. A reader should be able to follow cash without assuming every transfer immediately produces a completed asset.

Environmental transparency is part of the same programme but has a different standard of evidence. The public permit register identifies drilling and production authorizations, impact assessment dates, permit expiry, reportable incident definitions, measured emissions, significant enforcement decisions, available monitoring data and any material remediation obligations. The administrative record captures which regulator confirmed a fact and which claim rests on operator self-report.

Civil-society representation cannot be reduced to web access. The EITI Board's May 2026 decision recorded stakeholder concerns about civil-society selection and initiated a full validation with external participation. An implementation workbook therefore documents who selected representatives, applicable rules, whether affected groups had access to technical support, how dissenting submissions were considered and whether public records provide a traceable response. The 2026 process should be described according to its final verified status before publication. [S16, S17]

A release process protects analytical integrity. Each dataset includes version, publication date, source agency, applicable period, methodological changes and a clear correction notice. A monthly series that changes definitions halfway through the year should not appear as a continuous trend without reconciliation. Differences between production, exports, oil sales and deposits should be shown rather than obscured.

7. Local participation and the value-added question

Governments use several forms of local participation: supplier qualification, procurement preferences or reservations, mandatory reporting, technical training, joint ventures, domestic processing, public education and access to finance. The domestic economic outcome depends on the mix of actual productive inputs rather than one gross procurement figure. Guyana's 2021 Local Content Act and its 2026 procurement reporting offer a case for examining the difference between an administratively eligible supplier and independently measured domestic value added. [S05, S07]

The workbook separates business registration, beneficial ownership, share of labour paid locally, imports embedded in purchased goods, plant and equipment owned within the jurisdiction, local subcontracting, professional accreditation, technology transfer and taxes retained. A service supplier may be a meaningful domestic employer even with imported equipment. Another company may book large contract value through a local subsidiary with most technical work completed abroad. Without disaggregation, both transactions look similar in gross purchase statistics.

A registry should permit independently checkable ownership and skills claims while respecting commercial confidentiality and individual privacy. Independent verification might compare procurement statements with audited supplier accounts, customs classifications, professional training completion and aggregate labour statistics. Alternative explanations for spending increases, including domestic construction inflation or reclassification of supplier categories, belong in the same analytical report.

Geography is equally important. Offshore supply chains often concentrate near ports and the capital, while inland and remote districts face different infrastructure needs. A national local-content percentage says little about benefits to hinterland communities. A map of procurement distribution, pay and completed infrastructure is therefore more informative than a single national total. For a small island, limited land, port throughput, environmental carrying capacity and seasonal tourism affect local supply opportunities.

The analysis must also measure possible displacement. Large operator contracts may raise wage rates and absorb engineers, drivers and electricians previously employed by clinics, schools, municipal agencies and non-oil firms. The relevant question is whether new training and productive capacity expand the overall skilled workforce faster than demand, not merely whether the petroleum industry hires more people. Labour-force and wage data require source, sample period and inflation adjustment.

8. Indigenous peoples, coastal livelihoods and public participation

An ethical and analytical case study makes participation visible as a process and an outcome. Guyana's forest-carbon policy commits part of eligible proceeds to Indigenous village-led development plans. Offshore petroleum activities primarily expose marine and coastal interests, while national spending decisions affect interior communities. Both require independently documented benefits and grievances. A small island's customary marine tenure, coastal fishery governance and tourism land rights may present different institutions and legal obligations. [S22, S23]

The country profile maps statutory land tenure, customary resource rights, relevant constitutional provisions, applicable international obligations, agreed consultation procedures and environmental justice claims. The appropriate consultation and consent process depends on those instruments and the circumstances of the project. Evidence includes dates, disclosure formats, translations, support for independent technical review and a record of the response to documented objections. A signed attendance sheet alone does not establish informed participation.

Community-finance claims require four distinct records: funds legally designated, sums transferred, locally selected projects and independently observed outcomes. A documented transfer to a village account is one fact. Community consent to a project, completion of the project, continuing operation and resulting improvements are additional facts requiring their own evidence. The full white paper should preserve contrary testimony and avoid extrapolating from a handful of cases to every participating village.

Maritime impacts require similar attention. Fishers' exposure depends on where vessels operate, exclusion zones, changes in landings, port congestion, prices, compensation procedures and baseline variability. An increase in national GDP does not prove unaffected fishers benefited. A properly specified marine impact study accounts for weather, seasonal migration, fisheries management, other industries and changes in reporting. The state and operator's ecological claims need independent verification where data are available.

The publication protocol should provide a right of factual response for materially criticized institutions and stakeholder organizations without allowing any source to veto findings. It should protect respondents who face credible confidentiality or retaliation risks. A clear chronology of complaints, agency responses and court decisions is more useful than a general assertion of agreement.

9. Environmental assurance across the full project lifecycle

Environmental permitting and oil-spill financial assurance are related but distinct. Licensing decides whether and under what conditions exploration or production is lawful. Financial assurance addresses availability and enforceability of resources after a covered event. Emergency response identifies assets and decisions available within the first hours and days. Ecological restoration addresses losses long after the emergency ends. Confusing these categories can create false confidence or understate actual liabilities.

Guyana's 2025 oil-pollution legislation and May 2026 appeal over permit-linked financial assurances offer detailed subjects for legal verification. The 2026 mid-year report described implementing regulations as under development and commencement as a future event. At a new publication cut-off, enacted legislation, operational regulations, permit guarantees and appellate orders must each be verified separately. This handbook does not assume any quoted insurance total without the actual contractual documents. [S07, S19, S20]

A project-specific risk register records the hazard, expected pathways, sensitive receptors, probability model and its uncertainty, monitoring threshold, legal reporting trigger, escalation owner, containment strategy, finance instrument and external response partner. Where the country lacks credible dispersion modelling or ocean-current data, the register states the limitation instead of displaying unwarranted decimal precision. Regional assistance agreements require contact details, transit permissions, operational compatibility and periodic exercises.

Insurance due diligence examines covered activities, limit per occurrence and aggregate, exclusions, deductible, insurer creditworthiness, jurisdiction for claims, financial assurance beneficiary, rapid-access liquidity, environmental restoration scope, third-party claims and decommissioning. A parent-company guarantee raises separate questions about capacity, jurisdiction, enforceability and its interaction with an operator's continuing liability. Evidence of a guarantee's existence is different from evidence that public liabilities are fully secured.

Monitoring covers normal operations as well as exceptional spills. Discharge records, produced-water treatment, routine and non-routine flaring, methane measurement, drilling waste, ballast handling, vessel collisions, protected species, beach contamination, accidental chemical release and supply-base impacts need project-specific methods and reporting thresholds. Annual mean conditions should not conceal short high-intensity incidents.

A small-island adaptation module places reef health, mangroves, beaches, freshwater lenses, tourism seasonality and evacuation logistics into the response plan. Guyana's forest MRV and coastal flood adaptation experience contain relevant monitoring methods, but the appropriate ecological baseline differs. A transferability assessment states precisely which laboratory

methods, satellite products, emergency drills or public disclosure practices travel between jurisdictions and which require new expertise.

10. Public investment and durable service outcomes

Converting exhaustible revenue into durable national assets is a broader question than saving revenue in an investment fund. Infrastructure and human capital have different delivery schedules and lifecycle liabilities. Large petroleum receipts create an opportunity to finance capital projects but also expose a small economy to construction cost inflation, workforce shortages, poor maintenance planning and irreversible land-use changes. IMF staff's July 2026 mission highlighted Guyana's need for sound public investment appraisal and monitoring. [S08]

The public investment record begins with a published definition of the service problem, baseline conditions, alternatives, demand assumptions, environmental costs, construction schedule and lifecycle operation. For a hospital, the completion test includes staffing, power, maintenance, service hours and patient outcomes, not merely the ceremonial opening. For roads, the test covers load conditions, all-weather access, drainage and maintenance. For coastal barriers, the test covers projected flood risk, residual exposure and local ecological impacts.

The procurement register connects each project to tender design, qualified bidders where publicly available, evaluation criteria, contract award, change orders, schedule extensions, independent inspection and defects liability. It distinguishes price savings during tender from total cost after scope changes. A public project portfolio report should identify projects whose maintenance liabilities exceed planned operating budgets before those assets enter service.

Fiscal analysis compares infrastructure delivery capacity with projected annual petroleum cash under multiple prices and production paths. A legally available withdrawal is not evidence that domestic construction supply can absorb the resulting spending without delays or price escalation. A separate model estimates import demand, skilled labour, land requirements and potential crowding of non-oil investment. Uncertainty in domestic absorption is reported alongside uncertainty in the oil price.

Social outcomes need disaggregation. Regional living standards, access to water and electricity, education completion, maternal and child health, housing affordability, travel time and employment among women and younger workers are plausible indicators. Their applicability and target levels follow national development plans and community consultation. Surveys taken years before first oil should remain marked historical rather than be presented as current evidence.

11. Five-year observation schedule

The workbook groups observation by operational stage rather than guaranteeing a calendar date for oil production. Year 1 begins when a jurisdiction formally launches a resource-governance work programme, whether or not a commercial discovery exists. Every item below is a record requested for research and possible administrative review, not a mandatory legal prescription.

During months 0-3 the record covers petroleum licence and geological holdings, current law and commencement status, fiscal exposures, marine baseline, existing disaster protocols and the capacity of existing agencies. The finance ministry assembles a preliminary risk and revenue-uncertainty register. A ministry or regulator charged with petroleum maintains the licensing and contract inventory. Statutory audit bodies map their access to records. Representative stakeholder institutions identify consultation requirements under applicable law.

During months 4-6 the record includes independent technical advice procured, review of existing disclosure obligations, available fiscal and contract design alternatives, cost-audit capacity, ecological knowledge gaps and an initial incident communications exercise. Where petroleum prospects are not commercially proven, the budget scenario uses zero commercial oil revenue as its central binding reality rather than projected receipts. If a commercial discovery has occurred, engineering and fiscal scenarios identify range and timing.

During months 7-12 the country dossier details chosen regulatory and contract procedures, EIA scoping, transboundary notifications where required, future fund legislation options, sample reconciliation records, independent audit mandates and a funded skills plan. Publication may be staged in line with lawful data protection, but the record explains disclosure omissions. A tabletop exercise tests resource and environmental decisions across agencies with no assumptions of operational perfection.

During months 13-24 the observed documents reflect the actual development stage: environmental permit record, drilling or commissioning inspections, contract-cost audits, any effective resource fund legislation, validated production metering where applicable and reporting systems. A jurisdiction still in exploration does not need fictitious profit-oil tables; it records progress against exploration-stage tasks instead. Any delay in production updates the fiscal scenario and budget expectations.

During years 3-5 the portfolio review examines signed versus actual development schedules, year-by-year revenue reconciliations if production has begun, actual fund deposits and transfers, fiscal stress-test assumptions, public investment delivery, domestic-value measurements, environmental incidents, administrative appeal outcomes and independent community outcome evidence. Comparable historical series require a fixed base and disclose methodological revisions.

At each stage the workbook reserves a correction section. A delayed field, adverse environmental judgment, commodity shock or public procurement failure changes facts, which changes the scenario. Reviewers should be able to reconstruct the original assumptions and the date those assumptions were revised. The workbook does not erase an earlier forecast when a later publication supersedes it.

12. Ten cross-agency exercises for the implementation team

Exercise one asks the petroleum ministry and environmental regulator to trace a fictional exploration block from acreage designation through EIA scope and final approval. Record the public legal documents and any unresolved overlap of powers.

Exercise two asks independent technical counsel to quantify how a dry-well outcome changes the proposed administrative budget. Document the difference between spending on general-purpose geological data and spending that depends on an operating hydrocarbon industry.

Exercise three uses the executed PSA or concession terms to trace a fictional field's sales, allowable cost, royalty, taxation and state receipt. The revenue authority records how disputes and payment delays affect cash timing.

Exercise four asks the central bank, finance ministry and auditor to reconcile the same fictional oil payment across accounts, including a settlement posted one month after physical lifting. A mismatch creates a correction record, not a hidden balancing item.

Exercise five asks the parliamentary budget office to distinguish legally available fund transfers from cash already collected, projected year-end receipts and operational spending commitments. Run an illustrative low-price year and document different buffer designs without prescribing a single fiscal rule.

Exercise six asks the environmental regulator and coast guard to test a simulated spill with a neighboring state affected. Record who authorizes sampling, emergency procurement, shoreline access and claims intake. Repeat the same exercise under a power outage or severe coastal storm.

Exercise seven asks the procurement authority to evaluate a large petroleum-financed infrastructure project before and after a hypothetical 25% construction-cost overrun. Include maintenance, skill shortages and the consequences of delaying other projects.

Exercise eight asks the local supplier registry and an independent economist to reconcile a high gross procurement announcement with wage payments, imported equipment, local ownership, subcontracting and retained domestic profits.

Exercise nine asks a protected independent panel to document stakeholder participation in a fictional coastal development approval. The test is whether different views reach the decision-maker and receive reasoned responses, not whether every party agrees.

Exercise ten asks a mixed technical review team to assess a hypothetical halt in offshore output coinciding with flooding and a sharp oil-price fall. Recalculate cash, essential services, contract obligations, ecological emergency funding and the credibility of public communications.

Every exercise produces a date, participating institution, source documents, observations, corrective actions, verification method and retest date. An uncompleted exercise is shown as outstanding rather than assigned a favorable qualitative label.

13. What Guyana's case does not establish

Guyana's early NRF legislation is a historical observation. It does not by itself prove that a particular savings rate, withdrawal formula or investment portfolio has maximized intergenerational welfare. Such a claim would need long-run counterfactual analysis, accounting and household outcomes. A large fund balance and rapid production growth are not interchangeable with measured school performance, flood resilience or reductions in poverty. [S04, S07, S10]

The 2016 and later model PSAs document different fiscal terms in different commercial circumstances. These documents do not establish that the same contract works for a shallow-water field, a field with state equity, a landlocked onshore producer or an island whose public balance sheet cannot bear a large contingent spill liability. The exact risk allocation and counterparty financing capacity must be reconstructed for each case. [S02, S14]

Guyana's forest programme offers a separate climate-finance case, but a country without forests of comparable scale has no identical carbon asset. A country's petroleum revenues and forest-credit receipts have different legal rights, accounting bases and environmental effects. Carbon-credit eligibility and contract value do not establish measured nationwide carbon neutrality or a permanent right to petroleum expansion. [S22, S23]

The existence of enacted pollution legislation is not synonymous with commencement, funded enforcement or tested emergency operations. An appeal about financial assurance does not settle every possible ecological-loss scenario. The research must retain authenticated legal sources and independent insurance analysis as open requirements. [S19, S20]

Finally, country comparisons have different production histories and social baselines. Ghana's revenue law, Norway's mature financial fund, Timor-Leste's sovereign petroleum fund and Suriname's pre-first-oil preparations are documentary comparators rather than a league table. A useful successor study would collect consistent country-year data, label changes in accounting methodology, document the distinct welfare measures available and subject any causal claims to a defensible identification strategy. [S24-S27]

14. Compact reusable templates

Country opening record: Name of jurisdiction; programme initiation date; discovery status; resource classification; public debt observation year; existing royalty and tax base; existing environmental and marine protections; sensitive ecosystems; customary-rights framework; coastal and inland settlements; responsible institutions; independent audit mandate; unresolved treaty and maritime questions; last household-survey observation year; central-bank data accessibility.

Agreement comparison record: Contract identifier and version; signature and effective dates; contract model; legal approval; resource and price uncertainty at the time; development and abandonment obligations; royalty, tax and profit-sharing terms; cost eligibility and audit rights; stabilization and dispute resolution; assignment and beneficial ownership; environmental and financial assurance; public disclosure status; independent counsel and conflict review.

Fund reconciliation record: Fiscal period; audit status; opening cash and investment portfolio value; petroleum deposits by source; investment income and valuation adjustments; authorized withdrawal; receiving treasury account; budget-law reference; closing cash; reconciliation difference; publication and revision dates; public audit opinion.

Environmental assurance record: Project and licence identifier; baseline surveys; EIA date; permit conditions; discharge and emissions thresholds; incident reports; independent monitoring; response assets and mobilization results; insurance contract and exclusions verified; parent or other guarantee where applicable; claim jurisdiction; decommissioning security; statutory effective dates; public grievance and judicial review procedures.

Community outcomes record: Affected stakeholder class; legal consultation rule; disclosure and meeting dates; dissenting and supporting submissions; reasoned agency response; monetary entitlement and transfer; locally approved projects; construction and operation verification; measured benefit and distribution; continuing grievances; independent reviewer.

Public investment outcome record: Project identifier; approved business case; procurement method; financing mix; cost baseline; contract award and variations; completion evidence; maintenance provision; climate or environmental exposure; service baseline; actual beneficiaries; operation performance; auditor's findings; outstanding remedial work.

Correction and disclosure record: Dataset or legal assertion corrected; original publication date; conflicting source; confidence and reason for correction; old and new values; whether previous analytical conclusions change; approving researcher; next source verification date; cross-referenced exhibits.

The final 140-160 substantive-page monograph will integrate these templates with year-by-year datasets, a verified comparison of fund rules, legal text citations, multi-scenario modelling and independent external technical review. Until then, this playbook is an openly labelled research instrument attached to the working edition.

New producer implementation blueprint

New Producer Implementation Blueprint

A configurable exploration-to-production governance system derived from the Guyana case

How to use this blueprint

This document converts the Guyana research into a sequence of decisions for a country or island state. It does not prescribe a political choice, ownership structure, fiscal regime or spending ratio. Each government should adapt the sequence to its constitution, legal tradition, geology, marine ecology, population, debt position and institutional capacity.

Phase 0: Before a commercial discovery

A. Legal and institutional map

Create one public diagram showing:

- ministry responsible for petroleum policy;
- licensing authority;
- environmental regulator;
- revenue authority;
- central bank;
- auditor general;
- procurement authority;
- maritime safety authority;
- disaster-response command;
- local-government responsibilities;
- Indigenous or customary-rights authorities;
- parliamentary oversight bodies.

For every institution list statutory authority, decision rights, staffing, conflicts-of-interest rules and publication duties.

B. Geological information governance

Create:

- acreage map;
- licence register;
- seismic and well-data inventory;
- data-room access policy;
- relinquishment schedule;
- beneficial-ownership register for licence holders;
- retention and confidentiality schedule.

Avoid converting exploration forecasts into public-revenue forecasts.

C. Environmental baseline

Before drilling expands, record:

- currents and seasonal circulation;
- sensitive marine habitats;
- fisheries;
- mangroves, reefs and wetlands where relevant;
- coastal communities and water intakes;
- seabird and marine-mammal observations;
- baseline hydrocarbon chemistry;
- emergency ports and airfields.

Publish versioned datasets where safety and commercial restrictions permit.

D. Fiscal readiness

Build a transparent petroleum fiscal model with:

- price;
- production;
- development cost;
- operating cost;
- royalty;
- cost-recovery rule;
- profit sharing;
- tax;
- depreciation;
- decommissioning;
- financing;
- timing.

Run downside cases before upside cases.

Phase 1: Commercial discovery

Gate 1: Independent discovery review

Before public spending plans rely on oil:

- independently review the resource range;
- distinguish discovery from commerciality;
- publish uncertainty bands;
- require a development plan before treating reserves as production-ready.

Gate 2: Contract and fiscal review

Use independent petroleum fiscal expertise to model:

- investor breakeven;
- government revenue under low/base/high prices;
- delayed startup;
- cost overruns;
- lower-than-expected reservoir performance;
- decommissioning and abandonment.

Keep negotiation strategy confidential where law permits, but publish executed agreements unless a specific lawful exception applies.

Gate 3: Sovereign revenue architecture

Before first recurring petroleum receipt:

- enact the revenue-fund legal basis;
- specify deposit and withdrawal rules;
- define investment objectives;
- prohibit off-budget expenditure from the fund unless the constitution expressly provides otherwise;
- appoint accountable fund management;
- define audit frequency;
- specify public reporting;
- create conflict-of-interest rules;
- state how emergency deviations work.

The fund should integrate with the national budget, not become a parallel budget.

Phase 2: Development approval

Technical approval packet

Require the operator to submit:

- field development plan;
- reservoir depletion strategy;
- FPSO or platform capacity;
- metering design;
- gas handling;
- flaring controls;
- produced-water management;
- waste plan;
- decommissioning concept;
- local-content plan;
- emergency-response plan.

Environmental decision packet

Require:

- EIA;
- alternatives analysis;
- cumulative-impact analysis where multiple fields are planned;
- spill modelling;
- sensitive-receptor map;
- consultation record;
- financial assurance;
- monitoring plan;
- incident notification thresholds;
- restoration and compensation procedures.

Public revenue packet

Before sanction, publish a non-confidential fiscal sensitivity:

- \$40 oil;
- \$60 oil;
- \$80 oil;
- low/base/high production;

- cost-overflow case;
- delay case.

Do not publish a single-point revenue promise.

Phase 3: Twelve months before first oil

Revenue rehearsal

Run a simulated month:

1. operator reports production;
2. regulator verifies meters;
3. entitlement is calculated;
4. cargo is lifted;
5. sale price is determined;
6. royalty is invoiced;
7. state profit share is settled;
8. cash reaches the designated account;
9. central bank verifies receipt;
10. sovereign fund statement is published;
11. auditor tests the reconciliation.

Publish the exercise and corrections.

Emergency rehearsal

Run a live multi-agency spill exercise:

- detection;
- notification;
- command transfer;
- offshore containment;
- shoreline protection;
- fisheries closure;
- independent sampling;
- public communication;
- claims intake;
- cross-border notification.

Record mobilization times and corrective actions.

Public investment readiness

Create a ranked pipeline of projects, but require:

- economic and social appraisal;
- climate-risk review;
- procurement plan;
- land status;
- maintenance budget;
- delivery agency;
- measurable service output.

Do not equate the size of petroleum receipts with the size of the executable capital programme.

Phase 4: First oil through rapid scale-up

Quarterly production-to-cash statement

Publish:

- barrels produced;
- barrels lifted;
- government entitlement;
- realized sale price;
- royalty;
- profit oil;
- tax where applicable;
- sovereign fund deposit;
- investment income;
- withdrawal;
- closing balance.

Use the same definitions each quarter.

Cost-audit programme

Maintain:

- audit period;
- expenditures tested;
- amount challenged;
- amount accepted;
- amount recovered;

- disputed amount;
- arbitration or court status.

Do not announce proposed disallowances as recovered revenue.

Local-value ledger

Track:

- supplier beneficial ownership;
- domestic payroll;
- domestic procurement;
- imported component;
- training;
- certifications;
- technology transfer;
- export capability gained.

Environmental performance ledger

Track by project:

- permit;
- monitoring reports;
- flaring;
- discharges;
- spills;
- enforcement;
- financial-assurance status;
- emergency-drill date.

Phase 5: Production maturity

Fiscal transition

Move from startup assumptions toward:

- structural non-resource balance;
- long-term fund-return assumptions;
- public-asset maintenance;
- contingent liabilities;
- post-peak production scenarios.

Economic transition

Measure:

- productivity outside petroleum;
- non-oil exports;
- energy cost;
- food production;
- manufacturing;
- logistics;
- tourism where relevant;
- workforce participation;
- real household income.

Resource depletion plan

Before production declines:

- update abandonment funding;
- define post-production regulatory responsibilities;
- model reduced royalties;
- stress-test the national budget;
- determine required financial assets to sustain priority services.

A special module for small islands

Small islands face conditions that often differ from Guyana.

Marine concentration risk

A serious spill may affect a large share of fisheries, tourism coastline or desalination infrastructure. Financial assurance and regional response agreements should reflect this concentration.

Land and labour limits

A small workforce may be unable to supply both petroleum services and existing sectors. Local-content policy should prioritize capability and wage sustainability rather than gross percentage targets alone.

Fiscal scale

A small public administration may receive revenue far larger than its annual capacity to procure and manage projects. Offshore financial saving becomes more important when rapid domestic spending creates severe inflation or construction bottlenecks.

Natural-capital differences

Guyana's forest-carbon model depends on its forest endowment and monitoring history. Island comparators should assess mangroves, reefs, seagrass, fisheries and blue-carbon opportunities separately.

Disaster overlap

Hurricanes, storm surge, coastal erosion and petroleum accidents can coincide. Emergency planning should include compound events.

Twenty-four-month government action calendar after a commercial discovery

Months 0-3:

- confirm commercial-resource range;
- freeze unsupported revenue promises;
- publish licence chronology;
- appoint independent fiscal/legal/environmental advisers;
- launch baseline environmental update.

Months 4-6:

- publish fiscal-model methodology;
- draft revenue-fund framework;
- create inter-agency petroleum working group;
- publish beneficial-ownership requirements;
- establish contract and cost-data repository.

Months 7-12:

- enact fund and reporting rules;
- complete development EIA;
- define financial assurance;
- simulate first-oil revenue reconciliation;
- establish procurement and local-value baselines.

Months 13-18:

- appoint and train fund oversight bodies;
- conduct spill exercise;
- publish first public investment pipeline;
- test metering and cargo valuation controls;
- train audit teams.

Months 19-24:

- publish readiness audit;
- resolve material gaps before production;
- finalize quarterly disclosure templates;
- establish independent corrections process;
- publish downside fiscal scenarios.

Five-year scorecard after first oil

The scorecard records evidence, not political grades.

1. Revenue traceability: share of material petroleum receipts reconciled from cargo to sovereign fund.
2. Audit closure: percentage of audited costs with final disposition.
3. Fund transparency: statements published on schedule and externally audited.
4. Project delivery: planned versus completed public infrastructure, plus maintenance readiness.
5. Inflation and absorption: construction costs, wage pressure and execution delays.
6. Local value: domestic wages, ownership and higher-skill services retained.
7. Environment: permit compliance, incidents, spill drills and assurance coverage.
8. Social outcomes: current household survey, water, health, education and regional access.
9. Economic diversification: non-oil exports and productivity.
10. Intergenerational balance: financial assets, liabilities and decommissioning commitments.

Red flags

- budgeting against undiscovered or unsanctioned reserves;
- a sovereign fund with no binding publication rule;
- fund spending outside the normal budget;
- a single-point oil-price forecast;
- environmental permitting after commercial commitments become difficult to reverse;
- local-content targets without supplier ownership checks;
- gross procurement presented as domestic value added;

- carbon-credit contracts presented as immediate cash;
- production capacity presented as actual output;
- an enacted environmental statute presented as operational before commencement and regulations;
- an unresolved audit dispute presented as revenue already recovered;
- public investment measured by money spent rather than services working.

Final adaptation rule

Copy the decision discipline, not the exact Guyanese numbers.

Every country should ask which Guyana practice solved a general governance problem, which depended on Guyana's own natural endowment, and which remains under evaluation. The strongest blueprint is one that preserves this distinction.

Comparative cases and transfer limits

Comparative Cases Matrix

How Guyana should be compared without creating a political ranking

This matrix supports the flagship paper. It compares institutional mechanisms, not governments.

Comparative research protocol

For each comparator, the final paper will:

1. use the statute or official fund report as the primary legal source;
2. record the date the mechanism was enacted and the date commercial production began;
3. show actual fund flows for the latest comparable year;
4. separate official objectives from observed fiscal and social outcomes;
5. include at least one independent multilateral or peer-reviewed assessment;
6. identify a failure mode or constraint as well as a documented strength;
7. avoid scores, rankings, winners or political labels.

Ghana

Ghana's Petroleum Revenue Management Act provides a useful design comparison because it separates near-term stabilization from heritage savings and creates a dedicated public accountability mechanism. The research question is not whether this system is superior to Guyana's. The useful comparison is whether multiple-purpose funds or a single fund with layered rules generate clearer decision rights, fewer off-budget pathways and stronger public comprehension.

Norway

Norway is relevant for long-horizon financial discipline, integration between petroleum cash and the national budget, professional asset management and extensive disclosure. It is a poor direct template for early state capacity, infrastructure scarcity or immediate development gaps. The flagship will use Norway to illustrate institutional maturity and long-term asset conversion, while clearly stating why a new producer should not import a numerical spending guideline without its own macroeconomic model.

Timor-Leste

Timor-Leste is a small-state comparison for high petroleum-revenue concentration, public investment needs and the tension between saving and domestic development. Its experience helps test whether a strict benchmark remains credible when infrastructure deficits are large. The research will use updated Petroleum Fund reports before drawing conclusions about current sustainability.

Suriname

Suriname is the most useful regional pre-first-oil comparison at the 2026 cut-off. It offers a live test of which preparations can be completed before GranMorgu begins production and where national oil company arrangements, fiscal law and public-finance constraints differ. Planned production is not counted as observed output.

Small-island adaptation

For a small island state, the comparative chapter will add:

- reef and fisheries dependence;
- tourism concentration;
- hurricane and storm-surge overlap;
- desalination or freshwater intake exposure;
- limited ports and response vessels;
- narrow skilled-labour pools;
- smaller domestic supplier markets;
- limited land for shore bases;
- high import shares;
- regional mutual-aid requirements.

The island blueprint therefore emphasizes emergency cooperation, financial assurance, phased local-content targets, offshore savings during construction bottlenecks and independent ecological baselines before first development approval.

Expanded flagship research dossiers

The following sections are the current research-stage expansions for the 520-560-page flagship. They remain subject to integration, source-level editing, independent review and final publication QA.

PART I RESEARCH DOSSIER

Before the Discovery: Guyana's Geological, Institutional, Environmental and Fiscal Foundations

Working research dossier for Chapters 1-4 of the 520-560-page flagship publication. Evidence cut-off: 27 September 2026. Status: research-stage material. External review pending.

Purpose of Part I

Part I establishes what existed before the 2015 Liza discovery. This prevents a common analytical error: treating every later capability, law, institution or environmental policy as a consequence of oil.

The working hypothesis is narrower. Guyana entered the offshore petroleum era with several inherited legal, geological, public-finance and environmental institutions. Some were decades old. Others developed in the years immediately before commercial discovery. Their existence does not prove they were sufficient for the scale of later petroleum production. It does establish the institutional starting point against which post-2015 reforms should be measured.

Part I therefore asks four questions:

1. What economic, legal and institutional conditions existed before the commercial discovery?
2. What did Guyana already know about its petroleum geology and maritime acreage?
3. Which environmental and natural-capital institutions existed before first oil?
4. Which capacities had to be expanded after discovery because the scale and speed of petroleum development changed the demands on government?

The final chapters will distinguish inherited capacity from later adaptation.

Chapter 1 research dossier

Guyana before offshore petroleum

1.1 Economic starting point

The pre-discovery economy should be reconstructed from official national accounts, fiscal data, external-debt statistics, export structure, employment and infrastructure indicators. The chapter will not use present-day GDP as a proxy for historical administrative capacity.

The baseline dataset should include at least:

- nominal and real GDP for 2000-2015;

- export composition by gold, bauxite, rice, sugar, timber and other major categories;
- central-government revenue and expenditure;
- public debt and external debt;
- current-account position;
- inflation;
- electricity generation and reliability;
- road, port, aviation and telecommunications indicators;
- labour-force participation and skills data;
- poverty and household survey dates;
- regional population distribution;
- coastal flood exposure;
- public-sector employment in technical agencies;
- higher-education output in engineering, law, finance and environmental disciplines.

The reason for building a broad baseline is methodological. A country does not receive a petroleum windfall into an institutional vacuum. Its ability to negotiate contracts, supervise development, absorb capital investment and verify environmental performance depends on institutions and labour markets built for earlier economic conditions.

1.2 Commodity and extractive experience

Guyana's prior experience in mining, forestry and agriculture matters because extractive administration, land rights, environmental permitting and commodity-price volatility were not new concepts in 2015. The petroleum sector differed in scale, capital intensity, marine risk and contractual complexity, but several government functions had earlier analogues.

The final chapter should identify which capabilities transferred from mining and which did not. Examples include:

- licensing and concession administration;
- royalty assessment;
- environmental permits;
- occupational safety;
- customs and import controls;
- geological data management;
- community relations;
- Indigenous land issues;
- export documentation;
- international commodity-price exposure.

This section should avoid claiming that prior mining regulation automatically prepared the state for deepwater petroleum. It will instead map transferable administrative functions and identify where offshore petroleum required new expertise.

1.3 Geography and infrastructure constraints

A national map should show the concentration of population and economic activity along the coast, the location of major ports and airports, hinterland communities, mining corridors, protected areas and major drainage infrastructure. This provides context for later public-investment choices.

The final analysis should answer:

- Which infrastructure constraints predated oil?
- Which constraints later became more visible because rapid growth increased demand?
- Which petroleum-sector facilities were privately financed?
- Which supporting infrastructure involved public expenditure?
- Which investments primarily served the wider economy rather than the oil sector?

This distinction matters for transferability. A small island producer may have shorter internal transport distances but much higher exposure to port disruption, hurricanes, reef damage and tourism losses. Guyana's continental geography produces different infrastructure and environmental trade-offs.

Chapter 2 research dossier

A century of petroleum exploration

2.1 Exploration did not begin in 2015

The official Petroleum Management Programme records a long exploration history. It lists early signs of hydrocarbons, a first oil prospection licence in 1938, renewed offshore exploration activity in 1965, regulatory changes in the late 1960s, later exploration programmes and several interruptions caused by financing and boundary disputes. [S35]

This history is important because it shows the difference between petroleum potential and commercial success. A basin may attract exploration for decades before a commercially viable discovery. Governments designing institutions during exploration therefore face a difficult problem: they need sufficient capacity to protect national interests without committing large recurring public expenditures against an uncertain future industry.

The country blueprint should extract this lesson directly. Before commerciality is established, governments need scalable legal, geological, environmental and fiscal capability. They do not need to build an oversized production bureaucracy funded by hypothetical future revenue.

2.2 Public investment in geological knowledge

The final chapter should reconstruct the role of geological surveys, seismic work, public data, international technical assistance and promotional activity. The source register already records the government's summary of 1980s efforts to attract investors and synthesize petroleum data. [S35]

A stronger evidence package should add:

- original World Bank project documentation;
- Guyana Geology and Mines Commission records;
- historical licensing regulations;
- original seismic and well-data catalogues where public;
- records of government petroleum units;
- model agreements used before the modern offshore era.

The key question is not whether external assistance was good or bad. The question is what public capability remained after each programme ended. The final chapter should track data ownership, staff training, legislation and institutional continuity.

2.3 The 1986 statutory foundation

The Parliament of Guyana records the Petroleum (Exploration and Production) Act 1986. The law established a statutory framework for exploration and production and applied to petroleum in Guyana's land and maritime areas. [S38]

The full legal chapter should later determine:

- which provisions governed licensing before 2015;
- which regulations were in force at the time of the Stabroek agreement;
- what powers were held by the minister;
- how petroleum prospecting licences, production licences and agreements interacted;
- how confidentiality and data ownership worked;
- how environmental duties interacted with petroleum rights;
- which provisions have since been superseded or amended.

No 1986 provision should be quoted as current law without checking later legislation and commencement.

2.4 Maritime boundaries as a petroleum-governance issue

Offshore petroleum governance depends on uncontested or lawfully managed maritime rights. The Permanent Court of Arbitration's case record states that the Guyana-Suriname dispute involved oil concessions and a 2000 incident in which an oil rig and drill ship were ordered to leave a disputed maritime area. Guyana initiated arbitration in 2004. [S40]

This belongs in the petroleum history because boundary certainty affects exploration rights, company risk, seismic acquisition and development planning. The final chapter should use the award itself for legal propositions and the case record only for procedural history.

The transferable lesson is procedural. A prospective producer should map unresolved maritime boundaries before commercial development, identify applicable international dispute mechanisms and avoid treating acreage promotion as a substitute for clear sovereign title.

2.5 The Liza-1 discovery record

The official well register records Liza-1 as an exploration oil well in the Stabroek Block with a well date of 15 May 2015. [S36]

The final discovery chapter should separate:

- spud date;
- discovery announcement;
- well date;
- appraisal results;
- commerciality determination;
- reserve/resource estimates;
- development approval;
- final investment decision;
- first oil.

These are different milestones. Combining them into a single "discovery date" hides the development sequence.

2.6 Exploration evidence gaps

Before Part I is source-audited, the following items remain required:

1. Original licence instruments for historically significant offshore blocks.
2. A verified timeline of the 1999 Stabroek agreement and later contractual changes.
3. Original pre-2015 model agreements or fiscal templates where public.
4. Historical seismic acquisition and well chronology from regulator records.
5. Contemporary oil-price and frontier-basin investment conditions when material to negotiations.
6. A clear chronology of boundary disruptions and resolution.
7. Independent academic petroleum-geology literature for the Guyana-Suriname Basin.

Chapter 3 research dossier

Environmental and natural-capital foundations before oil

3.1 Environmental law predates first oil

Parliament records the Environmental Protection Act 1996, establishing the statutory foundation for environmental management, pollution control, impact assessment and sustainable use of natural resources. [S39]

This matters because environmental governance did not begin when oil production began. The analytical task is to determine how a general environmental framework designed before large-scale offshore petroleum was adapted to deepwater projects.

The final chapter should establish:

- EPA institutional authority;
- environmental authorization requirements;
- EIA triggers;
- public participation rules;
- permit enforcement powers;
- pollution and waste provisions;
- appeal rights;
- monitoring and reporting;
- interaction with petroleum approvals.

The existence of the 1996 act should not be treated as evidence of adequate later enforcement. The project-by-project chapter in Part V will examine permit implementation separately.

3.2 The 2009 Low Carbon Development Strategy

Guyana's original Low Carbon Development Strategy predates the 2015 commercial discovery and first oil by a decade. The official LCDS record describes the strategy as beginning in 2009 and the Guyana-Norway partnership as part of the first phase of results-based forest climate finance. [S22, S35]

This timing is central to the flagship argument. Forest conservation, climate finance and national natural-capital policy should not be described as petroleum-offset programmes invented after the oil discovery.

The final section should track:

- the original 2009 policy objectives;
- the 2009 Norway partnership;

- forest-monitoring system development;
- payment conditions;
- safeguards;
- investment channels;
- community and land-title components;
- later transition to LCDS 2030 and jurisdictional carbon markets.

The working source register records the government's 2024 LCDS update as reporting USD 227 million earned for forest climate services for 2009-2015. That figure is a government programme record and should be reconciled with Norway's published disbursement records before the final chapter. [S22]

3.3 Why natural capital belongs in a petroleum case study

A petroleum case study that measures only underground hydrocarbons and financial assets understates national wealth. Guyana's forests, biodiversity, fisheries, agricultural land, freshwater and coastal protection also produce economic and social value.

The final national wealth framework should classify at least:

- petroleum reserves;
- financial assets;
- public infrastructure;
- human capital;
- forests;
- biodiversity;
- fisheries;
- agricultural land;
- water resources;
- coastal protective systems;
- environmental liabilities.

The purpose is not to convert every ecosystem into a speculative monetary value. The purpose is to prevent petroleum revenue from being treated as the only national asset affected by development.

3.4 Coastal exposure

Guyana's coastal settlement pattern makes flood resilience and drainage material to national development. The final chapter should use official coastal-risk, drainage, sea-defence and climate-adaptation sources.

Required evidence includes:

- population distribution by region;

- coastal elevation;
- historical flood events;
- sea-defence coverage;
- drainage and irrigation systems;
- insured and uninsured exposure;
- projected climate impacts;
- adaptation expenditure and delivered outcomes.

This baseline becomes important later when petroleum-financed public investment is evaluated. A sea-defence allocation should be assessed against completed resilient infrastructure and reduced risk, not expenditure alone.

3.5 Environmental baseline for offshore petroleum

The flagship should identify which marine baseline datasets existed before large-scale offshore production and which were produced through project-specific EIAs.

The evidence register should include:

- currents;
- water quality;
- marine mammals;
- fisheries;
- seabirds;
- turtles;
- benthic habitat;
- mangroves;
- coastal protected areas;
- fishing grounds;
- shipping routes;
- sensitive shoreline receptors.

Where baseline data are operator-funded, the paper should record who collected the data, who owns it, whether raw data are public, and whether independent researchers have replicated the findings.

Chapter 4 research dossier

The state-capacity baseline before the petroleum windfall

4.1 Capacity is an institutional system

State capacity is not a single agency. For petroleum governance, it includes the ministry responsible for petroleum policy, licensing authorities, geological institutions, the environmental regulator, revenue administration, customs, central bank, finance ministry, auditor general, procurement bodies, parliament, courts, maritime authorities, emergency services and national statistics.

The final chapter should build a dated institutional map for 2014 and then compare it with 2019 and 2026.

4.2 Legal and fiscal capabilities before commercial production

The existence of the 1986 petroleum law, the 1996 environmental law and older public-finance institutions demonstrates an inherited legal base. [S38, S39]

The more difficult research question is operational depth. The flagship should estimate, where reliable records exist:

- petroleum-specialist staffing;
- environmental technical staff;
- reservoir and petroleum engineering expertise;
- transfer-pricing and international tax capability;
- cost-audit capacity;
- petroleum-law expertise;
- maritime response capability;
- procurement professionals;
- statistical staff;
- public investment appraisal skills.

This should use published staffing and budget evidence rather than retrospective assumptions.

4.3 The discovery-to-first-oil capacity race

The final publication will compare the institutional baseline with the four-year interval from the 2015 discovery to 2019 first oil.

Required evidence includes:

- technical assistance;

- new petroleum units;
- legal reforms;
- contract-administration systems;
- cost-audit arrangements;
- environmental review;
- sovereign-fund design;
- first-cargo measurement;
- central-bank custody;
- EITI implementation;
- local-content policy development.

The core question is where preparation occurred before revenue arrived and where reforms followed later.

4.4 The Natural Resource Fund as a pre-first-oil milestone

Parliament's official record lists the Natural Resources Fund Act 2019 as passed on 3 January 2019 and published on 23 January 2019. The act itself records presidential assent on 23 January 2019. [S37]

This gives the flagship a precise chronology point: the first statutory NRF framework preceded first oil later in 2019.

The analysis must still distinguish:

- legal enactment;
- commencement;
- operational account structure;
- first deposit;
- first investment return;
- first withdrawal;
- later replacement legislation;
- later withdrawal-rule amendments.

This is a central example of the paper's broader method. Early formal preparation is a documented fact. Long-term effectiveness is a separate empirical question.

4.5 Measuring institutional adaptation

The final state-capacity chapter should score no government and rank no political actor. Instead, it should present time-series measures where possible:

- petroleum regulator staffing;
- number of contracts or permits administered;

- audit completion time;
- publication lag;
- environmental inspection frequency;
- procurement processing time;
- investment project completion;
- statistical publication lag;
- unresolved legal or fiscal disputes.

The objective is to test whether institutional capacity kept pace with industry scale.

Part I exhibit programme

Exhibit I-1. Petroleum exploration chronology

Columns:

- date;
- licence/well/project;
- authority;
- operator;
- event type;
- evidence source;
- commercial result;
- legal significance.

Exhibit I-2. Pre-oil institutional map

Nodes:

- Ministry of Natural Resources;
- petroleum administration;
- GGMC/geological data functions;
- EPA;
- GRA;
- Ministry of Finance;
- Bank of Guyana;
- Audit Office;
- Parliament;
- courts;

- maritime and emergency authorities.

Each node should show statutory basis, key petroleum function and pre-2015 capability evidence.

Exhibit I-3. Environmental and natural-capital baseline

Layers:

- forests;
- protected areas;
- coastal population;
- fisheries;
- major rivers;
- oil blocks;
- shore-based petroleum support infrastructure;
- sensitive receptors.

Exhibit I-4. Before/after institutional capability matrix

Compare 2014, 2019 and 2026 for:

- petroleum law;
- revenue fund;
- local content;
- EITI;
- petroleum taxation;
- environmental permitting;
- spill legislation;
- production monitoring;
- public petroleum data;
- contract audit;
- public investment management.

No qualitative score should be assigned unless based on a transparent measurable indicator.

Exhibit I-5. What a newcomer should do before discovery

A staged checklist:

- resolve or manage maritime boundaries;
- maintain a public licence register;
- preserve state ownership of geological data;
- define environmental baseline standards;
- budget scalable technical expertise;
- model several fiscal regimes;
- design contract disclosure rules;
- establish conflict-of-interest controls;
- map emergency response partners;
- avoid debt or recurrent spending against unproven petroleum revenue.

Part I source package currently available

S22: official LCDS 2030 update and historical programme account. S28: pre-oil sovereign wealth fund Green Paper. S35: official petroleum exploration history. S36: official Liza-1 well record. S37: parliamentary NRF Act 2019 record. S38: parliamentary Petroleum Act 1986 record. S39: parliamentary Environmental Protection Act 1996 record. S40: Permanent Court of Arbitration Guyana-Suriname case record.

Part I next-source requirements

1. Original World Bank petroleum exploration promotion project records.
2. Original 1980s petroleum regulations and model agreements.
3. Guyana Geology and Mines Commission historical petroleum datasets.
4. Full PCA Guyana-Suriname award.
5. Norway's official disbursement records under the forest partnership.
6. Historical Bank of Guyana and Ministry of Finance macro series.
7. Historical EPA staffing, budget and authorization records.
8. Independent peer-reviewed Guyana-Suriname Basin geology.
9. Historical public-debt and external-sector records.
10. Coastal flood and sea-defence datasets.
11. Pre-2015 household and labour-force evidence.
12. Historical port, aviation and electricity capacity.

Part I research boundary

No statement in this dossier should be interpreted as a political rating of Guyana's governments before or after the petroleum discovery. The finished case study will document institutional sequence, measurable outputs, evidence gaps and transfer conditions. Disputed interpretations will be attributed to the relevant source.

PART II RESEARCH DOSSIER

Discovery, Contractual Architecture, Cost Recovery and Revenue Assurance

Working research dossier for Chapters 5-9 of the 520-560-page flagship publication. Evidence cut-off: 27 September 2026. Status: research-stage material. External legal and petroleum-fiscal review pending.

Research objective

Part II reconstructs the commercial and legal architecture governing the Stabroek Block without reducing the agreement to a single royalty rate or a single government-take figure.

The final analysis must distinguish:

- information available when the agreement was signed;
- geological and commercial risk at the time;
- signed fiscal terms;
- the mechanics of cost recovery;
- actual project costs submitted by the contractor;
- government audit findings;
- the status of disputed adjustments;
- later model fiscal terms developed under different geological conditions;
- later licences and agreements;
- petroleum revenues actually received.

No overall political score or contract ranking will be assigned.

Chapter 5 research dossier

Liza 2015 and the information shock

The Liza-1 well is recorded by the Petroleum Management Programme as an exploration oil well dated 15 May 2015. [S36]

The final discovery analysis should establish a chronology of information becoming available to both state and contractor:

1. pre-drill geological interpretation;
2. well spud and drilling;
3. hydrocarbon indications;

4. public discovery announcement;
5. appraisal drilling;
6. early resource estimates;
7. commerciality assessment;
8. concept selection;
9. development planning;
10. production-licence approval.

This chronology matters because a negotiating position changes as geological uncertainty declines.

The paper should not infer that information known after appraisal was available to negotiators at an earlier date. The correct counterfactual question is: what would a technically informed government and investor have known at each decision point?

Required evidence:

- original Liza discovery announcements;
- appraisal well records;
- resource-estimate revisions by date;
- oil-price environment during 2015-2016;
- deepwater development cost estimates available at the time;
- comparable frontier-basin fiscal terms;
- financing and execution risk.

Chapter 6 research dossier

The 2016 Stabroek petroleum agreement

The official Stabroek Block register identifies the petroleum agreement date as 27 June 2016 and the effective/licence date as 7 October 2016. [S41]

These dates must remain separate. The audit material for 2018-2020 also treats 7 October 2016 as the effective date for recoverability analysis. [S42]

6.1 Contract crosswalk required

The final flagship needs a clause-level table with at least:

- parties;
- contract area;
- term;
- exploration phases;

- relinquishment;
- work obligations;
- discovery notification;
- commerciality;
- development plans;
- petroleum production licences;
- royalty;
- cost recovery;
- profit oil;
- valuation;
- natural gas treatment;
- taxation;
- accounting procedure;
- audit rights;
- currency;
- imports;
- local purchasing;
- employment and training;
- assignment;
- change in control;
- abandonment;
- insurance;
- environmental duties;
- dispute settlement;
- stabilization;
- confidentiality;
- force majeure;
- effective date.

Every row should include:

- exact article/annex;
- short neutral description;
- fiscal or governance consequence;
- audit evidence;
- later interpretation or dispute;
- transferability note.

6.2 Headline fiscal mechanics

Existing official and audit sources establish the need to model the 75% cost-recovery ceiling as a ceiling, not as an assertion that actual economic costs always equal 75% of revenue.

The flagship should show at least four numerical states:

A. recoverable costs exceed the monthly ceiling; B. recoverable costs are below the ceiling; C. production exceeds sales; D. later project costs replenish the cost bank.

The simplified waterfall is pedagogical only. The final fiscal model must trace actual contract accounting.

6.3 Effective date and recoverability

The government-published 2018-2020 audit material records an exception involving insurance costs incurred before 7 October 2016 and states that, subject to specified pre-contract-cost treatment, costs before the effective date were not recoverable. [S42]

This is important because it demonstrates why the contract chapter must distinguish:

- signing date;
- effective date;
- pre-contract costs expressly recognized;
- costs incurred after effective date;
- costs later challenged by auditors.

The final manuscript should not infer final financial recovery from an audit exception. It must identify whether the contractor agreed, whether a credit was processed, whether government accepted the resolution and whether any dispute remained.

Chapter 7 research dossier

Cost recovery, accounting and the cost bank

Cost recovery is where fiscal theory becomes accounting practice.

The 2018-2020 audit published through the Petroleum Management Programme reviews cost and revenue components and includes detailed exceptions. [S42]

The final chapter should group audit issues by category rather than listing them randomly.

7.1 Proposed audit taxonomy

Allocation across blocks

Costs that benefit multiple licences or operations require a defensible allocation basis.

Timing

Costs incurred before or after the effective date, capitalized costs and costs recognized in later periods require correct treatment.

Duplicate or erroneous charges

Duplicate invoices, coding mistakes and costs assigned to the wrong block should be isolated.

General and administrative costs

Office, overhead and shared-services allocation should follow contractual accounting rules.

Emergency and environmental costs

Studies and response systems may benefit more than one project or licence and should be allocated consistently with the agreement.

Valuation

Cost oil and profit oil depend on volume and price rules. Valuation methods must be audited separately from cost classification.

Affiliate transactions

Related-party charges require particular attention to arm's-length principles and contract rules.

Decommissioning

Future abandonment and restoration obligations require explicit treatment.

7.2 The cost bank

A transparent cost-bank model should include:

Opening unrecovered costs + new recoverable costs + permitted adjustments

- cost oil recovered

= closing unrecovered costs.

The model needs project and category tags.

A single block-wide number can hide the effect of new project approvals. If development spending rises while older costs fall, government profit-oil entitlement can change materially without any amendment to the contract.

7.3 Audit status categories

Every audit item in the paper should carry one of these labels:

- identified;
- contractor response received;
- government response recorded;
- credit accepted;
- credit posted;
- dispute unresolved;
- arbitration;
- final settlement;
- no public evidence of disposition.

This prevents a proposed adjustment from being reported as money already recovered.

Chapter 8 research dossier

Later model terms and changing information

Later fiscal terms were developed after commercial success had reduced frontier uncertainty.

The flagship should compare agreement generations without implying that later terms were available under identical conditions in 2016.

Comparison fields should include:

- royalty;
- cost-recovery ceiling;
- profit sharing;
- corporate tax;
- signing bonus;
- acreage size;
- relinquishment;
- ring-fencing;
- local-content obligations;
- environmental obligations;

- decommissioning;
- audit rights;
- beneficial ownership;
- dispute settlement.

The core research principle is temporal fairness. Contract terms negotiated after multiple commercial discoveries should not be used as if they were contemporaneous market evidence for an earlier frontier negotiation.

At the same time, the historical agreement should be assessed against comparable agreements and available expertise at the time, not insulated from scrutiny by uncertainty.

Both propositions can be true.

Chapter 9 research dossier

Contract governance for a newcomer

A transferable contract-governance system should focus on process quality rather than copying Guyana's percentages.

Before negotiation

- establish a multidisciplinary negotiating team;
- preserve state control over geological data;
- obtain independent reservoir and petroleum-economics advice;
- model investor returns under several price and discovery outcomes;
- map fiscal interactions with tax law;
- identify environmental and decommissioning liabilities;
- publish conflict-of-interest rules;
- define confidentiality boundaries.

During negotiation

Maintain a decision log showing:

- clause proposed;
- government's objective;
- contractor position;
- independent technical assessment;
- fiscal effect;

- legal effect;
- environmental effect;
- final negotiated result;
- approval authority.

Before signature

Complete:

- fiscal model review;
- legal consistency review;
- environmental-law review;
- tax review;
- audit-rights review;
- beneficial-ownership check;
- dispute-resolution analysis;
- decommissioning-liability review;
- parliamentary/public disclosure assessment.

After signature

Create a contract administration calendar:

- work commitments;
- relinquishment dates;
- development submissions;
- production licences;
- annual budgets;
- cost statements;
- audit windows;
- royalty due dates;
- tax filings;
- environmental reporting;
- training obligations;
- decommissioning provisions.

Initial documentary findings from government-published audit material

The 2018-2020 audit provides useful evidence about the operation of the agreement in practice.

It includes exceptions relating to:

- costs coded to the wrong block;
- allocation of office-related construction;
- allocation of emergency-response study costs;
- insurance costs before the effective date;
- duplicate charges.

The paper should use these examples to demonstrate the role of audit rights rather than to generalize from individual exceptions.

The audit also discusses government profit-oil validation and cost-oil valuation. This gives the flagship a basis for explaining how volume, valuation and cost accounting interact.

A key publication control follows:

Audit findings are evidence of an audit process. They are not, by themselves, evidence that every exception was finally resolved or collected.

Contract chapter exhibits

Exhibit II-1. Agreement chronology

- original block agreement;
- 2015 discovery;
- June 2016 agreement execution;
- October 2016 effective date;
- production licence;
- later amendments if any;
- later model agreement;
- later block agreements.

Exhibit II-2. Fiscal waterfall

Show:

- gross petroleum available for allocation;
- royalty;

- cost recovery;
- residual profit oil;
- government share;
- contractor share;
- tax treatment;
- cash timing.

Exhibit II-3. Cost-bank bridge

Annual or quarterly:

- opening unrecovered costs;
- additions;
- disallowed/credited items;
- recoveries;
- closing balance.

Exhibit II-4. Audit exception register

Fields:

- audit period;
- exception;
- amount;
- contract basis;
- contractor response;
- government status;
- public resolution evidence.

Exhibit II-5. Agreement-generation crosswalk

2016 Stabroek versus later model terms, with a separate column describing the information environment at each date.

Required quantitative tests

1. Oil price at USD 40, 60, 80 and 100 per barrel.
2. Cost-recovery use at 25%, 50%, 65% and 75%, where contractually relevant.
3. New-project cost-bank additions.

4. Production downtime.
5. Sales timing versus production timing.
6. Government lift timing versus cash receipt.
7. Valuation difference under third-party and affiliate sale cases where permitted by evidence.
8. Disallowed-cost sensitivity.
9. Decommissioning liability scenarios.
10. Discounted cash flow under several project-cost assumptions.

All model outputs must be labelled illustrative unless calibrated to published actual data.

Source package currently available

S02: official Stabroek agreement publication page. S14: later model fiscal terms. S15: Guyana Revenue Authority royalty explanation. S29: IMF 2023 Article IV analysis of fiscal architecture. S41: Stabroek Block official register. S42: 2018-2020 cost-recovery audit publication. S43: regulatory document index including cost audits and licences.

Remaining Part II evidence gates

1. Machine-extract and independently review the complete executed 2016 agreement.
2. Build the full article-by-article crosswalk.
3. Obtain and review the 1999-2017 final audit.
4. Determine final public status of 2018-2020 exceptions.
5. Determine status of 2021-2022 and later audits.
6. Reconcile cost-oil volumes to monthly production reports.
7. Reconcile profit-oil entitlement to government lift records.
8. Verify later signed PSAs, not only model documents.
9. Obtain independent petroleum-fiscal expert review.
10. Obtain independent petroleum-law review.

Publication boundary

This dossier does not determine whether the 2016 agreement was "good" or "bad" as an overall political judgment.

The flagship will provide the contractual facts, fiscal mechanics, information available at relevant dates, actual audit evidence, later reforms, international comparison mechanisms and downside cases. Readers can assess policy choices from the documented record.

PART III RESEARCH DOSSIER

The Pre-First-Oil Window, First Revenue and Production Scale-Up

Working research dossier for Chapters 10-14 of the 520-560-page flagship publication. Evidence cut-off: 27 September 2026. Status: research stage. Monthly production and cargo-level reconciliation remain to be completed.

Research objective

Part III reconstructs the interval from commercial discovery through large-scale production.

The analytical sequence is:

commercial discovery → appraisal → development approval → environmental authorization → production licence → project construction → first oil → production ramp → cargo lifting → government entitlement → petroleum cash receipt.

Each stage has a different date and a different source.

The final publication will avoid a common error: using a single "first oil" date as if it represented the full development and revenue cycle.

Chapter 10 research dossier

Discovery to development decision

Liza-1 is recorded in the official well register with a well date of 15 May 2015. [S36]

The first development followed a sequence of:

- appraisal;
- field-development planning;
- EIA;
- environmental authorization;
- production licence;
- engineering/procurement;
- FPSO conversion/construction;
- subsea installation;
- drilling;
- commissioning.

The official Liza Phase 1 permit record states that the environmental application was submitted on 5 July 2016 and the environmental permit was issued on 1 June 2017. [S48]

The official petroleum-management site publishes the Liza Phase 1 production licence and Phase 2 addendum. [S86]

The final chronology must identify the exact production-licence issue date from the executed licence, rather than relying on later summaries.

Development-decision evidence

For each project the final table should include:

- commercial discovery/appraisal basis;
- field development plan;
- environmental application;
- EIA approval;
- petroleum production licence;
- final investment decision;
- sanctioned capital;
- FPSO name;
- design capacity;
- estimated wells;
- first oil;
- actual ramp-up;
- subsequent optimization.

Where company and government dates differ because one refers to FID and another to a production licence, both should be preserved.

Chapter 11 research dossier

The pre-first-oil governance window

The four-year interval between the 2015 commercial discovery and December 2019 first production gave government a finite preparation period.

A central documented milestone is the first Natural Resource Fund law, passed in January 2019 before recurring petroleum production. [S37]

The pre-oil Green Paper also documents government work on sovereign-wealth-fund design beginning before first oil, including external technical assistance and inter-agency work. [S28]

This chapter should build an institution-by-institution chronology:

Petroleum administration

- licensing;
- development-plan review;
- production measurement;
- operator reporting;
- contract administration.

Fiscal

- royalty assessment;
- cost audit;
- profit-oil entitlement;
- petroleum taxation;
- sales/marketing.

Sovereign revenue

- NRF legislation;
- central bank operational agreement;
- bank accounts;
- investment rules;
- reporting.

Environmental

- project EIAs;
- permits;
- spill plans;
- regulator staffing;
- monitoring.

Transparency

- contract publication;
- EITI;
- public petroleum data;
- parliamentary reporting.

The chapter should show which systems were operational before December 2019 and which were strengthened later.

Chapter 12 research dossier

First oil and the first revenue cycle

Production from Liza Phase 1 began in December 2019. Official and operator records consistently place the first project startup in that month. [S81, S83]

First production is not first government cash.

A cargo-based accounting chain should include:

1. production measured on the FPSO;
2. crude stored;
3. cargo nominated;
4. entitlement assigned;
5. tanker lifted;
6. cargo priced;
7. invoice settled;
8. royalty/profit-oil receipt recognized;
9. money deposited in the NRF.

The 2020 audited NRF statements show the practical timing difference. The fund recognized four 2020 profit-oil lifts while the fourth was still a receivable at year end and settled in 2021. [S45]

This makes the first-year case useful for newcomer countries.

A petroleum ministry, revenue authority, central bank and finance ministry must agree on:

- the production period;
- lift date;
- price;
- entitlement;
- invoice date;
- settlement date;
- accounting year.

Without this bridge, public reports can contain different numbers that are each correct under different accounting bases.

Chapter 13 research dossier

FPSO-by-FPSO production expansion

The production sequence verified so far is:

Liza Destiny

First production: December 2019. [S81, S83]

Liza Unity

First production: 11 February 2022. [S82]

Prosperity

First oil at Payara: November 2023. The government release dated 14 November 2023 states first oil had occurred and identifies Destiny and Unity as the earlier projects. [S83]

ONE GUYANA

Yellowtail first production: 8 August 2025. ExxonMobil stated that the fourth development increased combined installed capacity above 900,000 barrels per day. [S84]

In November 2025 the operator reported a point-in-time daily production milestone of 900,000 barrels. [S85]

The regulator's oil-production dataset covers actual production from 20 December 2019 through 31 July 2026. [S81]

The final production chapter should use the regulator's monthly data to calculate:

- monthly barrels;
- daily average;
- rolling 12-month production;
- FPSO share;
- production ramp;
- downtime;
- year-over-year growth.

Installed capacity and reported actual production must remain separate.

Project table to build

The final publication will update status only when a dated primary source establishes a startup.

Chapter 14 research dossier

Production data quality

Production data should be stored at the finest reliable time resolution available.

The core dataset needs:

- date;
- field/project;
- FPSO;
- oil produced;
- water produced;
- gas produced;
- gas reinjected;
- gas flared;
- downtime;
- oil lifted.

Five measures that must not be confused

Nameplate/design capacity

Engineering production capability under specified conditions.

Daily production

Observed output on a particular day.

Monthly daily average

Monthly barrels divided by days.

Annual average

Annual barrels divided by 365/366.

Cargo lift

A shipment event that draws from stored produced crude and need not match the same month's production.

A headline such as "900,000 barrels per day" must therefore identify whether it is:

- installed capacity;
- target;
- one-day milestone;
- period average.

Production and revenue bridge

The flagship should build a reconciled table with three linked but distinct layers.

Layer A. Physical production

- monthly barrels by FPSO;
- total field production;
- downtime.

Layer B. Commercial lifting

- cargo count;
- barrels lifted;
- entitlement owner;
- lift date;
- price.

Layer C. Public cash

- government profit-oil invoice;
- royalty;
- settlement date;
- NRF deposit.

The bridge will explain timing differences instead of forcing the three layers into the same calendar month.

Production-scale regulatory pressure

Production growth also changes administrative workload.

Each additional project creates:

- production reporting;
- cost statements;
- environmental monitoring;
- permit review;
- spill-response exposure;
- decommissioning obligations;
- local-content plans;

- cargo sales;
- tax filings.

The flagship should compare regulatory staffing and audit resources against:

- number of FPSOs;
- number of active wells;
- production volume;
- annual contractor spend.

This is a more meaningful capacity test than the number of laws enacted.

New-producer blueprint extracted from Part III

Before first oil, a newcomer should test:

1. production-meter verification;
2. crude-quality measurement;
3. cargo nomination;
4. sales/pricing protocol;
5. royalty invoice;
6. cost-accounting submission;
7. government entitlement calculation;
8. bank settlement;
9. sovereign-fund deposit;
10. public notification;
11. environmental incident reporting;
12. regulator emergency command.

The country should run a simulated first cargo before the actual first cargo.

Required Part III exhibits

1. Liza discovery-to-first-oil critical path.
2. Institution-readiness timeline 2015-2019.
3. FPSO commissioning timeline.
4. Monthly realized production by project.
5. Installed capacity versus actual output.
6. Cargo lift versus production bridge.
7. Profit-oil lift versus NRF deposit.
8. Revenue-recognition timing example from 2020.

9. Regulatory-workload growth.
10. First-oil readiness checklist.

Remaining Part III evidence gates

1. Extract exact Liza Phase 1 production-licence date and conditions.
2. Extract Liza Phase 2 addendum.
3. Extract Payara production licence.
4. Extract Yellowtail production licence.
5. Complete monthly production series through July 2026.
6. Build cargo-lift series.
7. Link government lifts to cash deposits.
8. Record project FIDs and sanctioned costs by date.
9. Compile downtime and production optimization.
10. Add independent petroleum-engineering review.

Publication boundary

Part III will not describe installed capacity as actual annual output.

It will not treat operator forecasts as achieved production.

It will not describe first oil as the date government first received cash unless the cash record proves that timing.

PART IV RESEARCH DOSSIER

The Natural Resource Fund, Fiscal Absorption, Public Investment and National Wealth

Working research dossier for Chapters 15-20 of the 520-560-page flagship publication. Evidence cut-off: 27 September 2026. Status: research stage. Annual audited account extraction and public-investment outcome review remain in progress.

Research objective

Part IV asks what happens after petroleum revenue becomes public financial wealth.

The central accounting sequence is:

petroleum entitlement → sale and settlement → NRF deposit → investment return → statutory withdrawal calculation → parliamentary appropriation → Consolidated Fund transfer → procurement → completed public asset or service → maintenance → measurable outcome.

Each stage has a different evidence source.

A country can have a well-reconciled sovereign fund and weak project delivery. It can also have urgent development needs and choose a larger legal withdrawal envelope while maintaining transparent accounting. The final paper will document choices and effects without assigning an overall political score.

Chapter 15 research dossier

NRF law through time

The first Natural Resource Fund law was enacted in January 2019 before first oil later that year. [S37]

The framework was subsequently replaced in 2021 and amended again in 2024. [S04, S05]

The flagship should treat these as separate legal generations.

Legal crosswalk fields

For each legal version:

- date passed;
- date assented;
- commencement;
- fund objective;

- petroleum revenue definition;
- custodian;
- board/governance bodies;
- investment mandate;
- reporting;
- audit;
- withdrawal formula;
- parliamentary authorization;
- emergency provisions;
- prohibited uses;
- public notification.

A later law should not be retroactively applied to earlier withdrawals.

Operational agreement

The Petroleum Management Programme publishes the December 2019 operational agreement between the Minister of Finance and the Bank of Guyana for management of the NRF. [S04/S43 context]

The final legal analysis should connect statutory responsibility to:

- bank custody;
- investment execution;
- records;
- reporting;
- expenses;
- internal controls.

Chapter 16 research dossier

Reconstructing fund cash flows

The NRF_RECONCILIATION_WORKBOOK.md is the accounting companion to this chapter.

The core distinction is between:

- cash receipts;
- revenue recognized;
- receivables;
- investment income;

- market value;
- withdrawals;
- closing net assets.

The 2020 audited statement demonstrates the distinction clearly. Four profit-oil lifts were recognized during 2020, while the fourth was still a receivable at year end. [S45]

This is why a government petroleum-revenue chart must state whether it uses:

- production;
- cargo lifts;
- invoiced entitlement;
- recognized accounting revenue;
- cash received.

Annual reconciliation standard

For each year: Opening audited balance + petroleum revenue recognized + investment income + other accounting movements

- withdrawals
- expenses

= closing audited balance.

A separate cash bridge will reconcile: Opening cash + cash deposits + cash investment receipts

- cash transfers
- cash expenses

= closing cash.

Restated comparatives from later audited statements take precedence over earlier presentation where the auditor explicitly restates the figures.

Chapter 17 research dossier

Saving, spending and intergenerational equity

Petroleum is a depleting asset.

When crude is extracted, the country exchanges an underground asset for:

- contractor/private income;
- public petroleum revenue;
- taxes;
- domestic activity.

The public-policy question is how public petroleum revenue is converted into durable wealth.

Potential destinations include:

- external financial assets;
- public infrastructure;
- human capital;
- debt reduction;
- social transfers;
- climate resilience;
- productive institutions.

Each has a different return profile and risk.

Intergenerational framework

The final paper should model:

- reserve life;
- production path;
- oil-price uncertainty;
- population growth;
- existing infrastructure gap;
- debt;
- public-investment return;
- financial-asset return;
- maintenance liabilities;
- disaster exposure.

No universal savings percentage follows from the existence of oil.

International comparison

Norway links normal-year fiscal use to long-run expected real returns from a mature external fund. [S74]

Ghana separates stabilization and heritage objectives. [S75]

Timor-Leste uses a petroleum fund with explicit liquidity management and substantial budget relevance. [S77, S78]

Guyana's architecture has its own withdrawal rules and development context.

The comparison should focus on fiscal mechanisms and conditions, not a ranked verdict.

Chapter 18 research dossier

The non-oil primary balance and absorption

An economy can face spending constraints before it faces a financing constraint.

The IMF July 2026 mission statement discusses absorption capacity, investment monitoring and a possible longer-run fiscal guide linked to the non-oil primary balance. [S64]

This is IMF staff analysis.

The flagship should explain the non-oil primary balance in plain terms:

Non-oil revenue

- non-interest public expenditure

= non-oil primary balance.

A larger deficit indicates more government spending financed by petroleum or borrowing rather than recurring non-oil revenue.

Absorption constraints

Rapid spending may encounter:

- limited contractors;
- limited engineers;
- imported materials;
- port congestion;
- land constraints;
- public-sector vacancies;
- procurement delays;
- inflation.

The correct response is an empirical policy question.

The research should measure:

- tender participation;
- contract variation;
- completion delays;
- construction inflation;
- wage pressure;
- import growth;
- project commissioning.

Chapter 19 research dossier

Public investment management

A withdrawal from the NRF does not identify which project ultimately receives the money.

Once transferred to the Consolidated Fund, public spending follows budget and procurement systems.

The flagship should therefore create a separate public-investment evidence register.

Project lifecycle

Need → appraisal → budget → procurement → contract → construction → completion → commissioning → operation → maintenance → service outcome.

A project should not be marked complete at contract award.

Outcome categories

For a road:

- travel time;
- vehicle operating cost;
- safety;
- flood resilience;
- maintenance condition.

For a hospital:

- staffed beds;
- waiting time;
- service availability;
- clinical outcomes.

For electricity:

- dependable capacity;
- generation;
- outages;
- tariff;
- losses.

For water:

- access;

- pressure;
- quality;
- continuity.

For schools:

- attendance;
- completion;
- learning;
- teacher availability.

Lifecycle liabilities

Every capital asset creates future costs:

- maintenance;
- staffing;
- utilities;
- insurance;
- replacement.

The national-balance-sheet chapter should record those liabilities.

Chapter 20 research dossier

National balance-sheet accounting

GDP is an annual flow.

A national balance sheet asks what assets and liabilities remain.

Assets

Financial

- NRF investments;
- reserves;
- other public financial assets.

Produced capital

- roads;
- bridges;

- ports;
- power systems;
- hospitals;
- schools;
- water systems.

Human capital

- education;
- health;
- technical capability.

Natural capital

- petroleum remaining;
- forests;
- biodiversity;
- fisheries;
- water;
- agricultural land;
- coastal protection.

Liabilities

- public debt;
- maintenance obligations;
- pensions;
- contractual guarantees;
- decommissioning exposure where borne publicly;
- environmental remediation exposure;
- disaster liabilities.

Petroleum depletion

Petroleum production reduces the remaining physical resource.

A comprehensive wealth measure should avoid describing gross petroleum revenue as a pure addition to wealth without recognizing depletion.

The country becomes wealthier over time when the converted assets and capabilities deliver durable value exceeding associated liabilities and resource depletion, subject to distribution and environmental outcomes.

The final research should avoid a single aggregate monetary valuation where evidence is too uncertain. Physical indicators and separate asset classes are acceptable.

NRF dashboard design

The public dashboard should contain at least:

Petroleum receipts

- royalty;
- profit oil;
- taxes where applicable;
- cash deposit date.

Fund performance

- opening balance;
- deposits;
- realized return;
- unrealized return;
- expenses;
- withdrawals;
- closing value.

Legal withdrawal

- formula;
- calculated ceiling;
- amount appropriated;
- amount transferred.

Public expenditure

This should link to a separate budget/project dataset rather than imply that fund money remains traceable to one project after consolidation unless the budget system supports that mapping.

Adverse fiscal scenarios

Price shock

A sustained price decline reduces petroleum receipts.

Test:

- legal withdrawal space;
- essential expenditure;
- project postponement;
- debt service;
- fund liquidity.

Production interruption

A technical outage reduces cargo availability.

Test:

- cash timing;
- budget assumptions;
- fund buffer;
- emergency expenditure.

Market loss

A sovereign fund invested in financial markets can experience negative returns.

Test:

- liquidity;
- withdrawal rule;
- forced-sale risk;
- rebalancing.

Construction inflation

High public investment raises project costs.

Test:

- nominal budget;
- physical output;

- contractor capacity;
- maintenance affordability.

Natural disaster

Test simultaneous:

- petroleum revenue disruption;
- emergency spending;
- infrastructure repair;
- fund withdrawal.

New-producer blueprint extracted from Part IV

Before first revenue, a newcomer should:

1. enact a lawful petroleum-revenue custody structure;
2. define petroleum revenue precisely;
3. define cash and accrual reporting;
4. publish deposit notices;
5. establish independent annual audit;
6. define withdrawal authority;
7. integrate withdrawals with the ordinary budget;
8. establish public-investment appraisal;
9. stress-test low-price and disaster cases;
10. publish a national-balance-sheet framework.

The numerical withdrawal rule remains a national political and fiscal choice.

Required Part IV exhibits

1. NRF legal-version timeline.
2. Annual audited stock-flow table.
3. Cash versus accrual reconciliation.
4. Withdrawal-rule crosswalk.
5. Petroleum deposits versus withdrawals.
6. Market-value versus cash balance.
7. Non-oil primary balance.
8. Absorption-capacity indicators.
9. Public investment lifecycle.

10. National balance sheet.
11. Oil-price/fund stress tests.
12. Disaster plus revenue-shock scenario.

Remaining Part IV evidence gates

1. Extract audited NRF accounts 2022-2025.
2. Verify legal commencement/version dates.
3. Reconcile annual receivables.
4. Reconcile all withdrawals to budget notifications.
5. Build non-oil primary-balance series.
6. Build project-level public investment register.
7. Verify lifecycle operating costs.
8. Build national wealth baseline.
9. Complete fund-stress model review.
10. Obtain independent accounting/public-finance review.

Publication boundary

Part IV will not equate a growing fund balance with complete development success.

It will not equate a large withdrawal with poor policy or a small withdrawal with good policy.

It will document the fiscal rule, conditions, trade-offs, accounting evidence and observed investment outcomes.

PART V RESEARCH DOSSIER

Environmental Governance, Offshore Risk, Financial Responsibility and Climate Coexistence

Working research dossier for Chapters 21-27 of the 520-560-page flagship publication. Evidence cut-off: 27 September 2026. Status: research stage. External environmental-law, marine-science and spill-response review pending.

Research objective

Part V tests the environmental governance system as a sequence rather than a label.

The sequence is:

baseline science → environmental application → environmental impact assessment → public participation → permit → project design controls → operating monitoring → incident reporting → regulator inspection → enforcement → remediation → financial responsibility → long-term ecological outcome.

The existence of one link does not establish performance of the full chain.

The final chapter will therefore distinguish legal requirements, permit conditions, operator implementation, regulator action, observed environmental outcomes and unresolved evidence.

Chapter 21 research dossier

The environmental authorization system

Guyana's environmental framework predates offshore oil production. The Environmental Protection Act 1996 established the general statutory basis for environmental management. [S39]

The Liza Phase 1 record gives a dated example of the environmental authorization chain. The official permit page records an application submitted on 5 July 2016, an approved EIA and management plan dated 1 June 2017, and an environmental permit issued on the same date. The management plan referenced a revised Oil Spill Response Plan and Wildlife Response Plan. The permit was modified in May 2021. [S48]

The final chapter should map statutory roles across:

- Environmental Protection Agency;
- Ministry responsible for petroleum;
- maritime authorities;
- Civil Defence Commission;

- coast guard and emergency services;
- public-health agencies;
- fisheries authorities;
- local government;
- neighbouring-state notification channels.

The chapter should also separate:

- environmental authorization;
- petroleum production licence;
- occupational safety;
- maritime navigation;
- oil-spill liability;
- emergency command.

These legal functions may interact but are not interchangeable.

Public participation

The Yellowtail public notice records an EPA determination that an EIA was required and invited written public submissions during the statutory review period. [S52]

A complete participation assessment should measure more than whether a notice appeared.

Required fields:

- notice date;
- consultation locations;
- language and accessibility;
- stakeholder groups;
- questions raised;
- written response;
- changes made to project design or permit;
- unresolved objections;
- appeal route.

The flagship should identify whether consultation occurred before material project decisions became irreversible.

Chapter 22 research dossier

Project-by-project environmental evidence

Environmental controls should be assessed by development, not generalized across the Stabroek Block.

The evidence register currently confirms:

- Liza Phase 1 original environmental permit and later modification;
- an official index containing Liza Phase 2 amended permit/EIA material;
- an official index containing Payara environmental permit/EIA material;
- Yellowtail EIA application/public notice;
- Uaru environmental permit;
- Whiptail environmental permit publication. [S48-S52]

The final project matrix should include:

- application date;
- EIA requirement decision;
- terms of reference;
- consultant;
- public consultation;
- final EIA;
- permit issue date;
- permit expiry;
- amendment date;
- production licence date;
- flaring condition;
- produced-water condition;
- waste condition;
- oil-spill plan;
- wildlife response;
- monitoring plan;
- independent audit requirement;
- financial assurance;
- inspection record;
- incident record.

Liza Phase 1 as an operating example

The 2021 EPA notice says the agency modified Liza Phase 1's permit as the project approached the EIA's estimated flaring level. The modified permit added emissions reporting requirements, flaring time constraints and a US\$30/tCO₂e payment beyond specified timelines. [S55]

This provides a testable regulatory chain:

- EIA estimate;
- observed operating issue;
- regulator action;
- modified permit;
- financial consequence;
- later performance.

The final paper should reconstruct actual monthly flaring and regulator payments before drawing conclusions about effectiveness.

Uaru as a later permit-generation example

The Uaru permit is effective from 27 April 2023 through 26 April 2028 and contains explicit EPA powers relating to sampling, analysis, annual review and permit modification. [S50]

The Uaru production licence separately requires environmental-law compliance and response capability. [S56]

A later-generation permit should not automatically be assumed stronger in practice. The flagship should compare measurable conditions and monitoring requirements across permit generations.

Chapter 23 research dossier

Oil-spill prevention, preparedness and response

Offshore spill governance has at least five layers:

1. prevention through engineering and operating controls;
2. operator emergency response;
3. national incident command;
4. regional/transboundary cooperation;
5. financial resources for cleanup and compensation.

A robust case study should test all five.

Response resources

The Uaru production licence requires expert personnel and equipment to prevent or respond to petroleum releases and includes capping-stack and dispersant provisions. [S56]

The final review must verify implementation:

- location of response equipment;
- ownership;
- service contracts;
- maintenance records;
- mobilization time;
- stock quantities;
- expiry and replenishment;
- trained personnel;
- drill history;
- regulator observation.

A written obligation is not the same as demonstrated readiness.

Worst-case discharge modelling

EIA spill models should be recorded with:

- release location;
- release duration;
- release rate;
- oil properties;
- season;
- wind;
- current data;
- weathering assumptions;
- response assumptions;
- shoreline probabilities;
- transboundary paths.

The final report should not repeat a model's output as a guaranteed prediction. It should compare assumptions against measured regional oceanographic data and explain uncertainty.

National command

The 2025 oil-pollution law adds a dedicated statutory layer to Guyana's oil-spill framework. Parliament records Act 6 of 2025 as passed and published in May 2025. [S53, S54]

The research register does not yet treat the full regime as operational. Before the final chapter, the project must verify commencement, regulations, appointment of the competent authority, forms, penalties, response-plan requirements and financial-responsibility rules.

Chapter 24 research dossier

Liability, insurance and financial assurance

Four concepts are commonly collapsed into one number:

- legal liability;
- insurance coverage;
- parent or affiliate financial assurance;
- cash available immediately after an incident.

They are not the same.

The final chapter should ask:

- Who is legally liable?
- For which categories of damage?
- Is liability capped?
- Which losses are insured?
- What exclusions apply?
- What is the aggregate limit?
- Is a parent guarantee callable directly?
- Who determines the amount due?
- Which jurisdiction governs disputes?
- How quickly is cash available?
- What public funds might be required before recovery?

Court decisions

A 2026 appellate dispute concerning financial assurance is material to the chapter, but the current source package includes only government characterization and secondary reporting. The final legal interpretation must use an authenticated judgment.

Until then, the manuscript should not state an independent conclusion about the legal adequacy of the financial-assurance regime.

Financial-stress framework

The flagship should present illustrative layers:

- operator working capital;
- insurance;
- parent support;
- contractual response services;
- public emergency expenditure;
- compensation;
- restoration;
- long-term monitoring.

Scenario values should be illustrative unless tied to public instruments.

Chapter 25 research dossier

Coastal, fisheries and biodiversity exposure

The ecological consequences of an offshore incident depend on where oil travels and what resources lie in its path.

The final environmental atlas should identify:

- artisanal and commercial fishing grounds;
- nursery and spawning areas;
- mangroves;
- coastal wetlands;
- marine protected areas;
- seabird areas;
- marine mammal habitat;
- turtle migration;
- ports;
- water intakes;
- tourism assets;
- neighbouring coastlines.

Fisheries

A strong fisheries analysis should separate:

- reported catch;
- landed value;
- employment;
- subsistence use;
- export markets;
- seasonality;
- vessel size;
- spatial distribution.

Compensation design requires evidence linking an incident to measurable loss. Baseline fishing effort and catch data therefore have value before an incident occurs.

Biodiversity

Guyana's biodiversity analysis should link offshore activity to national conservation policy without assuming either compatibility or incompatibility.

The paper should assess:

- project footprint;
- vessel traffic;
- underwater noise;
- light;
- discharge;
- spills;
- cumulative project effects;
- climate-related ecological pressures.

Chapter 26 research dossier

LCDS, forest finance and petroleum

Guyana's original Low Carbon Development Strategy began before commercial offshore discovery. This prevents the research from describing the strategy as a response created to offset petroleum production. [S22]

The later LCDS 2030 framework and jurisdictional forest-credit activity should be treated as a separate environmental-finance system.

A complete accounting bridge should show:

- forest-credit vintage;
- issuance;
- sale agreement;
- minimum contracted value;
- cash received;
- registry transfer;
- community allocation;
- national allocation;
- project spending;
- independently measured outcomes.

The working research already distinguishes a contracted purchase commitment from realized cash.

Petroleum and forest accounting boundaries

The final paper should separately disclose:

- upstream petroleum operational emissions;
- electricity and shorebase emissions;
- flaring;
- methane where measured;
- land-use emissions;
- forest-carbon credits;
- downstream combustion of exported petroleum.

No simple subtraction should be used to claim national carbon neutrality unless the underlying accounting standard supports the boundary and all material emissions are included.

Chapter 27 research dossier

Energy transition and emissions accounting

The environmental case includes more than oil-spill risk.

Offshore production affects:

- direct fuel combustion;
- flaring;
- methane;

- power use;
- shipping;
- aviation;
- material and construction emissions.

National energy policy separately affects:

- electricity generation;
- transmission;
- household energy cost;
- reliability;
- renewable integration;
- industrial demand.

The final paper should state which emissions are inside Guyana's national inventory and which occur after petroleum export.

Measurement hierarchy

For emissions:

1. direct measured fuel and flare data;
2. project-specific engineering calculations;
3. regulator-reviewed reports;
4. standard emission factors;
5. broad estimates.

Every chart should identify its measurement method.

Cumulative environmental governance

Six or more offshore developments cannot be assessed only as isolated projects.

The cumulative chapter should examine:

- vessel traffic;
- noise;
- air emissions;
- shorebase expansion;
- waste;
- cumulative spill probability;
- emergency-response concurrency;
- regulator workload;

- overlapping fishery effects;
- cumulative greenhouse-gas emissions.

A permit-by-permit system needs a parallel basin-wide view.

New-producer blueprint extracted from Part V

A newcomer country should establish the following before first production:

1. legally defined environmental authority;
2. marine baseline data owned or accessible by the state;
3. transparent EIA terms of reference;
4. documented public consultation;
5. project-specific permit conditions;
6. independent monitoring access;
7. incident-reporting deadlines;
8. spill command structure;
9. capping and containment capability;
10. regional response agreements;
11. enforceable financial assurance;
12. claims and compensation process;
13. restoration standards;
14. published annual compliance reporting;
15. rule for permit amendment when operating evidence changes.

For an island state, additional emphasis belongs on:

- reef systems;
- tourism dependence;
- hurricane disruption;
- single-port risk;
- limited storage;
- cross-border currents;
- imported response equipment.

Required Part V exhibits

1. Environmental-law chronology.
2. Project-by-project permit matrix.
3. EIA/public-participation timeline.

4. Flaring-control chronology.
5. Produced-water standards and volumes.
6. Spill command structure.
7. Response-equipment locations and mobilization time.
8. Financial-assurance layers.
9. Marine sensitivity atlas.
10. Oil-spill scenario pathways.
11. Environmental incident register.
12. Forest-credit and petroleum-accounting boundary.
13. Cumulative offshore-activity dashboard.

Remaining Part V evidence gates

- extract full Liza Phase 2 permit;
- extract full Payara permit;
- obtain and extract Yellowtail final permit;
- extract full Whiptail permit;
- collect environmental audit reports;
- collect inspections and regulator compliance notices;
- build observed flaring time series;
- build produced-water time series;
- verify Oil Pollution Act commencement;
- obtain authenticated 2026 appellate judgment;
- inspect public insurance/guarantee instruments;
- obtain marine and fisheries independent research;
- compare spill modelling with regional oceanography;
- obtain evidence of drills and response readiness;
- seek external environmental-law and marine-science review.

Publication boundary

Part V will not call Guyana's environmental system a success or failure as an overall political conclusion.

It will document the legal architecture, permit evolution, observed operating data, regulator actions, unresolved liabilities, environmental outcomes, stakeholder concerns and the conditions another country would need to adapt the mechanisms responsibly.

PART VI RESEARCH DOSSIER

Local Content, Domestic Value, Workforce, Household Welfare and Indigenous/Hinterland Outcomes

Working research dossier for Chapters 28-33 of the 520-560-page flagship publication. Evidence cut-off: 27 September 2026. Status: research stage. Independent labour, household-economics and Indigenous-rights review pending.

Research objective

Petroleum development affects national welfare through several channels:

- government revenue;
- direct employment;
- supplier contracts;
- wages;
- training;
- public investment;
- price changes;
- land and housing markets;
- infrastructure;
- community programmes;
- environmental effects.

These channels should not be collapsed into one local-content expenditure number.

The final flagship will distinguish:

- procurement booked to a Guyanese supplier;
- local beneficial ownership;
- imported content;
- domestic labour;
- domestic wages;
- domestic profit retained;
- domestic manufacturing or technical value added;
- training;
- permanent capability after a contract ends.

Chapter 28 research dossier

Local content law and implementation

Parliament records the Local Content Act 2021 as Act 18 of 2021, passed on 29 December 2021 and published on 4 January 2022. [S57]

The official Local Content Register provides a system for supplier certification and employment registration. [S58]

The final legal chapter should extract:

- statutory definition of Guyanese national;
- definition of Guyanese company;
- ownership and management requirements;
- scheduled services and targets;
- local-content plans;
- reporting frequency;
- supplier registration;
- employment requirements;
- capacity-development requirements;
- audit and inspection powers;
- offences and penalties;
- ministerial/regulatory powers.

Every statutory target should be recorded with the exact schedule and date in force.

Reporting evolution

Government mid-year reporting documents changes in implementation and reporting scope. The 2025 report states that half-year reporting guidelines were expanded to capture procurement activity in ten additional sectors and subsectors already identified in the legislation, with local-capacity assessment informing future targets. [S60]

The final paper should explain when reporting scope changes, because a rising reported procurement figure may partly reflect wider reporting coverage rather than only underlying economic growth.

Chapter 29 research dossier

Domestic value added versus local procurement

A supplier invoice does not reveal how much value stays in the country.

Consider four stylized cases:

Case A: imported item, local reseller

A Guyanese supplier imports a complete product and resells it. Domestic value may consist mainly of local wages, warehousing, financing and margin.

Case B: imported inputs, local fabrication

Raw material is imported but cutting, welding, engineering, testing and maintenance occur locally. Domestic value is higher.

Case C: local service

A locally owned engineering, logistics or professional-services firm supplies labour and expertise primarily from Guyana.

Case D: foreign-controlled local operation

A locally incorporated company employs Guyanese workers but profit, intellectual property and management services are largely external.

All four may contribute to the economy, but they should not be reported as economically identical.

Domestic value-added formula

The final research model should estimate:

Gross local-content procurement

- imported goods embedded in those purchases
- imported subcontracted services
- foreign management/IP charges attributable to delivery

= estimated domestic gross value added.

Then split domestic value into:

- wages;
- local operating surplus;
- taxes;

- domestic intermediate inputs.

This requires customs, tax, ownership and firm-level evidence.

Ownership verification

The local-content register should be sampled against:

- beneficial ownership;
- shareholder records;
- management control;
- bank/payment destination;
- payroll;
- customs imports;
- related-party contracts.

Certification status should remain separate from independent economic measurement.

Chapter 30 research dossier

Workforce and skills

Government petroleum reporting provides employment counts, but a durable skills analysis needs occupational detail.

The final workforce dataset should include:

- occupation;
- nationality;
- gender;
- age;
- region;
- permanent/temporary;
- offshore/onshore;
- employer;
- wage band;
- certification;
- training;
- promotion;
- retention;
- safety outcomes.

Skills pipeline

Map: secondary school → technical/vocational training → apprenticeship → entry role → technical certification → supervisor → professional/management role.

The oil sector may increase returns to technical training while also drawing engineers and tradespeople away from construction, mining, utilities and public agencies.

A national assessment should therefore measure both petroleum-sector employment and vacancies/wage pressure elsewhere.

Training outcomes

Training expenditure is an input.

Outcome indicators include:

- certificates completed;
- internationally recognized qualifications;
- post-training employment;
- wage progression;
- retention;
- local succession into positions previously expatriate;
- safety performance.

Chapter 31 research dossier

Household welfare and distribution

The Bureau of Statistics states that the Household Budget and Living Condition Survey fieldwork ran from November 2018 through November 2019 across all ten administrative regions. [S61]

This creates a major evidence boundary.

A pre-first-oil or first-oil-era household survey should not be reported as a 2026 poverty or living-standard measurement.

The final social chapter should build a timeline of available household evidence and label the latest observation year for every indicator.

Welfare dashboard

Required measures:

- real household consumption;

- income;
- food expenditure share;
- housing cost;
- electricity access and reliability;
- water and sanitation;
- internet;
- transport;
- school attendance/completion;
- health access;
- employment;
- regional inequality;
- gender;
- hinterland/coastal differences.

Macroeconomic GDP growth should appear separately.

Price effects

Rapid investment and public expenditure may affect:

- rent;
- land;
- construction materials;
- food;
- transport;
- skilled labour;
- services.

A household can face rising nominal income and rising costs at the same time.

The final analysis should therefore use real purchasing power rather than nominal income alone.

Chapter 32 research dossier

Amerindian and hinterland development

Indigenous and hinterland outcomes arise through more than petroleum.

Relevant systems include:

- land titling;

- public infrastructure;
- health and education;
- telecommunications;
- renewable/off-grid energy;
- carbon-credit benefit sharing;
- public employment;
- resource governance.

The LCDS programme states that 15% of carbon-credit revenues are made available for village-led plans and describes a village consultation/planning process. [S62, S63]

The final paper should distinguish:

- government programme design;
- village decision;
- funds allocated;
- funds received;
- project procured;
- project completed;
- project operating;
- measured community outcome.

Consent and participation

The research should seek evidence from:

- National Toshaos' Council;
- individual village councils;
- Indigenous organizations;
- legal cases;
- independent rights specialists;
- UN/other safeguard records.

Government programme descriptions should not substitute for affected-community evidence.

The chapter should also record disagreement where consultation, land rights or benefit-sharing processes are contested.

Chapter 33 research dossier

Social licence, grievance and participation

A large national development programme needs mechanisms for affected people to challenge decisions.

The final institutional map should include:

- EPA public-comment process;
- petroleum/ministry grievance mechanisms;
- judicial review;
- EITI multi-stakeholder engagement;
- village governance;
- procurement complaints;
- labour complaints;
- local-content supplier complaints.

For each mechanism:

- who has standing;
- submission process;
- deadline;
- decision maker;
- publication rule;
- appeal;
- observed case duration.

Evidence versus sentiment

Public opinion surveys, if methodologically sound, may measure attitudes.

They should not substitute for:

- statutory rights;
- environmental results;
- actual household outcomes;
- audited financial flows.

The flagship should report them as a separate evidence category.

Longitudinal local-content evidence

Government reports should be transformed into a consistent time series with fields for:

- reporting companies;
- sectors covered;
- gross procurement;
- training expenditure;
- local hires;
- total Guyanese employment;
- methodology changes.

For example, the 2024 Mid-Year Report described reporting from 40 companies, H1 procurement of US\$314 million, more than 454 new hires during the review period and 4,467 Guyanese employed by those reporting companies, while the government estimated total Guyanese oil-industry employment at about 6,500. [S59]

These figures are government-reported and should be retained with their stated population and reporting scope.

They should not be merged with later periods unless definitions are stable.

Local-value audit sample

The flagship should propose a stratified sample of suppliers across:

- engineering;
- fabrication;
- logistics;
- marine;
- aviation;
- catering;
- accommodation;
- transportation;
- legal/accounting;
- environmental services;
- manpower;
- equipment rental.

For each sampled contract:

1. contract value;
2. imported goods;

3. imported services;
4. Guyanese payroll;
5. domestic purchases;
6. taxes;
7. retained local profit;
8. training;
9. capital equipment left in country;
10. capability gained.

This produces a more meaningful measure of domestic value than invoice totals alone.

Distributional analysis

Petroleum benefits and costs should be mapped by:

- region;
- urban/rural;
- coastal/hinterland;
- income group;
- gender;
- age;
- occupation.

The final paper should avoid an unsupported statement that national gains were evenly distributed.

Likewise, absence of current household data should not be used to infer poor outcomes.

Where evidence is unavailable, state the gap.

New-producer blueprint extracted from Part VI

A prospective producer should establish before production:

1. legal definitions of local company and local worker;
2. transparent beneficial-ownership tests;
3. realistic sector-specific targets based on capacity;
4. supplier-development programmes;
5. tender visibility;
6. payment-timing rules;
7. workforce data standards;
8. training outcome measures;

9. independent domestic-value estimation;
10. updated household baseline;
11. regional distribution indicators;
12. grievance mechanisms.

For a small island state, targets need to reflect:

- labour-force size;
- tourism-sector competition for workers;
- limited industrial land;
- port capacity;
- import dependence;
- education pipeline.

Required Part VI exhibits

1. Local Content Act legal architecture.
2. Scheduled target matrix by service category.
3. Procurement and employment time series.
4. Domestic-value-added waterfall.
5. Supplier ownership map.
6. Occupational skills pipeline.
7. Wage and vacancy comparison across sectors.
8. Household-data timeline.
9. Regional welfare dashboard.
10. Indigenous/hinterland funding flow.
11. Village-plan outcome sample.
12. Grievance and remedy map.

Remaining Part VI evidence gates

- extract full Local Content Act schedules;
- compile reporting-company counts by year;
- reconcile official procurement data from 2022-2026;
- obtain supplier-register aggregate data;
- test beneficial ownership;
- obtain customs and labour evidence;
- build skills/training time series;

- identify latest representative household data;
- avoid updating old poverty data through inference;
- collect Indigenous/community sources;
- sample village-plan outcomes;
- collect grievance/appeal cases;
- external labour-economics review;
- external Indigenous-rights review.

Publication boundary

The paper will not equate gross local procurement with domestic value added.

It will not equate petroleum GDP with household welfare.

It will not equate a programme allocation with a verified community outcome.

Each will be measured on its own evidence.

PART VII RESEARCH DOSSIER

Economic Transformation, Public Investment, Energy, Diversification and Competitiveness

Working research dossier for Chapters 34-38 of the 520-560-page flagship publication. Evidence cut-off: 27 September 2026. Status: research stage. Macroeconomic and public-investment datasets require further reconciliation.

Research objective

Part VII asks whether petroleum income is being converted into durable productive capacity.

The analytical frame is broader than GDP.

The chapter separates:

- oil output;
- non-oil output;
- public investment;
- private investment;
- household welfare;
- infrastructure service quality;
- financial savings;
- human capital;
- natural capital;
- competitiveness.

No single indicator answers the development question.

Chapter 34 research dossier

Growth beyond headline GDP

IMF staff reported in July 2026 that real GDP grew by more than 19% in 2025 after average growth near 40% during 2023-2024. Staff also described broad non-oil expansion and similar high petroleum production during the first half of 2026. [S64]

These IMF observations provide a macroeconomic snapshot, but the final chapter should use official national accounts and revised historical series for quantitative decomposition.

GDP decomposition

Required time series:

- total real GDP;
- oil GDP;
- non-oil GDP;
- construction;
- agriculture;
- mining;
- manufacturing;
- transport;
- services;
- government;
- per-capita measures.

The dataset should preserve base-year changes and statistical revisions.

Why oil GDP requires separate treatment

Petroleum production can increase national output rapidly while:

- employing a small share of the labour force directly;
- importing substantial specialized equipment and services;
- generating profits accruing to foreign investors;
- generating government revenue on a different timetable.

This does not make oil GDP irrelevant. It means GDP must be linked to national income, fiscal receipts, domestic value added and household outcomes.

Non-oil growth

The final chapter should test:

- whether non-oil growth is concentrated in construction;
- whether tradable sectors expand;
- whether private productivity rises;
- whether public investment reduces logistics and power constraints;
- whether sector growth persists when oil-linked construction moderates.

Labour market

The IMF July 2026 statement reports unemployment at 6.2% at end-2025 and notes labour shortages as a policy issue. [S64]

The flagship should pair unemployment with:

- participation;
- vacancies;
- wage growth;
- migration;
- occupation;
- public/private sector;
- underemployment.

Chapter 35 research dossier

Infrastructure transformation

Petroleum revenue increases fiscal capacity, but infrastructure conversion has several stages:

appropriation → procurement → contract → construction → commissioning → operation → maintenance → service outcome.

The final project register should never mark an allocation as a completed asset.

Infrastructure classes

Transport

- national roads;
- bridges;
- community roads;
- hinterland roads;
- ports;
- airports.

Water and drainage

- potable water;
- drainage;
- canals;

- pumps;
- sea defence.

Social

- hospitals;
- schools;
- housing;
- digital infrastructure.

Productive

- power generation;
- transmission;
- industrial sites;
- logistics facilities.

Project register fields

For every major project:

- approved objective;
- implementing agency;
- original budget;
- revised budget;
- financing source;
- procurement method;
- contractor;
- award date;
- original completion date;
- actual completion;
- variations;
- maintenance plan;
- users;
- service performance.

This allows the paper to distinguish financial execution from delivery.

Lifecycle cost

Large petroleum-financed capital programmes create future recurrent costs.

The national balance sheet should estimate:

- maintenance;
- staffing;
- energy;
- insurance;
- replacement;
- debt service where applicable.

A project that requires high future subsidies has a different wealth profile from a financial asset earning external returns.

Chapter 36 research dossier

Energy security and gas-to-energy choices

The 2026 Budget Speech describes a 300 MW first phase of the Gas-to-Energy project and a broader programme of generation and grid investment. [S65]

The Guyana Energy Agency describes natural gas as part of a transition in the integrated grid alongside solar, wind, hydro and other resources. [S67]

These sources state government/agency plans. The flagship should separately measure actual commissioning, delivered megawatt-hours, reliability, cost and emissions.

Power-system metrics

Required:

- installed MW;
- dependable MW;
- peak demand;
- annual generation;
- system losses;
- outage frequency;
- outage duration;
- fuel consumption;
- fuel cost;
- average tariff;

- subsidy;
- renewable share;
- emissions intensity.

Gas-to-energy evaluation

The final analysis should identify:

- offshore pipeline capital cost;
- onshore NGL/power capital cost;
- financing;
- gas supply arrangements;
- operating cost;
- plant efficiency;
- grid upgrades;
- outage assumptions;
- tariff policy;
- LPG output;
- emissions relative to displaced HFO/diesel;
- methane leakage;
- project delay risk.

A government statement that a project will reduce prices is a policy claim until observed tariffs and full system costs are available.

Demand-side management

The GEA July 2026 update states the new action plan targets 516 GWh of energy savings by 2030 against projected generation of 2,508 GWh. [S66]

The flagship should track:

- implemented measures;
- verified savings;
- rebound effects;
- programme cost;
- household/business participation.

Chapter 37 research dossier

Private-sector development and diversification

Diversification should mean more than an increase in construction linked to petroleum.

The final chapter should test:

- export diversification;
- productivity;
- firm formation;
- domestic manufacturing;
- agriculture and food exports;
- services exports;
- mining value added;
- tourism;
- technology;
- regional trade.

Diversification metrics

Possible metrics:

- non-oil export share;
- Herfindahl export concentration;
- non-oil productivity;
- manufacturing value added;
- food import dependence;
- firm survival;
- formal employment;
- private investment outside petroleum support;
- research and technical capacity.

The paper should avoid a single composite political score.

Linkages

Petroleum can support non-oil sectors through:

- cheaper energy;
- logistics investment;

- skilled labour;
- demand;
- finance;
- public investment.

It can also create pressures through:

- higher wages;
- higher rents;
- construction demand;
- exchange-rate pressure;
- imported services;
- concentration of finance.

The net effect requires data by sector.

Chapter 38 research dossier

Inflation, competitiveness and Dutch-disease channels

IMF staff in July 2026 stated that available indicators did not show clear signs of overheating or resource-driven competitiveness pressure at that point, while noting that strong wage growth and wage-based real exchange rate indicators required monitoring. The same statement identified stronger foreign-exchange demand associated with import-intensive private investment. [S64]

These are dated IMF staff assessments, not permanent findings.

The flagship should construct its own indicator dashboard.

Indicator set

- CPI overall;
- food inflation;
- housing/rent;
- land prices;
- construction materials;
- nominal wages;
- real wages;
- unit labour cost;
- real effective exchange rate;

- imports;
- non-oil exports;
- vacancy rates;
- construction prices;
- public procurement prices.

Dutch-disease channels

The chapter should explain three mechanisms separately:

1. Spending effect: resource income increases domestic demand for non-tradables.
2. Resource movement: labour/capital shift toward petroleum-linked activity.
3. Exchange-rate/price effect: domestic costs rise relative to trading partners.

Guyana's stabilized exchange-rate framework changes how adjustment appears in observed indicators.

Policy response framework

The paper should describe, without prescribing a political choice:

- saving more petroleum revenue;
- pacing public investment;
- raising productive capacity;
- importing scarce inputs;
- labour training;
- migration policy;
- competition policy;
- productivity investment;
- exchange-rate/monetary options.

Each has trade-offs.

National wealth conversion framework

Petroleum extraction reduces an underground asset.

Long-run wealth conversion can be examined through changes in:

- sovereign financial assets;
- public infrastructure;
- human capital;

- private productive capital;
- environmental/natural capital;
- public debt;
- contingent liabilities.

A national wealth table should avoid double counting.

For example:

- petroleum revenue deposited into the NRF increases financial assets;
- a withdrawal reduces financial assets;
- a completed road increases produced capital;
- maintenance liabilities affect future public spending;
- environmental damage reduces natural capital.

The quality of wealth conversion depends on asset performance, not spending volume alone.

IMF 2026 evidence relevant to Part VII

The July 2026 mission statement provides several dated observations and recommendations:

- rapid oil and non-oil growth;
- strong construction;
- large public investment;
- increased FX demand with heavy import content;
- high petroleum inflows and NRF accumulation;
- low assessed risk of debt distress;
- need to align spending with absorptive capacity under sustained high oil prices;
- recommendation to consider a non-oil primary-balance guide;
- need for stronger spending-outcome monitoring;
- need to strengthen statistics. [S64]

The flagship should identify each as an IMF staff statement, not an independent fact where the claim is interpretive.

Required Part VII exhibits

1. Oil versus non-oil GDP.
2. Sector contribution to non-oil growth.
3. Public investment by sector.
4. Project completion versus appropriation.
5. Capital stock and maintenance liability.

6. Electricity capacity, generation and reliability.
7. Fuel mix and emissions intensity.
8. Gas-to-energy cost/reliability bridge.
9. Export concentration.
10. Wage and price dashboard.
11. Real exchange-rate indicators.
12. National wealth conversion table.

New-producer blueprint extracted from Part VII

Before spending large resource revenue, a new producer should build:

- public-investment appraisal;
- procurement transparency;
- cost and schedule control;
- lifecycle maintenance budgeting;
- project commissioning verification;
- national accounts capable of separating resource and non-resource sectors;
- household price monitoring;
- labour-market data;
- infrastructure service indicators;
- national balance-sheet reporting.

A small island state should add:

- hurricane resilience;
- port redundancy;
- imported construction inflation;
- tourism competitiveness;
- desalination/power exposure;
- limited contractor capacity.

Remaining Part VII evidence gates

1. Extract official 2026 national accounts.
2. Reconcile Ministry/IMF growth estimates.
3. Build 2010-2026 oil/non-oil series.
4. Build public-investment project register.
5. Verify completion and operating status for major assets.

6. Build electricity performance time series.
7. Complete gas-to-energy project cost and schedule record.
8. Calculate export concentration.
9. Build wage/price/real exchange-rate dashboard.
10. Build national wealth accounts.
11. Obtain independent macroeconomic review.
12. Obtain infrastructure/project-management review.

Publication boundary

Part VII will not infer household prosperity from GDP growth.

It will not infer infrastructure performance from budget size.

It will not infer economic diversification from non-oil construction alone.

Each proposition requires its own evidence.

PART VIII RESEARCH DOSSIER

Transparency, Cost Audits, Procurement, Parliamentary/Judicial Oversight and Public Data

Working research dossier for Chapters 39-43 of the 520-560-page flagship publication. Evidence cut-off: 27 September 2026. Status: research stage. Legal and public-finance review pending.

Research objective

A resource-governance system is not demonstrated by the number of reports published.

The flagship will test five linked questions:

1. Are material contracts, licences, production data and public receipts disclosed?
2. Are figures timely enough to support scrutiny?
3. Is independent challenge institutionally possible?
4. Are audit findings resolved and traceable?
5. Are public-finance and social outcomes visible after money leaves the petroleum account?

Transparency, oversight and enforcement will therefore appear as separate categories.

Chapter 39 research dossier

EITI and extractive disclosure

Guyana joined EITI in 2017. The EITI International Secretariat records an earlier 2022 Validation under the 2019 Standard and a new full Validation beginning 15 May 2026. [S70]

The EITI Board's 12 May 2026 decision records concern about continuing lack of full stakeholder confidence regarding the independence of the selection of civil-society representatives to the multi-stakeholder group. The Board chose to proceed with a full Validation and an external expert. [S69]

This is a formally attributed EITI concern.

The final flagship should not convert it into a broad judgment about political legitimacy.

Validation timeline to build

- EITI accession;
- first reports;

- 2022 Validation;
- corrective actions;
- report publication lags;
- 2025 multi-stakeholder representation dispute;
- deferred Validation;
- May 2026 Board decision;
- 2026 mission;
- external expert draft;
- stakeholder comments;
- final Board decision when available.

The current paper must state clearly if no final 2026 decision exists at the publication cut-off.

Disclosure quality

EITI performance should be assessed by category:

- licences;
- contracts;
- production;
- beneficial ownership;
- company payments;
- government receipts;
- state participation;
- subnational transfers;
- social/environmental payments;
- local content;
- commodity sales.

For each:

- data owner;
- observation year;
- release date;
- reconciliation method;
- source file;
- machine readability;
- unresolved discrepancy.

Timeliness

In a rapidly growing petroleum industry, a report covering fiscal year N but published in N+2 has different decision value from quarterly public data.

The paper should show disclosure lag explicitly.

Chapter 40 research dossier

Cost audits and revenue assurance

Cost-recovery audit is one of the clearest examples of why publication is not the same as final resolution.

The government-published IHS Markit final audit for 1999-2017 says the contractor submitted cumulative costs of about USD1.678 billion for review and that the audit team identified grounds to dispute about USD214.4 million plus overhead adjustments. [S71]

A later Ministry update says the government dispute was moved toward the agreement's sole-expert procedure and describes multiple audit workstreams. [S72]

These are distinct stages.

The final audit tracker should distinguish:

- amount submitted;
- amount audited;
- audit exception;
- government claim/dispute;
- contractor response;
- negotiation;
- sole expert;
- arbitration;
- final accepted adjustment;
- cost-bank credit;
- cash/revenue effect.

No audit exception will be reported as recovered revenue before the final accounting event is evidenced.

Revenue assurance institutions

The chapter should map:

- Ministry of Natural Resources;

- Guyana Revenue Authority;
- Ministry of Finance;
- Bank of Guyana;
- external petroleum auditors;
- Attorney General/legal teams;
- independent experts.

For each:

- mandate;
- staffing;
- data access;
- audit window;
- dispute authority;
- publication obligation.

Audit-cycle metric

Measure: audit period end → audit engagement → fieldwork → draft report → contractor response → final report → dispute → resolution.

A system with strong audit rights but prolonged unresolved findings has a different performance profile from a system with rapid final adjustments.

Chapter 41 research dossier

Procurement and public-spending oversight

Petroleum revenue becomes ordinary public-finance risk after transfer into the budget.

The public investment chapter must therefore connect:

- budget appropriation;
- procurement;
- contract award;
- variation;
- physical completion;
- payment;
- audit;
- service outcome.

The IMF 2026 mission statement encouraged continued strengthening of procurement practices and consistent compliance as public spending expands. [S64]

That is an IMF staff recommendation, not an author judgment.

Supreme audit institution

The Audit Office of Guyana identifies itself as the state's supreme audit institution and reports that the 2024 public accounts and performance report were presented to the National Assembly in September 2025. [S73]

The final review should extract findings relevant to major petroleum-financed expenditure without generalizing an individual audit issue across the public sector.

Project-audit register

Fields:

- ministry/agency;
- programme;
- funding source;
- contract;
- auditor observation;
- amount;
- management response;
- Public Accounts Committee response;
- corrective action;
- current status.

The final paper should differentiate:

- procedural noncompliance;
- unsupported expenditure;
- overpayment;
- delay;
- value-for-money concern;
- fraud allegation;
- proven fraud.

Chapter 42 research dossier

Parliament, judiciary and institutional checks

Formal oversight is distributed across:

- parliamentary appropriation;
- statutory reporting;
- parliamentary committees;
- independent audit;
- courts;
- administrative regulators;
- constitutional offices.

The flagship should map powers, not assign an overall political score.

Parliamentary petroleum-revenue trail

The final evidence chain should include:

- NRF receipt notice;
- annual report;
- maximum withdrawal calculation;
- budget estimate;
- Appropriation Act;
- transfer to Consolidated Fund;
- expenditure;
- Auditor General review.

Judicial review

Environmental permits and financial-assurance disputes have entered the courts.

The final legal chapter should use authenticated judgments.

For every case:

- parties;
- court;
- filing date;
- legal question;
- first-instance ruling;

- appeal;
- final order at cut-off;
- later proceedings;
- practical regulatory effect.

Government, company and civil-society descriptions of a judgment should be treated as positions unless they reproduce the authenticated text accurately.

Chapter 43 research dossier

Statistics and public data

A fast-changing petroleum economy places unusual pressure on statistical systems.

The IMF 2025 staff report described modernization work and a planned new household survey; the July 2026 mission statement again highlighted national accounts, price, external-sector, labour and household data as priority areas. [S64, S68]

The paper should distinguish:

- absence of a statistic;
- delay in publication;
- outdated base year;
- sampling limitation;
- incompatible definitions;
- actual adverse outcome.

Data-quality inventory

Required datasets:

Petroleum

- well;
- field;
- FPSO;
- production;
- flaring;
- water;
- lifting;
- price;
- government entitlement.

Fiscal

- royalties;
- profit oil;
- taxes;
- NRF;
- public debt;
- procurement;
- capital spending.

Economic

- national accounts;
- CPI;
- trade;
- balance of payments;
- wages;
- employment;
- firms.

Social

- household income;
- consumption;
- poverty;
- education;
- health;
- housing;
- regional indicators.

Environmental

- emissions;
- water quality;
- fisheries;
- biodiversity;
- incidents;
- inspections.

Reproducibility standard

Every quantitative exhibit should record:

- source;
- file version;
- download date;
- unit;
- transformation;
- missing values;
- code/formula;
- revision date.

Accountability architecture

The flagship should present accountability as six layers.

Layer 1. Rule

Law, contract, permit or regulation defines an obligation.

Layer 2. Measurement

A competent body measures performance.

Layer 3. Disclosure

The evidence is published in usable form.

Layer 4. Independent challenge

Auditor, court, parliament, civil society, affected community or expert can contest the record.

Layer 5. Correction

A substantiated problem leads to adjustment, sanction, remediation or rule change.

Layer 6. Outcome

The correction affects revenue, environmental performance, service delivery or public confidence.

An institution that performs strongly at one layer might perform differently at another.

Accountability case studies required

1. 1999-2017 cost recovery audit.
2. 2018-2020 cost recovery audit.
3. NRF annual audit and withdrawal reporting.
4. Liza Phase 1 permit modification after flaring.
5. Environmental financial assurance litigation.
6. EITI 2026 Validation.
7. Local-content supplier certification.
8. Major public-infrastructure procurement and completion.

Each case should follow the six-layer model.

New-producer blueprint extracted from Part VIII

A new producer should establish:

- contract publication policy;
- cost-audit calendar;
- beneficial-ownership register;
- production data standard;
- petroleum receipt publication;
- independent sovereign-fund audit;
- supreme-audit access;
- parliamentary reporting;
- regulator enforcement publication;
- civil-society/community participation;
- court access;
- corrections register.

For a small state with limited specialists, independent external expertise should transfer knowledge to permanent national institutions rather than remain a perpetual substitute for them.

Required Part VIII exhibits

1. Petroleum accountability system map.
2. Contract-to-cash audit trail.
3. Cost-audit timeline.

4. EITI chronology.
5. Disclosure lag dashboard.
6. Beneficial-ownership evidence flow.
7. Parliamentary petroleum-revenue flow.
8. Procurement oversight flow.
9. Environmental judicial-review chronology.
10. Statistical-data quality matrix.
11. Corrections register.

Remaining Part VIII evidence gates

- obtain final 2026 EITI decision if published by cut-off;
- extract all Guyana EITI report dates and covered fiscal years;
- build cost-audit resolution table;
- verify 2021-2025 audit workstreams;
- extract public-account findings tied to major flagship examples;
- obtain authenticated court judgments;
- map parliamentary receipt notifications;
- compile beneficial-ownership access;
- audit data publication lag;
- verify public procurement portal capability;
- obtain external public-finance and legal review.

Publication boundary

Part VIII will not treat disclosure as proof of performance.

It will not treat an audit finding as proof of recovered revenue.

It will not treat a contested EITI process as an overall judgment on a government or opposition.

It will document institutions, dates, evidence, disagreements and observable corrective actions.

PART IX RESEARCH DOSSIER

Comparative Petroleum Governance and Transferability

Working research dossier for Chapters 44-49 of the 520-560-page flagship publication. Evidence cut-off: 27 September 2026. Status: mechanism comparison in progress. No country ranking is intended.

Research objective

The purpose of comparison is not to identify a single "best" petroleum state.

The purpose is to isolate institutional mechanisms, identify the conditions under which they operate, and test whether any mechanism is relevant to an emerging producer.

The comparison uses six questions:

1. What resource and fiscal problem was the mechanism designed to solve?
2. What legal instrument governs it?
3. Which institution implements it?
4. What data are published?
5. What trade-off does it create?
6. Which national conditions limit transfer?

Chapter 44 research dossier

Norway

Norway provides a long-horizon comparison for converting petroleum cash into financial assets and integrating fund transfers into the national budget.

The Ministry of Finance's 2026 Government Pension Fund white paper describes the Government Pension Fund Global within a fiscal framework linked to an expected long-run real return of 3%. It says normal-year withdrawals should be below that level to preserve room for adverse periods. [S74]

The comparison should not reduce Norway to a 3% rule.

Relevant institutional components include:

- state petroleum taxation and ownership;
- transfer of petroleum cash to the fund;
- external portfolio investment;

- budget integration;
- parliamentary framework;
- central-bank asset management;
- transparent benchmark;
- long horizon;
- broad political continuity around the framework.

Transferability limits

Norway entered its modern fund era with:

- mature public institutions;
- high human capital;
- a developed tax system;
- a diversified economy;
- different infrastructure deficits;
- a larger historical production base.

A developing producer with major infrastructure and social gaps faces a different current-versus-future consumption trade-off.

The transferable mechanism is disciplined integration of petroleum revenue, investment and the budget.

The exact numerical spending rule is not automatically transferable.

Chapter 45 research dossier

Ghana

Ghana provides a closer institutional comparison as a developing petroleum producer with a legislated revenue-management system.

The Bank of Ghana describes two petroleum funds with distinct functions:

- Ghana Stabilisation Fund;
- Ghana Heritage Fund. [S75]

The Bank publishes financial statements and semi-annual reports. [S75, S76]

This creates a useful comparison with Guyana's single NRF architecture.

Mechanism comparison

Questions:

- Are stabilization and intergenerational objectives separated?
- How are petroleum receipts allocated before budget spending?
- What information goes to parliament?
- How often are reports published?
- How are the funds audited?
- How are transfers triggered?

The flagship should compare statutory design with observed implementation.

Transferability

A new state might choose:

- one integrated fund;
- separate stabilization and savings funds;
- a fund plus explicit fiscal rule.

The relevant design choice depends on:

- revenue volatility;
- reserve life;
- debt;
- budget size;
- financial market capacity;
- development gaps.

Chapter 46 research dossier

Timor-Leste

Timor-Leste is relevant because it is a small developing state with a petroleum fund that became central to national fiscal financing.

The Central Bank publishes annual and frequent performance reports. Its current management framework separates a Liquidity Portfolio, intended to meet expected withdrawals, from a Growth Portfolio designed for longer-term investment. [S77, S78]

This is analytically important for Guyana and island states.

A resource fund with large expected withdrawals needs a liquidity strategy separate from its long-run return objective.

Core comparison questions

- petroleum receipts versus investment return;
- annual budget withdrawals;
- fund balance;
- depletion of producing fields;
- estimated sustainable income framework;
- asset allocation;
- liquidity needs;
- domestic fiscal dependence.

The case is useful precisely because a large fund balance does not remove dependence risk.

The flagship should use current fund reporting and fiscal data before drawing conclusions.

Chapter 47 research dossier

Suriname

Suriname provides a nearby pre-first-oil comparison.

TotalEnergies stated on 19 September 2026 that the GranMorgu project has been progressing since final investment decision in October 2024 and that first oil remains targeted for 2028. [S79]

This positions Suriname in a different stage of the petroleum lifecycle from Guyana.

Guyana's most relevant lessons for Suriname concern the discovery-to-first-oil period:

- regulator capacity;
- revenue-fund readiness;
- environmental baseline;
- local-content expectations;
- contract administration;
- spill preparedness;
- public-data systems;
- public-investment planning.

Suriname's own institutions, state oil company, fiscal structure and political choices require separate analysis.

No Guyanese policy should be presumed suitable solely because the states share a basin.

Chapter 48 research dossier

Additional producer mechanisms

The flagship should add targeted mechanism cases rather than superficial country profiles.

Potential mechanisms include:

Trinidad and Tobago

The Heritage and Stabilisation Fund provides a mature Caribbean comparison for a hydrocarbon economy. The Ministry of Finance maintains annual and quarterly reporting. [S80]

Relevant issues:

- stabilization;
- heritage objective;
- mature production;
- economic diversification;
- small-state exposure.

Senegal

Potential relevance:

- new offshore production;
- domestic gas;
- local content;
- first-revenue institutional timing.

Namibia

Potential relevance:

- pre-production discovery;
- regulatory preparation;
- state participation;
- environmental baseline.

Mozambique

Potential relevance:

- large-resource expectations;
- project delay;

- security and institutional risk;
- avoiding expenditure against unrealized revenue.

Botswana

Not a petroleum case, but relevant to mineral wealth and public investment.

The final comparator set should be selected by specific analytical mechanism.

Chapter 49 research dossier

Island and small-state adaptation

A small island petroleum producer faces distinct constraints.

Marine concentration

A spill might affect:

- fisheries;
- reefs;
- beaches;
- tourism;
- desalination;
- ports;
- coastal infrastructure.

The same shoreline can support multiple core sectors simultaneously.

Infrastructure concentration

One major port, airport or power station may represent a large share of national capacity.

An emergency affecting one asset can therefore have systemic consequences.

Labour scale

A petroleum development can sharply alter wages in:

- construction;
- marine services;
- engineering;
- public administration;

- tourism.

Targets for local content should reflect the actual workforce and education pipeline.

Fiscal concentration

A small tax base can become heavily dependent on petroleum rapidly.

The blueprint should therefore stress:

- conservative revenue assumptions;
- liquidity reserves;
- multi-year budget stress tests;
- recurrent-cost discipline;
- catastrophe reserves.

Hurricane and climate exposure

An island response plan should test a spill coinciding with:

- tropical cyclone;
- port closure;
- airport disruption;
- telecommunications failure;
- evacuation.

Blue economy and natural capital

A forest-rich continental model cannot be copied literally.

An island should map:

- reefs;
- seagrass;
- mangroves;
- fisheries;
- tourism;
- blue carbon where scientifically credible;
- coastal protection.

These resources require their own evidence and rights systems.

Comparative mechanism matrix

This table compares institutional questions, not country performance rankings.

Transferability decision tree

Before adopting a mechanism from another country ask:

Resource condition

- commercial discovery?
- reserve range?
- production horizon?
- price exposure?
- development cost?

Fiscal condition

- debt?
- revenue base?
- recurrent expenditure?
- infrastructure deficit?
- disaster exposure?

Institutional condition

- central-bank capacity?
- audit capacity?
- environmental regulator?
- procurement?
- statistics?

Social condition

- population?
- regional inequality?
- Indigenous/customary rights?
- labour force?

- household baseline?

Environmental condition

- deepwater/nearshore?
- currents?
- fisheries?
- reefs?
- forests?
- hurricanes?
- transboundary risk?

Legal condition

- constitutional revenue powers?
- contract system?
- judicial review?
- maritime boundaries?
- treaty obligations?

Only after this screen should a foreign mechanism be adapted.

Comparative evidence standards

For each country:

- current legal text;
- current fund report;
- current fiscal data;
- historical context;
- independent assessment;
- observed problem;
- reform history.

Do not compare:

- one country's audited actual to another country's forecast;
- a mature fund balance to a new fund's annual inflow;
- gross petroleum revenue to government petroleum revenue;
- a policy target to observed performance.

Required Part IX exhibits

1. Petroleum lifecycle stage by comparator.
2. Fund objective and legal-design matrix.
3. Deposit/withdrawal rule matrix.
4. Reporting and audit-frequency matrix.
5. Public-investment interface.
6. Environmental governance mechanism comparison.
7. Local-content mechanism comparison.
8. Island-state risk matrix.
9. Transferability decision tree.
10. "Do not copy without adaptation" table.

New-producer conclusion from comparison

The principal transferable lesson is process discipline:

- date decisions;
- define fund objectives;
- separate cash and forecasts;
- audit petroleum costs;
- publish receipts;
- budget transparently;
- monitor environmental obligations;
- update household evidence;
- preserve institutional independence;
- revise rules transparently.

The numeric formula and institutional form remain national choices.

Remaining Part IX evidence gates

- extract Norway 2026 fund details;
- extract Ghana 2025/2026 fund reports;
- extract Timor-Leste 2025 fund report and 2026 fiscal data;
- add Suriname legal/fiscal architecture;
- extract Trinidad and Tobago HSF law and current reporting;
- select two additional new-producer cases;

- standardize all observations to common dates;
- independent comparative public-finance review.

Publication boundary

Part IX will not rank governments or declare a winning model.

It will show how different mechanisms work, what evidence exists, and what conditions control transfer.

PART X

New Producer Operating Manual

A staged implementation companion to the Guyana petroleum case study. Designed for emerging offshore producers, including island states. Evidence cut-off: 27 September 2026.

This manual translates research questions into administrative actions.

It does not prescribe a political choice, fiscal regime, spending ratio, state ownership model or energy policy.

The receiving country must adapt every action to its constitution, geology, marine ecology, debt, labour market, Indigenous/customary rights, climate exposure and international obligations.

Stage 0

Before a commercial discovery

Objective

Build enough public capacity to govern exploration without creating permanent spending commitments against hypothetical oil revenue.

0.1 Legal map

Publish a one-page legal map identifying:

- ownership of petroleum;
- licensing authority;
- environmental authority;
- maritime authority;
- tax authority;
- customs;
- central bank;
- national audit institution;
- parliament;
- courts;
- disaster-response command.

For every institution record:

- statute;

- date in force;
- responsible office;
- power;
- reporting duty;
- appeal route.

0.2 Acreage and geological-data register

Maintain:

- block coordinates;
- licences;
- expiry;
- work commitments;
- relinquishment;
- operator;
- beneficial ownership;
- well records;
- seismic ownership;
- confidentiality expiry.

The state should preserve institutional access to geological data after individual consultants or ministers leave office.

0.3 Environmental baseline

Before intensive drilling, collect:

- currents;
- bathymetry;
- water quality;
- fisheries;
- sensitive habitats;
- protected areas;
- shoreline receptors;
- tourism assets;
- marine traffic.

For an island:

- reefs;

- beaches;
- desalination intakes;
- cruise terminals;
- hurricane exposure.

Record:

- collector;
- method;
- date;
- raw-data ownership;
- public access;
- uncertainty.

0.4 Fiscal options

Model at least:

- royalty/tax licence;
- production sharing;
- state equity where legally feasible;
- hybrid options.

For each:

- dry-hole case;
- marginal discovery;
- large discovery;
- low price;
- high cost;
- development delay;
- abandonment.

Do not choose a fiscal system from headline government-take percentages alone.

0.5 Expert capacity

Create a roster for:

- petroleum law;
- reservoir engineering;
- petroleum economics;
- tax;

- audit;
- marine science;
- oil-spill response;
- insurance;
- public finance.

Every external contract should include:

- conflict check;
- scope;
- deliverable;
- price;
- knowledge transfer;
- data ownership;
- handover.

0.6 Maritime boundaries

Identify:

- agreed boundaries;
- unresolved claims;
- arbitration/treaty options;
- disputed blocks;
- notification obligations.

Do not market disputed acreage as if title risk does not exist.

Stage 0 gate

The country passes Stage 0 when it has:

- legal map;
- licence register;
- data policy;
- environmental baseline plan;
- fiscal-option model;
- independent adviser panel;
- conflict rules;
- boundary assessment;
- exploration emergency plan.

Commercial discovery is not required.

Stage 1

After a commercial discovery

Objective

Reduce information asymmetry before committing the country to a development framework.

1.1 Independent resource review

Obtain:

- discovery data;
- appraisal results;
- fluid characteristics;
- reservoir uncertainty;
- development concepts;
- recoverable-resource range.

Separate:

- petroleum in place;
- contingent resources;
- reserves;
- commercial development case.

1.2 Negotiation decision file

For each material contract clause record:

- state objective;
- operator proposal;
- independent adviser view;
- fiscal effect;
- legal effect;
- environmental effect;
- final text;
- approving authority.

Preserve draft/version history under lawful confidentiality.

1.3 Fiscal model

Model:

- production;
- price;
- capital cost;
- operating cost;
- royalty;
- cost recovery;
- profit share;
- tax;
- state equity;
- bonuses;
- decommissioning.

Stress:

- low price;
- delay;
- cost overrun;
- lower recovery;
- outage;
- future project cost additions.

1.4 Environmental decision file

Require:

- terms of reference;
- alternatives;
- baseline;
- cumulative effects;
- spill scenarios;
- climate effects;
- public consultation;
- mitigation;
- monitoring;
- financial assurance.

A development decision should show how material public comments were treated.

1.5 Decommissioning

Before production:

- estimate abandonment cost;
- define security mechanism;
- set contribution timing;
- establish residual liability;
- define asset removal and site restoration;
- audit the fund/security.

Stage 1 gate

Before final development approval:

- independent resource review complete;
- contract legally reviewed;
- fiscal model independently challenged;
- EIA complete;
- spill response tested;
- decommissioning plan funded;
- financial assurance documented;
- disclosure decision recorded.

Stage 2

First-oil readiness

Objective

Prove the revenue, environmental and emergency systems before commercial cash arrives.

2.1 Meter and production chain

Test: well → FPSO processing → storage → cargo measurement → lifting.

Independently verify calibration.

2.2 Sales and entitlement chain

Test: cargo → price → royalty → cost recovery → profit share → invoice → settlement.

Use a fictional cargo rehearsal.

2.3 Sovereign revenue chain

Test: receipt → sovereign account → public notice → accounting → audit.

If a sovereign fund exists, verify:

- legal account;
- custodian;
- signatories;
- investment rules;
- publication;
- audit.

2.4 Emergency drill

Test a serious hypothetical offshore event.

Measure:

- detection;
- notification;
- command activation;
- equipment mobilization;
- regional assistance;
- public communication;
- environmental sampling;
- claims intake.

Record actual exercise times.

2.5 Public transparency rehearsal

Ask an independent user to trace:

- licence;
- permit;
- production;
- sale;
- government receipt;
- fund deposit.

Missing definitions become a remediation item.

Stage 2 gate

First-oil readiness is not complete until:

- meter verification;
- fiscal rehearsal;
- bank reconciliation;
- environmental monitoring;
- emergency exercise;
- public data test;
- audit calendar

have each been documented.

Stage 3

First year of production

Objective

Demonstrate control under real transactions.

Month 1

Publish:

- actual production;
- first cargo;
- applicable fiscal terms;
- receipt method.

Verify:

- metering;
- invoice;
- bank receipt;
- accounting entry.

Quarter 1

Publish:

- production;
- cargo lifts;
- government receipts;
- environmental operating data;
- local employment/procurement with definitions.

Conduct:

- sample cost audit;
- permit inspection;
- spill-response review.

Quarter 2

Reconcile:

- physical barrels;
- sale barrels;
- royalty;
- profit oil;
- tax;
- fund deposits.

Review:

- contractor cost submission;
- local supplier data;
- grievances.

Quarter 3

Stress test:

- lower price;
- outage;
- spill;
- project delay.

Update:

- budget assumptions;

- liquidity;
- emergency capacity.

Quarter 4

Publish:

- audited fund accounts where timing permits;
- annual petroleum account;
- environmental annual report;
- local-content report;
- audit status;
- unresolved items.

No unresolved audit item should disappear from the public record merely because the year closes.

Stage 4

Years 2 to 5

Objective

Keep institutions growing at least as fast as the industry.

Annual institutional capacity review

Track:

- petroleum-regulator staff;
- engineers;
- auditors;
- environmental specialists;
- tax specialists;
- unresolved audits;
- permit workload;
- production volume.

Sovereign fund

Annually reconcile:

- opening balance;
- petroleum deposits;
- investment income;
- withdrawals;
- expenses;
- closing balance.

Separate market value and cash.

Public investment

For every major project:

- appraisal;
- contract;
- cost;
- completion;
- maintenance;
- service outcome.

Local value

Move from gross procurement to:

- domestic wages;
- domestic inputs;
- retained profit;
- local ownership;
- capability.

Social baseline

Update:

- household income/consumption;
- prices;
- housing;
- employment;
- education;
- health;

- regional indicators.

Do not rely indefinitely on a pre-oil survey.

Environmental system

Publish by project:

- emissions;
- flaring;
- discharge;
- incidents;
- inspections;
- permit changes;
- enforcement;
- spill drills.

Stage 4 gate

At year five conduct an independent public review of:

- legal framework;
- fiscal performance;
- fund accounting;
- public investment;
- environmental outcomes;
- local value;
- household evidence;
- institutional capacity.

The review should propose alternatives rather than assign a political score.

Crisis playbooks

A. Oil price falls sharply

Immediate:

- update realized price;
- update petroleum receipts;
- preserve cash/accounting distinction;

- recalculate lawful transfer ceiling;
- review budget exposure.

Within one quarter:

- prioritize essential recurrent spending;
- review project sequencing;
- update debt plan;
- stress fund liquidity.

Do not respond to a temporary price move using an unverified assumption of permanent decline.

B. Production outage

Immediate:

- establish technical cause;
- update production/cargo schedule;
- estimate cash timing.

Review:

- budget;
- fund deposits;
- contract obligations;
- environmental implications;
- repair logistics.

Separate capacity from actual production during the outage.

C. Major offshore spill

Immediate:

- protect life;
- activate incident command;
- stop source where feasible;
- notify affected jurisdictions;
- preserve evidence;
- initiate independent sampling.

Financial:

- activate operator liability;
- insurance;

- guarantee;
- emergency public funding if lawfully required.

Public:

- publish verified facts;
- sampling methods;
- response zones;
- claims process.

Do not publish speculative volume as established fact before measurement.

D. Cost-audit dispute

Maintain:

- amount claimed;
- auditor finding;
- contractor response;
- legal basis;
- procedure;
- final result.

Do not budget a proposed adjustment as recovered money.

E. Construction overheating

Measure:

- bid counts;
- project delays;
- construction prices;
- wages;
- imports;
- contractor concentration.

Options for public decision include:

- sequencing;
- imported capacity;
- training;
- procurement reform;
- slower rollout.

F. Natural disaster plus oil-revenue shock

Stress:

- emergency spending;
- lost tax revenue;
- infrastructure damage;
- petroleum downtime;
- fund withdrawals;
- insurance.

This combined case is especially relevant to island producers.

Island-state adaptation module

Marine economy

Baseline:

- tourism share;
- fishery share;
- reefs;
- seagrass;
- mangroves;
- beaches;
- desalination.

A marine incident can affect several sectors at once.

Single-point infrastructure

Map:

- one primary airport;
- one container port;
- one fuel terminal;
- one power station.

Create contingency for concurrent disruption.

Labour

A small labour force should not adopt unattainable local-content targets.

Prioritize:

- safety;
- technical roles;
- transferable certification;
- supervisory succession;
- supplier capability.

Revenue concentration

A small state should model petroleum receipts as a share of:

- total revenue;
- GDP;
- exports;
- foreign exchange.

The higher the concentration, the more important a low-price and production-interruption buffer becomes.

Publication and data blueprint

A newcomer should maintain one public evidence portal with linked datasets.

Legal register

- act;
- regulation;
- licence;
- agreement;
- permit;
- court decision.

Operational register

- wells;
- production;
- flaring;
- cargoes;
- incidents.

Fiscal register

- royalty;
- profit oil;
- tax;
- audit;
- sovereign fund.

Environmental register

- EIA;
- permit;
- monitoring;
- inspection;
- spill drill;
- enforcement.

Social/local register

- employment;
- supplier data;
- training;
- household indicators;
- community programmes.

Each dataset needs:

- observation date;
- publication date;
- unit;
- source;
- revision history.

24-month pre-first-oil action calendar

Months 24-18

- legal gap review;
- institutional map;

- regulator staffing;
- fiscal model;
- environmental baseline;
- maritime response agreements.

Months 18-12

- sovereign revenue framework;
- central-bank procedures;
- cost-audit procurement;
- local-content capacity study;
- household baseline;
- emergency equipment contracts.

Months 12-6

- first-cargo rehearsal;
- spill drill;
- public portal;
- production-meter assurance;
- petroleum-sales procedure;
- fund accounting test.

Months 6-3

- repeat reconciliation exercise;
- close audit gaps;
- verify insurance/guarantees;
- parliamentary reporting rehearsal;
- public communication protocol.

Months 3-0

- final readiness audit;
- regulator staffing check;
- first-cargo roles signed;
- bank accounts tested;

- public reporting calendar issued.

Five-year dashboard

Track annually:

- actual petroleum production;
- realized price;
- public petroleum receipts;
- fund balance;
- withdrawals;
- public debt;
- major project completion;
- real non-oil GDP;
- inflation;
- real wages;
- local value added;
- household indicators;
- environmental incidents;
- audit resolution time.

The dashboard should retain missing values rather than replace them with assumptions.

Final decision principles extracted from the Guyana study

1. Prepare legal and financial architecture before recurring revenue.
2. Keep legal versions dated.
3. Keep capacity, production, cargoes and cash separate.
4. Audit recoverable costs.
5. Keep sovereign-fund accounting separate from public-investment outcomes.
6. Keep petroleum emissions and forest/carbon accounting separate.
7. Verify environmental readiness, not permit existence alone.
8. Measure domestic value, not procurement labels alone.
9. Update household evidence after the economic transition.
10. Preserve stakeholder challenge and corrections.
11. Adapt foreign mechanisms to national conditions.
12. Stress-test adverse cases before relying on high-price projections.

Acceptance checklist for a country using this manual

A country should not describe itself as ready solely because every box is ticked.

The evidence behind each box should be auditable.

Minimum evidence package:

- laws;
- contracts;
- permits;
- audited accounts;
- production data;
- response drill;
- source register;
- stakeholder record;
- corrections log.

The objective is an institutional system capable of learning as evidence changes.

NATURAL RESOURCE FUND RECONCILIATION WORKBOOK

Companion technical annex for Part IV of the flagship publication. Working status: partially populated from audited and official period reports. Evidence cut-off: 27 September 2026.

Why this workbook exists

Natural Resource Fund figures appear in several valid but different forms:

- petroleum revenue recognized under financial reporting;
- cash received into the fund;
- revenue receivable at period end;
- investment income;
- government withdrawals;
- transfers to the Consolidated Fund;
- cash and cash equivalents;
- market value;
- closing net assets;
- annual forecasts.

The flagship must never place these values in one time series without reconciling their accounting basis.

Control equation

For a cash-based period bridge:

Opening cash + cash petroleum receipts + cash investment receipts + other cash inflows

- transfers/withdrawals
- cash expenses

= closing cash.

For audited accrual statements, add or subtract receivables, payables, unrealized gains/losses and any classification changes required by the financial statements.

Verified early-year audit observations

2020

Source: Bank of Guyana audited NRF financial statements. [S45]

Audited statement of income, G\$'000:

- Royalties: 2,694,223
- Profit oil: 38,651,798
- Interest on deposits: 1,581
- Net investment gain: 993
- Total income: 41,348,595

Audited cash-flow statement, G\$'000:

- Net cash flow from operating activities: 31,060,827
- Government withdrawal: 0
- Closing cash and cash equivalents: 31,060,827

The notes identify revenue receivables of G\$10,287,768, representing the fourth 2020 profit-oil lift, which was due in January 2021.

The notes report three royalty cash deposits during 2020 totaling USD 12,921,933.23.

The audited profit-oil note recognizes four 2020 lifts totaling USD 185,380,324.44, while the fourth lift was a year-end receivable rather than 2020 cash.

This produces the first key reconciliation rule:

2020 recognized petroleum revenue is not identical to 2020 cash petroleum deposits.

2021

Source: Bank of Guyana audited NRF financial statements. [S46]

Audited statement of changes in equity, G\$'000:

- Opening balance at 31 December 2020: 41,348,595
- 2021 royalties: 10,841,067
- 2021 profit oil: 74,478,746
- 2021 income: 25,902
- Deposits to Consolidated Fund: 0
- Closing balance at 31 December 2021: 126,694,310

Audited cash-flow statement, G\$'000:

- Net cash flow from operating activities: 10,313,670
- Royalties: 10,841,067
- Profit oil: 74,478,746
- Government withdrawal: 0
- Closing cash and cash equivalents: 126,694,310

The 2021 statement carries restated 2020 comparatives. The final dataset should use the latest audited comparative where later statements explicitly restate an earlier year.

Withdrawal history verified from Bank of Guyana quarterly reporting

Source: Bank of Guyana June 2025 NRF Quarterly Report. [S47]

The 2022-2025 transfer figures above describe cash withdrawals. They do not show the final use, project completion or social outcome of the transferred funds.

June 2025 fund snapshot

Source: Bank of Guyana June 2025 Quarterly Report. [S47]

For Q2 2025 the report states:

- opening market value at end-March: USD 3,339.38 million;
- end-June market value: USD 3,192.15 million;
- Q2 petroleum inflows: USD 617.95 million;
- Q2 outflows: USD 800 million;
- Q2 net return: USD 34.82 million.

The financial summary reports in G\$'000:

- Q1 opening balance: 676,725,568
- Q1 royalties: 17,549,019
- Q1 profit oil: 78,273,383
- Q1 outflow: 83,400,000
- Q1 investment result: 7,112,071
- Q1 closing balance: 696,260,041
- Q2 royalties: 17,810,764
- Q2 profit oil: 111,032,905
- Q2 outflow: 166,800,000
- Q2 investment result: 7,260,398
- Q2 closing balance: 665,564,108

This reporting should be kept separate from Table 6's transaction-by-transaction cash-receipt chronology until receivable timing is reconciled.

Cash-receipt chronology control

Bank of Guyana Table 6 supplies a dated petroleum cash-receipt history from inception through 30 June 2025.

The final reproducible dataset should store every transaction with:

- deposit date;
- petroleum source;
- FPSO/project where identified;
- receipt type: profit oil or royalty;
- USD amount;
- cargo/lift identifier where public;
- production period;
- sale/lift date;
- invoice date;
- receivable-recognition period;
- fund-deposit period.

The paper should then generate two separate annual charts:

1. **Cash deposited by calendar year.**
2. **Petroleum revenue recognized by accounting year.**

Differences between the charts are timing differences unless other accounting adjustments apply.

2026 H1 government reporting bridge

Existing source S07 records the September 2026 Ministry of Finance mid-year report as stating for the first half of 2026:

- government profit-oil cash receipts: USD 1,778.6 million;
- royalties: USD 218.4 million;
- withdrawals: USD 1,020 million;
- end-June NRF cash: USD 4,294.2 million.

These figures are official mid-year reporting, not yet a final audited 2026 annual account.

The final publication should reconcile them against:

- Bank of Guyana Q1 and Q2 2026 reports;
- monthly NRF reports;
- later audited 2026 statements when available;
- any period-end receivables.

Annual master table

The table below deliberately leaves fields blank until the relevant audited statement is extracted and reconciled.

Withdrawal-rule legal crosswalk to build

The final legal table will compare:

- 2019 Act;
- 2021 Act;
- 2024 amendment;
- annual maximum withdrawal calculations;
- parliamentary approval;
- notification requirements;
- actual transfer dates.

The table will not state a preferred political spending ratio.

Public-investment separation

The NRF workbook ends when money reaches the Consolidated Fund.

A separate public-investment dataset must follow funds into:

- appropriation;
- procurement;
- contract award;
- physical completion;
- commissioning;
- maintenance;
- service delivery;
- beneficiary outcomes.

This separation is required to avoid treating a lawful withdrawal as proof of a completed national-development result.

Audit controls

Before flagship publication:

1. Extract audited 2022-2025 statements.
2. Normalize currencies and accounting units.
3. Reconcile every annual opening balance to prior closing balance.
4. Reconcile receivables across year boundaries.
5. Reconcile Bank of Guyana cash-deposit dates to Ministry petroleum-lift records.
6. Reconcile withdrawals to parliamentary notifications and Consolidated Fund records.

7. Record investment income separately.
8. Preserve all restatements.
9. Explain differences between market value, cash, net assets and accumulated capital.
10. Have an independent accountant review the final annual bridge.

Publication rule

No NRF figure enters the final executive summary unless its accounting basis, date and currency are explicit.

ENVIRONMENTAL PERMIT AND RISK REGISTER

Companion evidence register for Part V of the flagship publication. Evidence cut-off: 27 September 2026. Status: partially populated. Permit existence, permit conditions and observed compliance are recorded separately.

Control rule

Each project is tracked through six different evidence classes:

1. application and public notice;
2. environmental impact assessment and management plans;
3. environmental permit;
4. production licence environmental obligations;
5. monitoring, incidents and regulator enforcement;
6. spill response, insurance and financial assurance.

A project is not described as environmentally compliant merely because a permit exists.

Project register

Liza Phase 1 permit chronology

The official permit record states:

- application submitted 5 July 2016;
- approved EIA dated 1 June 2017;
- Environmental and Socioeconomic Management Plan dated 1 June 2017;
- revised Oil Spill Response Plan and Wildlife Response Plan dated 1 June 2017;
- environmental permit issued 1 June 2017;
- permit modified 12 May 2021. [S48]

This sequence gives the flagship a dated example of environmental assessment occurring after the 2015 commercial discovery but before production began.

The 2021 EPA notice says the permit modification responded to resumed flaring and projected exceedance of the EIA's estimated flaring volume. The agency said the modification introduced emissions reporting, technical flaring requirements, time limits and payment of US\$30 per tonne of CO₂e for flaring beyond the specified timelines. [S55]

The publication must still test observed performance. The next dataset needs:

- monthly flaring volume;
- duration and cause;

- permitted versus exceptional flaring;
- tonnes CO2e;
- payments assessed;
- payments received;
- equipment failures;
- corrective action;
- inspection findings.

Uaru permit controls

The official Uaru environmental permit states an effective period from 27 April 2023 through 26 April 2028. The permit text gives the EPA authority to order sampling and analysis of project discharges, effluent or waste by an approved independent laboratory at the permit holder's expense. It also reserves the authority to suspend, modify or cancel the permit under specified conditions and links annual review to the previous year's required audit. [S50]

Those provisions establish regulatory powers. They do not establish the frequency with which powers have been exercised or the findings of any audit.

The Uaru production licence separately requires compliance with environmental law, adequate expert personnel and equipment for spill response, dispersant arrangements and access to capping-stack capability. [S56]

The final flagship should map every response obligation to evidence of:

- equipment physically in-country;
- subscription contracts;
- mobilization time;
- drills;
- regulator observation;
- stock inspection;
- personnel competence;
- transboundary notification;
- replacement after use or expiry.

Oil Pollution Act 2025 status control

Parliament records the Oil Pollution Prevention, Preparedness, Response and Responsibility Act 2025 as Act 6 of 2025, passed 16 May 2025 and published 30 May 2025. The Official Gazette archive records an extraordinary gazette dated 28 May 2025. [S53, S54]

The register does not yet mark the full regime as commenced.

Before publication, obtain:

- commencement order;
- implementing regulations;
- designation of the competent national authority;
- prescribed financial-responsibility requirements;
- response-plan regulations;
- reporting forms;
- enforcement procedures.

If no commencement instrument is verified by the evidence cut-off, the final paper will state only that the law was enacted/published and separately describe government statements about planned commencement.

Financial assurance register

Financial assurance must be decomposed into separate legal instruments:

- operator primary liability;
- commercial insurance;
- parent/company guarantee;
- affiliate guarantee;
- capping and containment service subscriptions;
- pollution-response contracts;
- government contingency resources;
- compensation and claims procedure.

Required fields:

- instrument name;
- parties;
- effective date;
- maximum amount if any;
- scope of covered loss;
- exclusions;
- attachment point;
- currency;
- claims jurisdiction;
- proof of financial capacity;
- termination rights;
- regulator access;
- public disclosure status.

A headline "coverage" number will not be presented unless these terms are independently verified.

Environmental incident taxonomy

Every event should be coded as:

- flaring exceedance;
- produced-water issue;
- hydraulic-fluid spill;
- crude-oil spill;
- chemical release;
- vessel incident;
- well-control event;
- air-emission exceedance;
- waste-management incident;
- wildlife impact;
- permit/reporting noncompliance;
- near miss.

Fields:

- date;
- project/FPSO;
- material;
- volume;
- regulator notification;
- immediate response;
- enforcement;
- monetary sanction;
- corrective action;
- independent sampling;
- closure evidence.

Marine ecological evidence

The final paper requires primary or independently reviewed data on:

- fisheries;
- marine mammals;

- turtles;
- seabirds;
- benthic habitat;
- coastal mangroves;
- water quality;
- currents;
- sediment;
- shoreline sensitivity;
- protected areas;
- spawning and nursery grounds.

For each dataset record:

- collector;
- funder;
- method;
- observation dates;
- raw-data availability;
- geographic coverage;
- peer review;
- limitations.

Spill response stress tests

The flagship will test institutional preparedness under at least four scenarios:

Scenario A: controlled surface release

Test local containment, boom availability, skimming, temporary storage, shoreline protection and waste disposal.

Scenario B: subsea loss of well control

Test capping stack mobilization, intervention capability, relief-well assumptions, dispersant logistics, aerial surveillance and sustained response.

Scenario C: adverse weather plus spill

Test port closure, aviation limits, responder safety, communications redundancy and delayed external support.

Scenario D: transboundary movement

Test notification, evidence-sharing, neighbouring-state coordination, compensation, fisheries closures and jurisdiction.

These are preparedness tests, not predictions of event probability.

Evidence hierarchy for environmental claims

Highest weight:

- signed permits;
- enacted/commenced law;
- regulator inspections;
- laboratory results;
- audited monitoring reports;
- authenticated judgments;
- independently verified incident records.

Supporting evidence:

- EIA models;
- operator monitoring;
- government statements;
- academic studies;
- civil-society submissions;
- media investigations.

Where sources conflict, the final chapter will state the disagreement and identify what evidence would resolve it.

Remaining environmental evidence gates

1. Extract Liza Phase 2 amended permit.
2. Extract Payara permit and amendments.
3. Obtain final Yellowtail permit.
4. Extract Whiptail permit conditions.
5. Build all-project EIA metadata table.
6. Compile flaring by project and month.
7. Compile produced-water volumes and standards.
8. Compile regulator inspection and audit records.
9. Verify Oil Pollution Act commencement and regulations.

10. Obtain authenticated 2026 Court of Appeal judgment concerning financial assurance.
11. Obtain actual insurance/guarantee documents where public.
12. Compile spill equipment and drill evidence.
13. Add independent fisheries and biodiversity analysis.
14. Record stakeholder and community submissions.
15. Have an external environmental-law specialist review the final chapter.

Publication boundary

The environmental chapter will neither assume environmental success from permit issuance nor infer regulatory failure from the existence of incidents. It will test the full chain from assessment through permit, monitoring, enforcement, remediation and financial responsibility.

MODEL METHODS AND LIMITATIONS

Working technical annex for the flagship Guyana petroleum case study. Evidence cut-off: 27 September 2026.

Purpose

The repository contains reproducible arithmetic for contract sensitivity, production shocks and sovereign-fund stress testing.

These models are decision aids. They are not forecasts of Guyana's petroleum revenue, operator returns, future production, future oil prices or future Natural Resource Fund balances.

Contract waterfall

The simplified model uses five principal inputs:

- illustrative barrels;
- illustrative realized price;
- royalty fraction;
- cost-recovery ceiling;
- requested recoverable-cost fraction.

The model then calculates:

Gross sales = barrels x price.

Actual cost recovered = gross sales x the smaller of requested cost fraction and the cost-recovery ceiling.

Profit pool = gross sales - actual cost recovered.

Government profit oil = profit pool x illustrative government profit share.

Royalty = gross sales x royalty fraction.

Government total in the teaching model = government profit oil + royalty.

The current teaching implementation uses the published 2% royalty, 75% cost-recovery ceiling and equal residual profit split associated with the 2016 Stabroek agreement's public fiscal explanation. Contract-specific accounting, valuation, taxation, timing, recoverability, decommissioning and dispute provisions require clause-level review before any result is described as an actual entitlement.

Why a percentage of gross sales is not a permanent government take

When recoverable costs use the full ceiling, the residual profit pool is smaller.

When actual recoverable costs fall below the ceiling, the residual profit pool is larger.

Later investment can add new recoverable costs.

For this reason, the 14.5% simplified binding-cap illustration should not be applied as a constant percentage across every field, month or development stage.

Contract sensitivity dataset

ILLUSTRATIVE_SCENARIOS.csv contains 16 synthetic combinations:

- oil price: USD 40, 60, 80 and 100 per barrel;
- requested recoverable cost fraction: 20%, 50%, 65% and 75%;
- illustrative barrels: 10 million in each scenario.

The purpose is to show arithmetic sensitivity while holding volume constant.

No scenario is labelled as a Guyana forecast.

Combined production shock dataset

ILLUSTRATIVE_PRODUCTION_SHOCKS.csv contains 18 synthetic combinations:

- production volume factor: 70%, 85% and 100% of an illustrative 10-million-barrel period;
- price: USD 40, 60 and 80;
- requested recoverable cost fraction: 50% and 75%.

This demonstrates that a price shock, production interruption and high cost recovery can occur together.

The cases are arithmetic stress tests. They do not estimate the probability of any operational outage.

Sovereign-fund stress model

The fund model uses a simplified annual identity:

Closing balance = opening balance + deposits + illustrative investment return - withdrawals.

The return assumption is applied to the opening balance only.

Actual sovereign-fund performance depends on intra-year cash timing, portfolio weights, market values, realized/unrealized gains, expenses and accounting standards.

The synthetic fund cases are:

1. illustrative baseline;
2. lower deposits following a low-price condition;
3. higher spending;

4. market loss followed by recovery.

Each case begins with an illustrative USD 4 billion opening balance.

The figures deliberately do not reproduce the legal Guyana NRF withdrawal formula or predict actual future petroleum deposits.

Why the fund model is separate from the NRF reconciliation

NRF_RECONCILIATION_WORKBOOK.md reconstructs observed historical public accounts.

ILLUSTRATIVE_FUND_STRESS.csv models hypothetical future paths.

Observed and hypothetical values must never appear in one chart without explicit labeling.

Model review gates

Before the fiscal models enter the final flagship:

1. Extract the executed PSA's relevant fiscal and accounting clauses.
2. Review the accounting annex and valuation provisions.
3. Reconcile simplified formulas with actual petroleum entitlements.
4. Validate cost-bank treatment.
5. Separate royalty and profit-oil timing.
6. Review tax treatment.
7. Review decommissioning treatment.
8. Test unit consistency.
9. Test extreme cases and zero cases.
10. Obtain independent petroleum-fiscal review.

Before the fund model enters the final publication:

1. Reconcile audited annual NRF statements.
2. Verify statutory withdrawal rules by legal version.
3. Use transparent deposit assumptions.
4. Use transparent return assumptions.
5. Model liquidity separately from long-horizon return.
6. Include adverse market-return periods.
7. Include oil-price and production shocks.
8. Identify any public-debt interaction.
9. Identify disaster-related fiscal exposure.
10. Obtain independent public-finance review.

Reproducibility

The source code is `illustrative_model.py`.

The generated datasets are:

- `ILLUSTRATIVE_SCENARIOS.csv`;
- `ILLUSTRATIVE_PRODUCTION_SHOCKS.csv`;
- `ILLUSTRATIVE_FUND_STRESS.csv`.

The final publication should record the Git commit associated with every model output and include a checksum manifest.

Interpretation rule

A model result establishes only what follows from its stated assumptions.

It does not establish:

- future oil prices;
- future production;
- actual recoverable costs;
- future government receipts;
- optimal fiscal policy;
- expected spill loss;
- household welfare.

Any final decision statement must return to observed evidence, legal authority and disclosed assumptions.

Source register

Dated source IDs correspond to citations in the manuscript. Public URLs link to originals. Key unresolved verification tasks appear after the register.

S01 ExxonMobil Guyana 2023 Annual Report

Early seismic and onshore exploration, May 2015 Liza commercial discovery, 2019 first oil and FPSO dates

Limit: Operator is an interested party; cross-check public licence register

<https://corporate.exxonmobil.com/locations/guyana/publications/2023-annual-report>

S02 Guyana Ministry of Natural Resources, Stabroek petroleum agreement

Official text of the 2016 agreement, contract terms and baseline obligations

Limit: Distinct from future PSAs and public summaries

<https://petroleum.gov.gy/docs-and-pubs/petroleum-agreement-stabroek-block/>

S03 Ministry of Natural Resources, block register

June 27 2016 agreement / October 7 2016 effective date and licence information

Limit: Different contract milestone dates refer to different acts

<https://petroleum.gov.gy/oil-blocks/stabroek-block/>

S04 2024 NRF Annual Report, Ministry of Finance, notes/legal framework

NRF law assented January 23 2019; replaced December 30 2021; withdrawal rule amended February 6 2024

Limit: Full audited report must be checked for changing cash period

https://finance.gov.gy/wp-content/uploads/2025/11/2024_NRF_Annual_Report_Final_Draft-JD0715_GE1507-Final-Print.pdf

S05 Parliament of Guyana, Acts index

Local Content Act 2021 and NRF Act 2021 parliamentary dates

Limit: Parliamentary passing, assenting and commencement dates differ

<https://www.parliament.gov.gy/publications/acts-of-parliament/P80>

S06 Finance, 2026 Budget Speech, January 26 2026, 124-page PDF

January production/revenue/NRF assumptions and policy descriptions

Limit: January projections were superseded by September mid-year update

https://finance.gov.gy/wp-content/uploads/2026/01/Budget_Speech_2026.pdf

S07 Finance, 2026 Mid-Year Report, issued September 2026, printed pp. 21-25 and 104

H1 2026 21 government lifts, USD 1,778.6m profit-oil proceeds, USD 218.4m royalties, USD 4,294.2m end-June fund cash, USD 1,020m withdrawals; revised forecast USD 6,497.6m petroleum deposits

Limit: Government estimates and projections, not audited end-year results

<https://finance.gov.gy/wp-content/uploads/2026/09/Mid-Year-Report-2026.pdf>

S08 IMF, July 31 2026 mission concluding statement

IMF staff views on balance/withdrawal rule, absorption, audit, procurement, AML/CFT, statistical gaps and LCDS

Limit: Staff preliminary findings, not IMF Executive Board Article IV report

<https://www.imf.org/en/news/articles/2026/07/31/mcs073126-guyana-staff-concluding-statement-of-the-2026-article-iv-mission>

S09 IMF, 2025 Article IV report and press

Detailed macro, fiscal and debt context, external baseline

Limit: Historical to 2024; do not use as September 2026 snapshot

<https://www.imf.org/en/publications/cr/issues/2025/05/07/guyana-2025-article-iv-consultation-press-release-staff-report-and-statement-by-the-566712>

S10 World Bank, Guyana country overview

Social indicators, non-oil growth and lack of current household poverty data

Limit: Explicitly labels last cited poverty estimate 2019, not 2026

<https://www.worldbank.org/ext/en/country/guyana>

S11 ExxonMobil, Yellowtail startup, August 8 2025

Fourth FPSO, installed capacity exceeded 900k bopd

Limit: Installed capacity is not sustained actual output

https://corporate.exxonmobil.com/news/news-releases/2025/0808_exxonmobil-guyana-begins-production-at-fourth-offshore-guyana-project

S12 ExxonMobil, November 12 2025

900k bopd daily output milestone

Limit: A single-day threshold is not 2025 annual average

<https://corporate.exxonmobil.com/locations/guyana/news-releases/11122025-daily-oil-production-hits-900000-barrels-in-guyanas-stabroek-block>

S13 Petroleum Management Programme oil-production chart

Government monthly production chart through July 31 2026

Limit: Preserve date and distinguish unit from capacity

<https://petroleum.gov.gy/data-chart/oil-production/>

S14 Government of Guyana, 2023 model fiscal terms

Announced 10% royalty, 65% cost ceiling and 10% corporate tax for future licensing

Limit: Do not apply revised terms retroactively to Stabroek

<https://petroleum.gov.gy/article/govt-announces-wave-of-enhanced-fiscal-terms-for-first-offshore-oil-and-gas-competitive-licensing-round-highlights/>

S15 Guyana Revenue Authority, royalty treatment, 2026

2016 2% royalty added to simplified 12.5% government profit share when cap binds

Limit: 14.5% example is not constant government take across the life of the block

<https://gra.gov.gy/royalty-payments-to-guyana-government/>

S16 EITI, May 12 2026 Board decision

Second validation from May 15; concern over civil-society selection and stakeholder confidence

Limit: No 2026 final validation result has been established in this work

<https://eiti.org/es/node/28602>

S17 EITI, 2026 call for stakeholder views

2022 validation and 2026 validation scope

Limit: Distinguish earlier assessment from later validation

<https://eiti.org/offers/guyana-2026-eiti-validation-call-views-stakeholder-engagement>

S18 Guyana EITI country secretariat

December 2025 report covers fiscal year 2023; validation visits June 2026

Limit: Secretariat claims require triangulation with EITI global board

<https://eiti.gy/about/>

S19 Parliament, Oil Pollution Prevention, Preparedness, Response and Responsibility Act 2025

Enacted Act No. 6 of 2025

Limit: September mid-year report says draft regulations progressing and commencement expected end-2026; enactment is not commencement

<https://parliament.gov.gy/publications/acts-of-parliament/oil-pollution-prevention-preparedness-response-and-responsibility-act-2025-act-6-of-2025>

S20 Government statement about May 7 2026 Court of Appeal decision

Appeal set aside High Court unlimited financial-guarantee order

Limit: Obtain authenticated judgment and independent legal commentary before final jurisprudential analysis

<https://dpi.gov.gy/press-release-ag-hails-coas-ruling-on-exxonmobil-epas-financial-guarantees/>

S21 EPA environmental authorisation services

Environmental authorisation/permit process and regulated-project categories

Limit: Online application portal alone does not prove monitoring quality

<https://eservices.epa.gov.gy/permits>

S22 Guyana LCDS 2030, October 2024 update

2009 framework, Norway USD 227m for 2009-15, 2022 policy and forest-credit programme; 15% community allocation

Limit: LCDS describes government strategy and self-reported expenditure; impact audit separate

<https://lcds.gov.gy/guyanas-low-carbon-development-strategy-lcds-2030-october-2024-update/>

S23 LCDS, Guyana ART TREES fact sheet, 2022

Hess minimum USD 750m purchase agreement, credit volume and minimum prices

Limit: Contracted value is not money already received

<https://lcds.gov.gy/fact-sheet-on-guyanas-engagement-in-art-trees/>

S24 Ghana Petroleum Revenue Management Act 2011

Stabilisation Fund, Heritage Fund, publication cadence and public accountability

Limit: Comparator institutional design, not a performance ranking

<https://ghalii.org/akn/gh/act/2011/815/eng%402011-04-15>

S25 Norway, 2025 Government Pension Fund white paper

Entire net state petroleum cash transferred to GPF; 3% real-return fiscal guideline

Limit: Norway's long accumulation and mature state capacity differ from a new producer

<https://www.regjeringen.no/contentassets/185dca55ef5649d1bc08db2699bb861f/en-gb/pdfs/stm202420250022000engpdfs.pdf>

S26 Timor-Leste central bank, 2005/2011 Petroleum Fund statutes

Statutory fund structure and amendments

Limit: Cross-check 2026 depletion pressures independently before claims about outcomes

https://www.bancocentral.tl/publication/laws_np/

S27 TotalEnergies Suriname, September 19 2026

GranMorgu 2024 FID, first oil planned 2028

Limit: Planned start is not operating output

<https://totalenergies.sr/press/suriname-granmorgu-project-takes-shape-major-production-equipment-arrives>

S28 Guyana pre-oil sovereign wealth fund Green Paper

Documents work beginning in late 2016 with Commonwealth Secretariat support, inter-agency working group in 2017, IMF missions and FARI training before first oil

Limit: Historical government policy paper. Treat forecasts in the paper as superseded by actual later events

<https://petroleum.gov.gy/wp-content/uploads/2024/10/Green-Paper-Final.pdf>

S29 IMF 2023 Article IV staff report

Explains two-rule system: Stabroek PSA revenue sharing plus NRF withdrawal framework; notes first oil receipts deposited in 2020 and first budget transfer in 2022; records 2023 model-PSA changes and governance priorities

Limit: IMF analysis reflects information available in 2023 and should not substitute for 2026 data

<https://www.elibrary.imf.org/view/journals/002/2023/379/article-A001-en.xml>

S30 Natural Resource Governance Institute, natural-resource fund governance guidance

Clear objectives, deposit/withdrawal rules, investment rules, independent oversight, broad consensus and budget integration; warns a fund alone does not produce good governance

Limit: Comparative policy guidance, not jurisdiction-specific legal advice

https://resourcegovernance.org/sites/default/files/NRF_Complete_Report_EN.pdf

S31 NRGi sovereign wealth fund guidance for policymakers

Highlights oversight, disclosure, fiscal-rule fit, dangers of direct domestic investment from funds and the need

to fit rules to national conditions

Limit: Older cross-country guidance; current country practices require fresh verification

<https://resourcegovernance.org/articles/sovereign-wealth-funds-guidance-policy-makers>

S32 LCDS homepage, accessed September 2026

Current official payment dashboard lists cumulative Phase II carbon-credit payments of USD 353.5m, including March 2026 USD 57.9m

Limit: Official programme reporting; independent audit of final use and outcomes remains separate

<https://lcds.gov.gy/>

S33 EITI validation schedule, accessed September 2026

Guyana validation commenced 15 May 2026 and remains listed among ongoing validations

Limit: Do not state a final 2026 validation score until EITI publishes a decision

<https://eiti.org/validation-schedule>

S34 EITI Board decision 2026-19, 12 May 2026

Full validation launched with external expert; Board expressed concern over continuing lack of full stakeholder confidence in civil-society representative selection

Limit: Board concern is attributed to EITI, not adopted as the author's independent conclusion

<https://eiti.org/es/node/28602>

S35 Petroleum Management Programme, official exploration history

Records the long pre-2015 exploration sequence, 1938 licence history, 1965 offshore licences, 1980s investment promotion and later exploration interruptions

Limit: Government historical summary; individual dates should be cross-checked against original licences where material

<https://petroleum.gov.gy/history/summary-history-petroleum-exploration-guyana/>

S36 Petroleum Management Programme, Liza-1 well register

Records Liza-1 as an exploration oil well in the Stabroek Block with well date 15 May 2015 and mapped coordinates

Limit: Administrative well register; commerciality and reserve estimates require separate sources

<https://petroleum.gov.gy/well/liza-1/>

S37 Parliament of Guyana, Natural Resources Fund Act 2019 record

Parliament lists passage on 3 January 2019 and publication on 23 January 2019; the official act records presidential assent on 23 January 2019

Limit: Statutory establishment does not establish later implementation quality or unchanged withdrawal rules

<https://parliament.gov.gy/publications/acts-of-parliament/natural-resources-fund-act-2019>

S38 Parliament of Guyana, Petroleum (Exploration and Production) Act 1986

Establishes the 1986 petroleum-law foundation for exploration, exploitation, conservation and management, including maritime areas

Limit: Later legislation and regulations must be version-checked before applying historical provisions to 2026

<https://parliament.gov.gy/publications/acts-of-parliament/petroleum-exploration-and-production-act-1986/>

S39 Parliament of Guyana, Environmental Protection Act 1996

Establishes the pre-oil statutory environmental framework and environmental-impact mandate

Limit: Existence of statutory authority is separate from project-level permitting, enforcement and outcomes

<https://parliament.gov.gy/publications/acts-of-parliament/environmental-protection-act-1996>

S40 Permanent Court of Arbitration, Guyana v. Suriname case record

Documents the 2000 oil-concession incident, 2004 commencement of arbitration and maritime-boundary dispute framework

Limit: Use the final award itself for detailed legal propositions; case page provides procedural history

<https://pca-cpa.org/en/cases/9/>

S41 Petroleum Management Programme, Stabroek Block register

Records June 27 2016 petroleum agreement date, October 7 2016 effective/licence date and June 15 2017 production licence date

Limit: Administrative register; contract clauses must be checked against executed text

<https://petroleum.gov.gy/oil-blocks/stabroek-block/>

S42 Petroleum Management Programme, VHE initial 2018-2020 Stabroek cost-recovery audit page

Government-published audit record covering cost and revenue review and identifying the operator as responsible for cost-recovery accounting

Limit: Initial audit status and later government resolution of exceptions must be distinguished

<https://petroleum.gov.gy/docs-and-pubs/vhes-initial-audit-report-for-the-stabroek-block-cost-recovery-audit-2018-to-2020/>

S43 Petroleum Management Programme, regulatory-news document index

Lists public cost-recovery audits, production licences and environmental permits, including 1999-2017 and 2018-2020 audit records

Limit: Document availability does not establish final settlement of every audit finding

<https://petroleum.gov.gy/documents-categories/regulatory-news/>

S44 Bank of Guyana, NRF audited financial statements index

Provides audited NRF financial statements for 2020 through 2025 in one official series

Limit: Each annual statement must be read on its own accounting basis and checked for restatements

<https://bankofguyana.org.gy/bog3/publications/natural-resource-fund/audited-financial-statements>

S45 Bank of Guyana, NRF audited financial statements 2020

Records 2020 royalties, profit oil, receivables, cash, first-year accounting policies and no government withdrawal during the year

Limit: GYD reporting and accrual receivables must be reconciled to USD cash deposits before cross-year comparisons

https://bankofguyana.org.gy/bog/images/accounts_budgeting/natural_resource_fund/financial_statements/nrf_audited_financial_statements_2020.pdf

S46 Bank of Guyana, NRF audited financial statements 2021

Records 2021 royalties, profit oil, cash flows, no government withdrawal and replacement of the 2019 Act by the 2021 Act

Limit: Contains restated comparative 2020 figures; use audited restatement rather than mixing earlier presentation

https://bankofguyana.org.gy/bog/images/accounts_budgeting/natural_resource_fund/financial_statements/nrf_audited_financial_statements_2021.pdf

S47 Bank of Guyana, NRF Quarterly Report June 2025

Provides quarterly inflows, market value, interest, transfers to the Consolidated Fund and cumulative withdrawal history through June 2025

Limit: Quarterly market value differs from annual audited closing balances and should not be merged without date alignment

https://www.bankofguyana.org.gy/bog/images/accounts_budgeting/natural_resource_fund/quarterly/nrf-june2025-quarterly.pdf

S48 Petroleum Management Programme, Liza Phase 1 Modified Environmental Permit

Records original permit dated 1 June 2017, the approved EIA/ESMP and revised oil-spill/wildlife plans, and modification on 12 May 2021

Limit: Permit existence and conditions do not establish independent compliance; monitoring evidence must be reviewed separately

<https://petroleum.gov.gy/docs-and-pubs/liza-phase-1-modified-environmental-permit/>

S49 Petroleum Management Programme, EPA-EEPG project document index

Official index lists Liza Phase 1 permit, Liza Phase 2 amended permit and EIA materials, Payara permit/EIA/OSRP

documents and related environmental records

Limit: Index establishes document availability, not project performance; each permit must be reviewed individually

<https://petroleum.gov.gy/projects-initiatives/environmental-protection-agency-esso-exploration-and-production-guyana-limited/>

S50 Petroleum Management Programme, Uaru Environmental Permit

Permit Ref. 2023-023-EEPGL, effective 27 April 2023 to 26 April 2028; includes EPA powers for sampling, permit modification and annual review linked to audit

Limit: Project-level compliance and audit results require separate evidence

<https://petroleum.gov.gy/wp-content/uploads/2024/10/Uaru-Signed-Permit.pdf>

S51 Petroleum Management Programme, Whiptail Environmental Permit page

Official publication dated 12 April 2024 for the Whiptail environmental permit

Limit: Detailed permit conditions require document-level extraction and monitoring evidence

<https://petroleum.gov.gy/whiptail-environmental-permit/>

S52 Petroleum Management Programme, Yellowtail public notice

Records the 12 May 2021 environmental-authorization application and EPA determination that an EIA was required before approval or rejection

Limit: Application/EIA scoping is not evidence of a final permit or later compliance

<https://petroleum.gov.gy/notice-to-the-public-yellowtail-development-project/>

S53 Parliament of Guyana, Oil Pollution Prevention, Preparedness, Response and Responsibility Act 2025

Parliament records Act 6 of 2025, passed 16 May 2025 and published 30 May 2025

Limit: Enactment/publication must be kept separate from commencement and implementing regulations

<https://parliament.gov.gy/publications/acts-of-parliament/oil-pollution-prevention-preparedness-response-and-responsibility-act-2025-act-6-of-2025>

S54 Official Gazette of Guyana, Act 6 of 2025 publication

Official Gazette archive records the petroleum pollution Act as an extraordinary gazette dated 28 May 2025

Limit: Gazette publication alone does not establish that every provision had commenced by the research cut-off

<https://officialgazette.gov.gy/index.php/publications/2778-extraordinary-gazettes-28th-may-2025-act-no-6-of-2025-the-oil-pollution-prevention-preparedness-response-and-responsibility-bill-2025>

S55 Petroleum Management Programme, Liza 1 flaring permit modification notice

EPA states the modified permit added emissions reporting, flaring time limits and a US\$30/tCO₂e payment for flaring beyond specified timelines

Limit: Agency statement documents the regulatory requirement; actual emissions, invoices and payments need period evidence

<https://petroleum.gov.gy/epa-issues-a-modified-environmental-permit-to-eevgl-for-the-liza-1-project/>

S56 Petroleum Management Programme, Uaru Petroleum Production Licence

Licence requires compliance with environmental law, expert personnel/equipment for spill response and specified capping-stack/dispersant arrangements

Limit: Licence requirements are obligations, not evidence that all readiness conditions were demonstrated in practice

<https://petroleum.gov.gy/index.php/documents/uaru-petroleum-production-licence>

S57 Parliament of Guyana, Local Content Act 2021

Parliament records Act 18 of 2021, passed 29 December 2021 and published 4 January 2022

Limit: Enactment establishes the statutory framework; measured domestic value added requires separate evidence

<https://www.parliament.gov.gy/publications/acts-of-parliament/local-content-act-2021-no.-18-of-2021>

S58 Guyana Local Content Register

Official supplier/employment registration system and Local Content Secretariat portal

Limit: Certification establishes registry status, not independent proof of domestic value added, capability or retained profit
<https://lcregister.petroleum.gov.gy/>

S59 Ministry of Finance, 2024 Mid-Year Report

Government reported H1 2024 procurement and employment information from contractor/subcontractor reports, including reporting scope and projected training/procurement

Limit: Self-reported sector figures require independent ownership, customs, payroll and value-added checks
<https://finance.gov.gy/wp-content/uploads/2024/08/Mid-Year-Report-2024.pdf>

S60 Ministry of Finance, 2025 Mid-Year Report

Describes expansion of local-content reporting to additional sectors/subsectors and ongoing assessment of local capacity

Limit: Government implementation update; not an independent measure of outcome quality
<https://finance.gov.gy/wp-content/uploads/2025/09/Mid-Year-Report-2025.pdf>

S61 Guyana Bureau of Statistics, Household Budget Survey 2018/2019

Official survey page states HBS/Living Condition fieldwork ran from November 2018 through November 2019 across all ten administrative regions

Limit: Historical household evidence must not be represented as current 2026 welfare data
<https://statisticsguyana.gov.gy/surveys/>

S62 LCDS, consultations and land-tenure fact page

Official programme account describes 15% community revenue share, village-plan process and Amerindian Land Titling programme

Limit: Government account requires triangulation with Indigenous/community sources and independent safeguards evidence
<https://lcds.gov.gy/facts-on-consultations-and-land-tenure-guyanas-art-trees-programme/>

S63 LCDS 2030 October 2024 Update

Government reported more than 240 villages receiving carbon-credit revenue and village-led projects under the 15% allocation

Limit: Programme spending/participation figures require independent outcome and consent verification
<https://lcds.gov.gy/guyanas-low-carbon-development-strategy-lcds-2030-october-2024-update/>

S64 IMF, 2026 Article IV mission concluding statement

IMF staff reports 2025 growth, continued H1 2026 expansion, fiscal/absorption issues, procurement/audit priorities, data gaps and diversification observations

Limit: Preliminary staff findings, not an Executive Board Article IV report; policy recommendations are attributed to IMF staff
<https://www.imf.org/en/news/articles/2026/07/31/mcs073126-guyana-staff-concluding-statement-of-the-2026-article-iv-mission>

S65 Ministry of Finance, Budget Speech 2026

Official 2026 budget policy, public-investment allocations and announced energy/infrastructure programme

Limit: Budget allocation and target do not establish project completion or delivered outcome
https://finance.gov.gy/wp-content/uploads/2026/01/Budget_Speech_2026.pdf

S66 Guyana Energy Agency, Demand Side Management Action Plan handover, July 2026

GEA reports a 2030 target of 516 GWh energy savings against projected generation of 2,508 GWh

Limit: Government/agency plan and target; implementation and achieved savings require later measurement
<https://gea.gov.gy/2026/07/01/6793/>

S67 Guyana Energy Agency, Demerara-Berbice Integrated System transition page

Official description of the electricity mix, gas-to-energy concept and planned renewable complement

Limit: Some project dates on the page are historical planning dates and must be updated against 2026 delivery

evidence

<https://gea.gov.gy/renewable/demerara-berbice-integrated-system/>

S68 IMF, 2025 Article IV staff report

Describes official-statistics modernization, planned new household survey and macroeconomic context before the 2026 mission

Limit: Historical to 2025 and superseded where 2026 evidence exists

<https://www.elibrary.imf.org/view/journals/002/2025/103/article-A001-en.xml>

S69 EITI Board Decision 2026-19

Board launched a full Guyana Validation from 15 May 2026 and recorded concern about stakeholder confidence in civil-society representative selection

Limit: Validation remained a process at the research cut-off; do not state a final score without a later Board decision

<https://eiti.org/es/node/28602>

S70 EITI 2026 stakeholder consultation

International Secretariat sought views on Guyana implementation from April 2022 to May 2026, including civil-society participation and barriers

Limit: Consultation questions and submissions are inputs to Validation, not a final assessment

<https://eiti.org/offers/guyana-2026-eiti-validation-call-views-stakeholder-engagement>

S71 Petroleum Management Programme, IHS Markit final 1999-2017 cost-recovery audit

Government-published audit states audited cost-recovery submission totaled USD1.6778bn and identified USD214.4m plus overhead adjustments as having grounds for dispute

Limit: Audit recommendation and government dispute are not identical to final cash recovery or settlement

<https://petroleum.gov.gy/documents/ihs-markit-final-audit-report-stabroek-block-cost-recovery-audit-1999-2017>

S72 Ministry of Natural Resources, audit-status update

Government states the 1999-2017 audit moved to the PSA sole-expert process and describes status of multiple cost audits

Limit: Ministry status statement requires later verification of settlement, expert decision or arbitration outcome

<https://petroleum.gov.gy/article/update-on-status-of-stabroek-block-cost-recovery-and-verification-of-profit-oil-audits/>

S73 Audit Office of Guyana, 2024 public-accounts report portal

Supreme audit institution states the 2024 public-accounts/performance report was presented to the National Assembly in September 2025

Limit: Public-account audit observations must be tied to specific entities/projects; do not generalize individual findings across government

<https://www.audit.org.gy/site/>

S74 Norway Ministry of Finance, Government Pension Fund 2026 white paper

Describes current GPFG framework, 3% expected-real-return fiscal guideline and long-run role of petroleum wealth in public finance

Limit: Norway's mature institutions, large accumulated assets and economic structure differ materially from a new producer

<https://www.regjeringen.no/en/documents/meld.-st.-7-2025-2026/id3155109/>

S75 Bank of Ghana, Ghana Petroleum Funds

Official fund page describes Stabilisation and Heritage objectives, semi-annual reporting and annual/quarterly audit framework

Limit: Ghana's statutory allocation formula and production history differ from Guyana; compare mechanisms rather than outcomes by headline balance

<https://www.bog.gov.gh/ghana-petroleum-funds/>

- S76** Bank of Ghana, 2026 Ghana Petroleum Funds reporting index
 Official reporting index includes 2025 audited statements and 2026 semi-annual report
 Limit: Use individual statements for numerical comparisons and preserve reporting dates
<https://www.bog.gov.gh/ghana-petroleum-funds/gpf-financial-statements/>
- S77** Central Bank of Timor-Leste, Petroleum Fund reporting portal
 Official portal publishes annual, quarterly and monthly performance reports including 2025 annual reporting and 2026 updates
 Limit: Fund size and investment returns must be analysed alongside budget withdrawals, petroleum depletion and domestic fiscal dependence
https://www.bancocentral.tl/publication/pf_rep/
- S78** Central Bank of Timor-Leste, Petroleum Fund management framework
 Describes Liquidity and Growth portfolios and publication requirements
 Limit: Current portfolio structure reflects Timor-Leste's withdrawal needs and should not be copied without national cash-flow modelling
https://bancocentral.tl/stats/about_pf/
- S79** TotalEnergies EP Suriname, GranMorgu update 19 September 2026
 Operator states project FID occurred in October 2024, construction is progressing and first oil remains targeted for 2028
 Limit: Operator project update; planned 2028 first oil is not an observed production result
<https://totalenergies.sr/press/suriname-granmorgu-project-takes-shape-major-production-equipment-arrives>
- S80** Trinidad and Tobago Ministry of Finance, Heritage and Stabilisation Fund reporting portal
 Official portal publishes annual and quarterly HSF reports through 2026
 Limit: Mature producer and dual stabilization/heritage design offer mechanism comparison, not a direct blueprint for Guyana or a new island producer
<https://www.finance.gov.tt/publications/heritage-and-stabilisation-fund/>
- S81** Petroleum Management Programme, Oil Production dashboard
 Official production series covers 20 December 2019 through 31 July 2026
 Limit: Monthly realized production must remain separate from installed/nameplate capacity
<https://petroleum.gov.gy/data-chart/oil-production/>
- S82** ExxonMobil, Liza Phase 2 first oil, 11 February 2022
 Operator records first production from Liza Unity on 11 February 2022 and planned ramp to 220,000 bpd
 Limit: Operator statement is suitable for startup date; realized monthly production should use regulator data
https://corporate.exxonmobil.com/news/news-releases/2022/0211_exxonmobil-starts-production-at-guyanas-seacond-offshore-development
- S83** Petroleum Management Programme, Payara first oil, 14 November 2023
 Government records first oil at Prosperity in November 2023; Liza Destiny started December 2019 and Liza Unity February 2022
 Limit: Government media release includes future production estimates that must not be treated as observed output
<https://petroleum.gov.gy/article/govt-welcomes-first-oil-at-prosperity-fpso/>
- S84** ExxonMobil, Yellowtail first production, 8 August 2025
 Operator records startup of the fourth project and combined installed capacity above 900,000 bpd
 Limit: Installed capacity is not annual average production
https://corporate.exxonmobil.com/news/news-releases/2025/0808_exxonmobil-guyana-begins-production-at-fourth-offshore-guyana-project
- S85** ExxonMobil, 900,000 bpd milestone, 12 November 2025
 Operator reports a daily production milestone after Yellowtail ramp-up
 Limit: One-day/point-in-time output is not a full-year average

<https://corporate.exxonmobil.com/locations/guyana/news-releases/11122025-daily-oil-production-hits-900000-barrels-in-guyanas-stabroek-block>

- S86** Petroleum Management Programme, Liza Phase 1 Production Licence and Phase 2 Addendum
Official production-licence publication provides legal project authorization documents for Liza developments
Limit: Licence conditions require document-level review and do not establish realized performance
<https://petroleum.gov.gy/docs-and-pubs/liza-phase-1-petroleum-production-licence-and-phase-2-addendum/>
- S87** Parliament of Guyana, Petroleum Activities Act 17 of 2023
Establishes the newer statutory petroleum framework for licensing, regulation, data, safety, decommissioning and administration
Limit: Legacy contractual rights and transitional application require agreement-specific legal analysis
<https://parliament.gov.gy/publications/acts-of-parliament/petroleumactivitiesact17of2023>
- S88** Bank of Guyana, audited NRF financial statements 2025
Audited financial statements for the year ended 31 December 2025, including income, financial position, cash-flow and notes
Limit: Must be reconciled to Ministry reporting using consistent currency, timing and accounting basis
https://bankofguyana.org.gy/bog3/images/accounts_budgeting/natural_resource_fund/financial_statements/nrf_audited_financial_statements_2025.pdf.pdf
- S89** EITI Validation schedule, checked 27 September 2026
Lists Guyana among ongoing Validations and notes commencement on 15 May 2026
Limit: Ongoing status is not a final Validation decision or score
<https://eiti.org/validation-schedule>
- S90** International Court of Justice, Guyana v Venezuela case record
Official case docket for the controversy concerning the 1899 Arbitral Award, including jurisdiction, preliminary objection and provisional-measures material
Limit: The merits remain before the Court; this publication must not predict or decide the outcome
<https://icj-cij.org/case/171>
- S91** International Court of Justice, Order of 1 May 2025
Official provisional-measures order in Guyana v Venezuela
Limit: Interim judicial measure, not a final merits judgment
<https://www.icj-cij.org/node/205403>
- S92** CARICOM statement, 28 April 2026
Regional statement supporting peaceful resolution through the ICJ and addressing conduct concerning the Guyana-Venezuela controversy
Limit: Political/diplomatic statement; use for regional context, not for a merits determination
<https://caricom.org/statement-from-the-caribbean-community/>
- S93** Forest Peoples Programme, ART Guyana carbon-credit case study
Documents criticisms concerning ART certification, Indigenous safeguards and complaint mechanisms
Limit: Advocacy research; contested claims require triangulation with ART, government and community evidence
<https://www.forestpeoples.org/pt/publications-resources/reports/article/arts-controversial-certification-of-carbon-credits-to-the-government-of-guyana-a-case-study-on-challenges-for-high-integrity-labels-in-carbon-markets/>
- S94** Office of the President, Guyana Oil & Gas Capacity Building Project SESA terms of reference
Identifies strategic offshore and onshore environmental and social risk categories including fisheries, spill risk, marine habitats and coastal facilities
Limit: Terms of reference identify risks to assess; they are not measured outcome evidence
<https://op.gov.gy/guyana-oil-gas-capacity-building-project-p166730-terms-of-reference-tor-strategic-environmental-social-assessment/>

S95 LCDS, CORSIA-eligible ART TREES credits, 28 February 2024

Records issuance of 7.14 million 2021-vintage credits and CORSIA eligibility claims, including corresponding-adjustment context

Limit: Programme statement; registry and UNFCCC records should be used for final accounting verification

<https://lcds.gov.gy/guyana-announces-worlds-first-carbon-credits-for-use-in-un-airline-compliance-programme-corsia/>

S96 Ministry of Finance, NRF monthly reports 2026

Official monthly NRF reporting through July 2026 available at the publication cut-off

Limit: Monthly reports need reconciliation to quarterly, annual and audited accounts before aggregation

<https://finance.gov.gy/monthly-reports/>

S97 Stabroek News, Anand Goolsarran analysis of PSA revenue provisions, August 2025

Independent public commentary using NRF audited accounts to discuss profit-sharing, royalty and taxation

Limit: Commentary, not primary law or audit evidence; use only as an attributed critical interpretation

<https://www.stabroeknews.com/2025/08/04/features/accountability-watch/revisiting-the-profit-sharing-royalty-and-taxation-provisions-of-the-petroleum-sharing-agreement-part-ii/>

S98 Official Gazette of Guyana, Extraordinary Gazette, 28 May 2025, Act No. 6 of 2025

Confirms formal publication of the Oil Pollution, Prevention, Preparedness, Response and Responsibility Act 2025 in the Official Gazette

Limit: Publication confirms enactment/gazetting, not necessarily commencement of every provision or issuance of implementing regulations

<https://officialgazette.gov.gy/index.php/publications/2778-extraordinary-gazettes-28th-may-2025-act-no-6-of-2025-the-oil-pollution-prevention-preparedness-response-and-responsibility-bill-2025>

S99 Petroleum Management Programme, Uaru Environmental Permit, financial assurance conditions 14.3-14.5

Permit states financial assurance should be guided by reasonably credible costs, requires liability/environmental-liability insurance, and requires parent/affiliate guarantee agreements to provide resources for environmental obligations after default

Limit: Permit terms establish obligations and architecture; they do not by themselves prove current insurance limits, guarantee amount, liquidity or enforceability in a future loss event

<https://petroleum.gov.gy/wp-content/uploads/2024/10/Uaru-Signed-Permit.pdf>